MERCHANDISING MANAGEMENT SYSTEM (HUMAN RESOURCE 1): ONBOARDING AND OFFBOARDING

TASK MANAGEMENT, AND ONLINE JOB

POSTING WITH AI RESUME

EVALUATION

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THE RESEARCHERS

DEDICATION

This business research study is wholeheartedly dedicated first and foremost to the researchers, for executing dedication, time, effort, motivation, sacrifice, and courage to make this conducting study a fruitful and successful piece of work.

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And lastly, above all, to our Almighty God, for giving guidance, strength, power of mind, protection, skills and for giving us a healthy life. All of these we offer to you.

THE RESEARCHERS

ABSTRACT

Title: **MERCHANDISING MANAGEMENT SYSTEM (HUMAN RESOURCE 1): ONBOARDING AND OFFBOARDING, TASK MANAGEMENT, AND ONLINE JOB POSTING WITH AI RESUME EVALUATION**

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In the era of business expansion and digitalization, HR management has emerged as the backbone of organizational efficiency, especially for companies that heavily rely on effective recruitment and onboarding. The recruitment and onboarding processes are delayed, thereby resulting in more errors and overall reduced productivity by traditional recruitment processes and manual paper work. With the vision to address the issues,

this project integrates critical HR functionalities, such as an AI-based Resume Evaluation, Online Job Posting, and Task Management in a Merchandising Management System. This system is designed to optimize job posting, candidate tracking, and employee task management by offering an end-to-end, technology-enabled solution, thus enabling seamless transition from hiring to onboarding and organizational effectiveness and employee engagement.

The agile scrum technique used in this project enab

les incremental system modifications based on input from stakeholders. Surveys and interviews are used by the researchers to collect information on the efficacy and usability of the system. The Applicant Tracking System, Onboarding Management, and Employee Portal were all integrated into the system, which was created in a modular fashion.

The Merchandising Management System has significantly improved HR operations. The company decreased the amount of manual labor needed for applicant screening by improving the posting of job hiring and application tracking. The use of AI-based resume review sped up the recruiting process by finding a qualified candidates based on a predetermined criteria. By managing tasks efficiently, the employee portal ensured that new recruits completed the required onboarding tasks. Additionally, the offboarding module made sure that exit procedures were professional and well-documented. The solution increased overall employee engagement, reduced processing time, and improved HR efficiency.

Compared to conventional HR practices, utilization of AI-based recruitment and online task management yielded several benefits. The capability of the system to automate basic HR processes resulted in less human error and workload. The addition of an interactive onboarding and offboarding system also facilitated an organized and employee-oriented system. However, issues of user adaptation and data security problems had to be resolved, including additional streamlining such as AI-based learning programs and upgraded information security protocols.

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Chapter 1

INTRODUCTION

1.1. Background of the Capstone Project

Great Wall Arts is a wholesaler and retailer upstart company that is widely focused on selling local hand-made products. The company has established a reputation for quality and uniqueness in its services by promoting local artists and encouraging traditional craftsmanship. The need for a much better, more systematic approach to human resource functions and, especially in such areas as recruitment and task management comes with expansion of the organization.

Merchandising Management System: Human Resource 1 addresses the problems of the inefficient recruitment process, and difficulties in tracking employees' requirements and progress. This system provides online job posting, which allows the company to advertise jobs online to easily reach a wider range of people that might be interested in applying. It includes an online application form where the applicants can fill in their basic information and upload a résumé. The system also provides an Interactive Onboarding Timeline, where the employee can upload their required documents online, whereas HR will be notified in real-time and will go through the submitted documents. It can set up a resubmission process when there are any errors or mistakes in the submitted document that were discovered by HR. This helps to ensure the HR and employees complete their important tasks on time.

Additionally, this project will help enhance the efficacy and transparency of HR operations inside the company. The features of Merchandising Management System: Human Resource 1 may also be able to help in providing an accessible experience for both applicants and employees.

1.2. Context and Scope

One of the challenges that the Great Wall Arts encounter in the merchandising industry is the delays and communication gaps between employees and the HR department, which may be a result from a slow and inefficient usage of traditional HR practices.

The Merchandising Management System: Human Resource 1 operates in the retail industry, a project that focuses on enhancing human resource management. This system includes online job posting, allowing interested applicants to apply online, employee onboarding which features an employee portal for task management, and offboarding, where employees are allowed to submit their resignation letters online. However, the system's scope is restricted to the Applicant Tracking System, and Employee Onboarding and Offboarding; it does not include Payroll Management, as well as other broader HR services.

1.3. Problem Statement

The company faces problems such as:

1. Lack of the number of applicants reached by using traditional recruitment methods, such as job posting on social media or employee referrals;
2. Manually tracking applicants could result in discrepancies when handling a huge number of candidates, especially during the applicant screening; and
3. Time-consuming and vulnerable to error manual data entry and record-keeping using Excel and Google Sheets.

The project aims to solve these problems by integrating an online job posting, applicant tracking system, and long-term data storage with search functions in our system, easily allowing the HR department to access or retrieve previous candidate profiles or employee performance data.

1.4. Goals and Objectives

The main goals and objectives of Merchandising Management System: Human Resource 1 project are as follows:

1. Create an online application form within the job posting feature to reach a wider range of applicants for easier recruitment;

2. Develop an interactive onboarding timeline for employees to track and complete the tasks with notifications for pending or overdue tasks;
3. Implement an employee portal for real-time task monitoring and personal information updates; and
4. Providing online resignation submission feature to reduce the need for personal meetings unless required.

1.5. Significance and Relevance

The significance of Merchandising Management System: Human Resource 1 lies in its capacity to streamline HR processes such as online job posting, task management, and offboarding. By reducing the need for traditional methods, these features improve operational efficiency allowing both employees and HR teams to handle tasks more effectively. This allows efficient decision-making and gives the company an improved understanding of HR activities. This project is highly relevant in a modern business environment, where efficient HR processes can significantly impact a company's operational success.

For the Great Wall Arts, this project has several advantages since it decreases manual workload and improves data accuracy by automating key HR procedures like job posting, employee onboarding and offboarding.

These advantages include efficiency in the recruitment process, improved employee satisfaction and organisational transparency.

For the HR Department, this project encourages a more organized and accessible method to monitor the status of applicants as well as current employees, ensuring smoother transitions during hiring and resignations. By automating these tasks, HR team can focus on more strategic activities which will contribute to the company's success.

For the Employees, this system is a user-friendly portal that facilitates task management, document uploads, updates regarding the employment status, and provides clear communication between employee and HR team resulting in an enhanced overall employee experience and satisfaction.

For the Future Researchers, this project will become a basis for future studies relating to streamlining HR processes. This will create opportunities to research how technology may further enhance HR operations, improve organizational efficiency and adapt to evolving requirements for labor.

1.6. Structure of the Document

This paper is divided into five sections, including the following: the different aspects of the capstone project. The system's design, development process, and implementation details, testing and evaluation of the final system.

Chapter 1: Introduction

Provides the background of the study, statement of the problem, objectives, significance, scope and limitations, and the definition of terms. This chapter establishes the foundation and purpose behind developing the Merchandising Management System: Human Resource 1.

Chapter 2: Review of Related Literature and Systems

Discusses existing systems, theories, and previous studies related to human resource management, applicant tracking systems, onboarding, and offboarding. This chapter helps justify the need for a more efficient and integrated HR platform.

Chapter 3: Methodology

Details the research design, the Agile Scrum methodology applied, roles and responsibilities, sprint cycles, system modeling (such as Use Case Diagrams, Sequence Diagrams), and tools used for the project's development.

Chapter 4: System Development and Implementation

Focuses on the technical aspects of the system. It presents the Requirement Analysis, Business Process Architecture (BPA), Application

Architecture, Data Architecture, and Technology Architecture. This chapter also covers the Development Process, Implementation stages, Testing and Quality Assurance methods, and Results and Evaluation, supplemented with detailed tables, figures, and diagrams to illustrate system workflows and structures.

Chapter 5: Summary, Conclusions, and Recommendations

Summarizes the findings of the project, discusses the project achievements, lessons learned, and provides recommendations for future improvements. It reflects on how the system enhanced HR processes and suggests potential enhancements to further optimize performance.

At the end of this paper, readers will have an extensive knowledge of the Merchandising Management System: Human Resource 1 and its impact on company's HR processes' efficacy.

Chapter 2

RELATED STUDIES AND LITERATURE REVIEW

2.1. Agile Scrum Methodology Overview

By breaking down development into smaller and easier to manage sprints, Agile Scrum methodology allows for a flexible and iterative development and guarantees that each system feature is being developed progressively. Regular sprint reviews and retrospective maintain satisfactory outcomes, while reducing risks by identifying and resolving issues early in the development stage. This approach fosters active collaboration between the development team and HR, which results in a system that meets the needs of the client and adapts quickly to changes.

Agile Scrum Related Studies and Research

This section provides five related studies and research about agile scrum methodologies that were implemented in their project successfully:

Adkins, J. K., & Tu, C. (2019). A capstone course can be beneficial for students and potential employers, according to research. Included are the actions done to set up the capstone course and get ready for the course. Students used Scrum, an agile approach that consists of five sprints and frequent meetings. *Applying an agile approach in an Information Systems capstone course. Information Systems Education Journal*, *17*(3), 41–49.

Zayat, W., & Senvar, O. (2020). Kanban is a scheduling system that uses visual cues for work-flow management; and Scrum is an agile project completion framework based upon division of activities into sprints that are more manageable chunks. It is very evident that the results are quite flexible both towards the Agile goals. Scrum is ideal for even newer and more complex projects while focusing on making it involving with a continuous flow environment by constantly trying to approach system improvement wherein such corporations among customer and development teams bring all such essential skill, like planning and organizational capacity; presentation, as well as review requirements. *Framework Study for Agile software Development via SCRUM and KANBAN. International Journal of Innovation and Technology Management*, 17(04).

Rush, D. E., & Connolly, A. J. (2020). With Scrum serving as the organizational logic for completing coursework, this article offers a methodology for teaching an entire semester-long IT project management course that includes standard PMI-based curriculum (sans software development). In order to facilitate the organization of student teams, assignments, and activities, this framework adapts popular Scrum practices from the industry for use in the classroom. The goal of integrating Scrum into an existing course is to optimize student learning of both traditional project management material and the socially complicated, hard-to-teach "soft" skills that contribute to the success of Scrum teams. Without consuming too much time or resources, this thorough incorporation of Scrum into a conventional, predictive IT project management course goes far beyond individual tasks or units*. An Agile Framework for Teaching with Scrum in the IT Project Management Classroom. The Journal of Information and Systems in Education*, *31*(3), 196–207.

Jiménez, V., et al. (2020). The most effective costing system is the one that is most tailored to the needs of the company, not necessarily the most comprehensive or accurate. This process relied heavily on the iterative process, many sprints, and the establishment of the product backlog. The performance and financial sustainability of organizations were improved by the adoption of the Scrum methodology, which made it possible to design and implement an ABC system that was better suited to the needs of the company and to identify issues early on, when they are typically only noticed at the end of the implementation process. *Using agile project management in the design and implementation of Activity-Based costing systems. Sustainability*, 12(24), 10352.

Duarte, V., et al. (2023). This is among the first studies to assess the use of Service-Learning approaches in conjunction with an Agile framework (such as Scrum) in educational settings. The Service-Learning methodology, which allows students to gain practical experience by working with real-life issues and firms, is another instrument that would help improve the quality of educational delivery. Results from a case study in the course demonstrate that this combination of the agile framework and service-learning improved teamwork, time management, and general job satisfaction. *Information system development by using agile teamwork and Service-Learning. IEEE Transactions on Education*, 66(5), 431–441.

2.2. Enterprise Architecture Concepts

This section outlines how the system achieves its goals and objectives through key functionalities such as Application Tracking System (ATS), Onboarding Task Management, Employee Portal, and Offboarding. Enterprise Architecture (EA) concepts guide the design and development of the system, ensuring alignment with client expectations and business needs.

It utilizes the The Open Group Architecture Framework (TOGAF) Architecture Development Method to guarantee smooth integration with constituents such as Application Tracking System (ATS), Onboarding, as well as the Employee Portal. In addition to the benefits of planning at the front end, it supports interaction between system components and other internal applications during development and implementation. Interoperability, scalability, and security would thus be improved.

By following Enterprise Architecture principles, the system is designed to be scalable to accommodate future growth, secure to protect sensitive data, and flexible to support future expansions and

Business Architecture

The system is designed to optimize the recruitment and employment process in merchandising. The following modules represent the core business process the system supports:

Applicant Tracking System (ATS): Online Job Posting, Applicant Evaluation, Managing Candidates, Interview Scheduling

Onboarding: Pre-employment Requirements, Task management.

Employee Portal: Task Board, Pre-Employment Requirement Submission, Document Uploads

Offboarding: Resignation Tracking, Exit Interview Scheduling, Account Termination Request and Notification

Application Architecture

The system possesses a modular architecture in its design of the application where different interdependent components will be working in perfect harmony. All modules; Applicant Tracking System (ATS), Onboarding, Employee Portal, and Offboarding, are isolated but fully integrated in such a way that they interact with each other. The system was done with Laravel Blade and Livewire for semi-Single Page Application (SPA).

User Interface: Built using Tailwind CSS interface meant to be user-friendly on all major devices compatible.

Backend System: Powered by Laravel, handling business logic, data processing, and system integration.

Data Management: The system data is stored in a centralized MySQL database for secure and efficient data handling.

Data Architecture

The system follows a centralized data architecture using a database and conceptual data model. The design was done as an abstracted representation of data connected with various functions such as Recruitment, the Applicant Tracking System (ATS), Onboarding, the Employee Portal, and Offboarding.

The model outlines how these entities interact and relate to one another through their attributes, which include:

Applicant Details: Such information includes names, contact information, email, and other basic information about the applicant relating to their application status.

Job Title: Titles of positions being applied for, along with their descriptions and requirements.

New Hire Employee Details: Includes personal details of new-hired employees, employment histories, documents, and positions in the job.

Departments: Departments of an organization consisting of staff that execute different functions with related processes.

The use of Universally Unique Identifiers (UUID) in the application database ensures there is a safe storage such that none other than him can perform manipulations involving data pertaining to the Job posting section of the Applicant Tracking System.

Technology Architecture

Technology architecture in an enterprise system refers to the blueprint for the application of various technologies that support the functionality, scalability, performance, and security of a web application. It is regarded as the base of enterprise architecture and forms the basis for the establishment of the technical structure, guidelines, and standards that should govern an organization's IT environment to effectively meet its business needs today and into the future. Technology architecture is such a significant part of an enterprise architecture, outlining its technical structure and requirements or the IT environment of a company. It defines exactly what software and hardware system development requires, including every detail of the techstack, which involves frontend UI/UX design, rear-end infrastructure such as on-cloud servers and domain hosts, and network topology required to support the organizational flow.

Core Components of Technology Architecture

Tech Stack: Technology Architecture-Definition of the overall tech stack powering your web application, that is, the entire system layers from the frontend to the backend.

Frontend: System are developed from frontend frameworks, which are Laravel Blade for dynamic functions, Tailwind CSS for an interactive user interface, and Alpine.js and JavaScript that give dynamic functionalities to the web application.

Backend: One of the most featured features of the Laravel framework-elegant syntax, feature set, and developer tools. It is applied in core application logic management, data processing, business rules for smooth interactions, and data flow. All in-built features like routing, authentication, and caching make the complexity of backend operations reduced and security and performance improved. It can further be used in the development of REST APIs. This further makes easy communication between the frontend as well as other systems and with MySQL databases.

Infrastructure Layer: This architecture defines a set of definitions to be run by an application on the infrastructure. The domain hosting services are what link an application to a web domain so it may be accessed by end-users. Domain hosting often involves supports for load balancing and anti-Distributed Denial of Service (DDoS) attack to ensure safe, fast content delivery around the world.

Network Topology: This system's Star Topology was chosen for its affordability and suitability for smaller configurations. The concept offers centralized network management through the network server's provision of straightforward resource allocation, centralized storage, and printer access. Department-specific workstation connections reduce the likelihood of unauthorized access, and a router with built-in firewalls enhances security by protecting against external threats. Data redundancy is provided via local backup and network storage, preventing data loss.

Furthermore, the architecture is scalable, allowing additional workstations and storage to be installed as needed. The system offers high-speed performance by connecting devices directly to a central switch, especially for essential wired equipment, even though wireless connectivity makes portable devices like laptops and phones possible. Services that require a dedicated web hosting connection are supported by are accessible to the general public. This configuration is ideal for the reliable, secure, and efficient operations of small and medium-sized businesses because of its simple, hierarchical structure, which facilitates troubleshooting and reduces the likelihood of network congestion.

Integration with Other Architecture Domains

Integration across architecture domains ensures alignment with organizational goals.

Business Architecture: Technology supports recruitment and employee management goals, securing and streamlining resource access.

Application Architecture: Modular components using Laravel, Tailwind, and APIs enable smooth interactions across Applicant Tracking System (ATS), Onboarding, and Employee Portal, promoting flexibility.

Data Architecture: Centralized MySQL ensures secure, consistent data handling across modules, using Universally Unique Identifiers (UUID) to maintain unique identifiers and data integrity.

This integration creates a scalable, secure, and cohesive system that meets current and future organizational needs.

TOOLS	DESCRIPTION
1. Laravel	Laravel is a free, open-source PHP framework for building web applications. It's designed to make web development more enjoyable and creative.
2.Tailwind CSS	Tailwind CSS is a free, open-source framework for building websites using CSS. It lets you build layouts directly in HTML markup without writing additional CSS.
3. Livewire	Livewire is a framework for building dynamic web interfaces using Laravel. It allows developers to create interactive components, like forms and tables, that update in real time. Livewire is part of the TALL stack, a modern approach to building web applications.

4. Alpine.js	Alpine.js is a JavaScript framework that helps you add interactivity to web pages. It's a lightweight alternative to larger frameworks like Vue or React, and it's easy to use.
5. MySQL	An open-source relational database management system that stores information in tables composed of rows and columns, just as other relational databases. Structured Query Language, or SQL as it is more often known, allows users to define, manipulate, control, and query data.
6. Indevfinite Web Hosting	Indevfinite was founded to offer flexible system integration, with an emphasis on website design, maintenance, and development as well as a range of software development for your company's requirements.

Figure 1. Technology Architecture

Enterprise Architecture Concepts Related Studies and Research

Dumitriu, D., & Popescu, M. A. (2020). Technical descriptions of an architecture in a particular business domain can be analyzed, comprehended, and applied with the help of an Enterprise Architecture Framework (EAF). For many businesses, creating and/or modifying a suitable EAF has been one of the most difficult problems in recent years. As a result, the research presented in this paper highlights the advantages and disadvantages of the major Enterprise Architecture Frameworks through a comparative comparison. The results mostly pertain to what and how the Enterprise Architecture Framework should be designed in order to maximize business benefits and streamline operations. *Enterprise Architecture Framework Design in IT Management. Procedia Manufacturing,* 46, 932-940.

Niemi, E., & Pekkola, S. (2020). As a planning and governance strategy, enterprise architecture (EA) has been widely used to manage complexity and ongoing change while bringing the business together around a shared objective. This article explains how EA benefits are realized in order to study the EA benefit-realization process. In particular, the emphasis is on the methods, materials, and procedures that provide the EA with advantages. The results of a thorough case study demonstrate that the EA benefit-realization process is a lengthy, intricate series of actions. Businesses gain from EA in a number of ways, from the beginning, when a thorough knowledge begins to take shape, to years later, when quantifiable results, such cost reductions, become apparent. *The Benefits of Enterprise Architecture in Organizational Transformation. Business and Information Systems Engineering,* 62, 585–597.

Kotusev, S., Kurnia, S., & Dilnutt, R. (2022). The hypothesis claims to address a number of various enterprise archictecture (EA) artifact characteristics and their interrelationships. The first comprehensive theory explaining how various enterprise archictecture (EA) artifacts are used in organizations is presented in this paper. This study advocates for more research on various enterprise archictecture (EA) artifacts and their use in various enterprise archictecture (EA) practices, and it proposes that the various enterprise archictecture (EA) research community should concentrate on examining individual various enterprise archictecture (EA) artifacts rather than investigating various enterprise archictecture (EA) in general. *The practical roles of enterprise architecture artifacts: A classification and relationship. Information and Software Technology,* 147.

Grave, F., et al (2023). There is no denying the importance of enterprise architecture (EA) in strategic renewal initiatives like digital transformations. Businesses depend on enterprise architecture (EA) to help with strategic renewal, particularly strategic renewal driven by digital technology. Information regarding the enterprise architecture's (EA) present, future, and transitional states is recorded and disseminated through enterprise architecture (EA) artifacts. They make it easier to synchronize and bridge the communication gap between IT and business stakeholders. The mechanism by which enterprise architecture (EA) artifacts contribute to organizational performance improvement and strategic renewal is yet unknown, despite the fact that there is a wealth of study in this field. *Enterprise Architecture Artifacts' Role in Improved Organizational Performance. Lecture Notes in Business Information Processing,* 483.

Halawi, L., et al (2019). From the mid-1980s onward, enterprise architecture has been evolving steadily. Despite 35 years of research and application, there are still no universally accepted definitions or standards. The expansion of the number of enterprise architectural frameworks is evidence of this. Nonetheless the substantial amount of research, terminology needs to be standardized based on a meta-analysis of the literature. To be successful, enterprise architecture initiatives need support from all levels of the company and be seen as valuable additions. *Where We Are with Enterprise Architecture. Journal of Information Systems Applied Research,* 12(3).

2.3. Micro-services Architecture

Microservices is one architecture pattern where individual applications are broken down into separate services. Typically, all the services run in a separate environment and interplay with other services via API. This architecture thus benefits more than a typical traditional monolithic architecture majorly because of flexibility and scalability and also because of independently developed, deployed, and maintained services.

The microservices architecture plans to separate modules into independent microservices using REST API such as:

Applicant Tracking System (ATS): The Applicant Tracking System is implemented as a standalone microservices to handle overflowing applications from Job posting to ensures smooth application and mass hiring process. The other services of Applicant Tracking System (ATS) is interview scheduling where allowing the HR to schedule interviews seamlessly, it integrates with calendar (Google Calendar) and sends email interview for applicants.

Onboarding: The Employee Portal and Task Management System will have the support of REST APIs to handle request processing so that people may get onboarded more efficiently.

Offboarding: The microservices shall support in managing notifications and updates related to account termination requests.

The Benefits of Implementing Microservices Architecture include:

Flexibility: This system consists of multiple modules and features. Microservices allow different developers to work on the system using various frameworks, programming languages and databases that best suit the specific features being developed.

Scalability: The other big advantage of microservices is that it allows the separate modules and features to be scaled up or down on demand without affecting other services, which would make the system more effective in handling varied kinds of workloads.

Independent Development: Every module and feature has been developed, published and also controlled separately. In this, therefore, a development team works on different independent modules and features.

Reduced Downtime: With the flexibility of independent development and scaling of modules, microservices lessen the deployment time for the new feature as well as the scale time of existing ones. Others can remain up, while one remains down during maintenance; thus, this raises the availability of a system.

The challenges of microservice architecture include:

Complexity of Management: Managing multiple microservices can be complex, especially as the number of services grows. This includes monitoring, logging, and maintaining dependencies between services.

Security: Microservices require additional security due to the increased system complexity. In a microservices architecture, the system is broken down into modular components, each with its own features, running in separate containers, and often communicating over a network. Each service also manages its own databases, making them individual points of potential attack.

Service Communication: Inter-service interactions are essential to microservices. It can be difficult to manage several modules with separate APIs and protocols.

Testing: Testing is challenging in microservices due to its nature, it requires comprehensive and varied strategies, as traditional testing methods may not be sufficient. Each module needs to be tested independently, as well as in integration with other features.

2.4. DevOps and CI/CD

DevOps is a methodology that combines “Development” and “Operations,” emphasizing collaboration between software developers

(Dev) and IT operations (Ops) teams. This approach aims to automate, integrate, and enhance the software development and deployment processes.

The primary goal of DevOps is to shorten the systems development life cycle while ensuring software quality and security. By enhancing the ability to release new features, quickly detect bugs, and respond to reports, DevOps promotes the continuous operation and monitoring of systems, ultimately improving the reliability of software releases.

This section explains how the Developers and the IT teams deliver Continuous Integration (CI) and Continuous Deployment/Delivery (CD) throughout the project. The Developers use the CI workflow in that they trigger GitHub Actions after committing and pushing code to the repository. This process initiates the build, testing, and staging processes, preparing the application for deployment into production. IT teams monitor and run it in production, so to assure continuous and smooth development and deployment; manage all the tasks connected with performance and reliability of the system.

CI Pipelines Used by Developers

The following CI pipelines and commands are utilized by the development team:

Commit code process: Developers commit their code changes to the local repository (GIT) and push them to the remote repository on GitHub. This action triggers the GitHub Actions workflows defined for the repository.

Laravel CI/CD Pipeline: After the code was committed the push on main repository will Trigger the GitHub Actions and staging branches and on pull requests.

This pipeline consists of multiple jobs:

Build: The Build fetches code from a repository, gets your PHP setup, installs the project with its dependencies, creates your.env file, generate your application key, compilation of frontend assets, then notifies on status of build.

Test: This tests the repository, PHP configuration, installation of all the dependencies, environment variables setup, and database migrations ran by Pest and PHPUnit relative to the test. And it ends reporting the test status.

Deploy to Staging: This will run as code is pushed to the staging branch. Deploy an application to the staging server through SSH while running necessary commands.

Deploy to Production: It is automatically initiated whenever code gets pushed to the main branch and does the same as it does when it is in the production server.

2.5. Relevant Studies and Research

Research that have been done regarding challenges and opportunities on Applicant Tracking System and Human Resource Management System can give good knowledge for the development of our Merchandising Management System: Human Resource 1. Here, we enumerate some of the local and foreign research that focus on the importance of integrated solutions and streamlined HR procedures:

Local Studies

Grimaldo, J., et al. (2020). The study indicated that both groups of respondents found the tools simple to use, which had an impact on their attitude and intention to use or reuse them. While the hiring managers thought the current tools were restrictive, the job seekers considered the tools helpful in influencing their attitude and intention to utilize. Nonetheless, the intention of recruitment officers to utilize the instrument is influenced by their attitude regarding its utilization. Lastly, more features for online hiring tools were suggested. *Utilization of e-recruitment tools as perceived by recruiters and job applicants.*

Pineda, C., & Cayabyab, L. A. (2023). Labor market intermediaries are firms that provide a meeting venue for both employers and employees. They therefore facilitate the process of transaction facilitation and networking. LMIs that exist only in a digital platform are called DLMIs. DLMIs have been in increasing demand as a method of posting job offers and conducting job search in the Philippines. Because of their significant impact on the labor market, DLMIs are not adequately covered by the present recruitment and placement rules due to the vagueness in the application of Article 13, Section B of the Labor Code of the Philippines (LCP) of digital platforms. *Facilitating employment opportunities in the digital space: A study on the activities and services of digital labor market intermediaries in the Philippines*.

Baconawa, K. (2020). Online Task Management System assists in navigating the tasks they need to do and will be alerted when a task is still not completed by the given timeline. By using this system, it eliminates all the time-consuming and needless meetings, thus maximizing the time in doing their job efficiently. *Efficient Online Task Management System for NORSU Offices.*

Fetalvero, S. (2024). An online poll on HRM practitioners' perspectives of e-recruitment was conducted using a customized questionnaire. According to the results, the respondents' opinions about e-recruitment are overwhelmingly positive. The primary factors influencing their attitude toward e-recruitment were their opinions on its usability and convenience of use. *Predictors of Attitude and Intention to Use E-recruitment among Human Resource Management Practitioers in the Province of Romblon. Romblon State University Research Journal, 6(1), 9-18.*

Pontillo, H. R., et al. (2024). Recruitment managers have encouraged to adopt social media recruitment (SMR) as it gained recognition due to its cost-effectiveness and quick vacancy fulfilment capabilities. *The Mediating Role of Credibility and Satisfaction in Social Media Recruitment Outcomes. International Journal of Science and Management Studies (IJSMS),* 7(2).

Foreign Studies

Gupta, S. (2024). Through the automation of the tracking, management, and organization of job applications, the Application Tracking System (ATS) project seeks to improve the candidate experience, decrease the time and effort required for the hiring process, and offer a centralized platform to handle all recruitment-related tasks. *Application Tracking System*. *Global International Management Research Journal,* 12(8), 77.

Spezie, M., & Bragantini, D. (2023). Without specialized software to manage various tasks and schedules, it becomes challenging to remember all of the sub-activities of a given project, assign the appropriate importance to them, and then balance them with the daily routine of the office: if time is wasted, resources must be added to finish the necessary work, which will unavoidably result in higher costs and penalize the entire company in terms of effort and performance object achievement. *The Development of a Task Management Software (TMS): A bridge between project management's sub-activities (especially in multiproject context) and "ordinary" assignments to follow. PM World Journal*, 12(4).

Chen, H., & Cui, X. (2022). The traditional HRM mode is ineffective and lacks management ability, which cannot meet the needs of the current information society. With the continuous improvement of information technology, it is imperative to apply computer technology and network technology to HRM. The number of small and medium-sized enterprises is not only greatly increasing these days, but their scale is also expanding rapidly, making HRM more and more difficult and the corresponding management cost rising. *Design and implementation of human resource management system based on B/S Mode. Procedia Computer Science*, *202,* 949-954.

Tiwari, A., et al. (2019). Businesses find it challenging to as it takes a long time to sift resumes based on their needs as a result of the increasing competition and the increasing skill levels of candidates. Using a program called ATSS which helps HR team in sorting resumes based on various criteria, the business may obtain a sorted list of candidates who meet the company's requirements rather than having to go through each application. *Applicant Tracking and Scoring System*. *International Research Journal of Engineering and Technology (IRJET),* 6(4).

Idris, M. (2023). The manual recruitment system is no longer effective to hire applicants or to fill vacancies in the organization, as it is negatively impacting HR department's performance. However, online recruitment helps HR department by facilitating a huge number of job applicants and it also provides candidates a chance to apply for more than one vacant job that meets their skills. *Design and Implementation of an Online Recruitment System for Youth Graduates.*

2.6. Integration of Information Systems in Enterprise Environments

The system was developed to automate and streamline HR processes for Great Wall Arts, functioning as an enterprise resource planning (ERP) by integrating key modules such as Applicant Tracking System (ATS), Onboarding, Employee Portal, and Offboarding. By adopting Agile methodology, the system enhances integration flexibility and adaptability, allowing teams to modify strategies based on new information and changing requirements, ensuring ongoing relevance and effectiveness.

Agile Principles for Integration

Iterative Development: Making the development of the system incremental allows for minimal modifications. This makes it simpler to test and identify integration issues early on by allowing teams to integrate modules gradually.

Client Feedback Summary: Regular communication with stakeholders throughout the development process ensures that the system aligns with user needs and business goals. This ongoing interaction validates integration effectiveness and confirms that all requirements are met before full deployment.

Utilization of CI/CD: To automate testing and deployment procedures using CI/CD pipelines ensures that integrations are continuously tested and deployed, reducing manual effort and errors before deployment.

Benefits of Integration

Streamlined operations: Integrating and automating the recruitment and employment process in the HR system, reducing manual tasks and decreasing HR workload. This contributes to faster processing times and greater productivity.

Improved Data Accuracy: Automating HR processes reduces human error, ensuring that data is recorded accurately and minimizing discrepancies in applicant data management.

Refined User Experience: Establishing a comprehensive, user-friendly system provides real-time access to information across integrated modules and services, improving user workflows and overall satisfaction.

Challenges of Integration

While Agile development offers numerous advantages for system integration, challenges remain such as:

Integration Complexity: Connecting multiple modules and ensuring seamless data flow can be technically challenging and time-consuming, requiring careful preparation and the implementation of appropriate integration methods, such as utilizing middleware and APIs to make connections.

Realtime Data Synchronization: Providing real-time access to integrated modules is vital for a smooth user experience, but keeping data consistently updated across systems without delays or discrepancies is challenging.

Managing Changing Requirements: Frequent changes in system requirements or business needs can complicate the integration process. When making incremental modifications, careful coordination is essential to ensure that existing functionalities continue to operate smoothly without interruptions.

Microservices architecture will be used with the whole system being divided down into smaller, isolated modules that can exist independently in just one environment yet coordinate with each other. With this, all intercommunications of these microservices occur through REST APIs by having an API gateway for safety and security, so excellent performance will happen.

The development team will break down the process into manageable tasks by using an agile methodology. Here, the objective is to produce a standard and efficient business process and workflow by optimizing the entire HR recruitment process-that is, from initial recruitment to employment.

Chapter 3

METHODOLOGY

3.1. Agile Scrum Methodology in the Project

Using the Agile Scrum method, the project aims to construct a comprehensive system that includes features in Applicant Tracking System, Onboarding and Offboarding for a Merchandising Management System: Human Resource 1 such as:

a. Document Upload
b. Application Tracking
c. Employee Portal
d. Real-time Task Management
e. Direct Offboarding Alerts

The Agile Scrum methodology enables constant feedback and development which is crucial for adjusting the system to meet the specific demands of the client.

Key Phases of Development

1. **Planning Phase:** The project will begin with the construction of a product backlog where all desired features such as online job posting, the ability to track applications in real-time, and the ability to upload documents will be included.

2. **Sprint Execution:** The development team will carry out their tasks in sprints, that last at intervals of two to four weeks. Specific features will be developed, tested, and evaluated during each sprint.
3. **Feedback and Improvement**: Regular sprint reviews and retrospectives will be conducted. These sessions will evaluate the current situation of the project, analyze challenges that may encountered, and take feedback from stakeholders into consideration.

3.2. Roles and Responsibilities

Specific roles and responsibilities are essential for efficient project execution and delivery in the Agile Scrum method used to design the project.

a. **Product Owner** – responsible for outlining the project’s requirements and prioritizing the product backlog to make sure the system meets the client’s demands.

b. **Scrum Master** – responsible for ensuring the team adopt Agile principles and scrum practices and remove any challenges that may be encountered by the development team.

c. **Development Team** – in charge of designing and implementing the functionalities of the system. They are composed of developers, designers, and testers.

3.3. Sprint Cycles

The Agile Scrum methodology will be utilized for the development of Merchandising Management System: Human Resource 1, with the project organized into numerous sprint cycles, that last two to four weeks each, to guarantee steady progress and regular enhancements.

Sprint	Assignee	Duration	Goals and Objectives
Kick-off meeting	Noora	2-3 hours	To communicate with the team and plan for the project.
UI Dashboard Figma Sample Designing	Rosario	1 week	Creating UI design sample for the system in Figma
Interview with client	Noora, Dela Torre, Camo, Cullo, Rosario	1 week	Look for a possible client and interview

UI/UX Design Layout for Feature 1 (Online Job Posting)	Dela Torre	1 week	Creating UI/UX layout for dashboard of the first feature
Feature 1 Functionalities and Database	Camo	1-2 weeks	Implementing Online Job Posting functionalities with database
Chapter 1	Rosario	1 week	Completion of Chapter 1 of the document for checking
Developing UI/UX for Feature 2 (Applicant Tracking System)	Dela Torre	1 week	Creating UI/UX layout for dashboard of the second feature
Meeting with client	Noora	2-3 hours	Talking with the client about additional demands or changes

UI/UX Layout Modifications	Dela Torre, Noora	1 week	Adjusting the UI/UX layout design with the client’s demands.
Feature 2 Functionalities and Database	Camo	2 weeks	Implementing ATS functionalities in the system with database included
Chapter 2	Cullo	1 week	Completion of Chapter 2 of the document for checking and revisioning
Integration of Features 1 & 2 (Online Job Posting and ATS)	Camo	1-2 weeks	Integrating the two system features; Online Job Posting and Applicant Tracking System
Chapter 3	Rosario, Cullo, Dela Torre, Noora, Camo	1-2 weeks	Completion of Chapter 3 of the document with the collaboration of developers for diagrams

Revision of the document: Chapter 1, 2, & 3	Rosario, Cullo	1 week	Revising of the document to ensure it is aligned with the system
UI/UX Layout Design for Feature 3 (Onboarding Management)	Dela Torre	1 week	Designing UI/UX Layout for Onboarding Management in HR portal
Feature 3 Functionalities and Database	Camo	2 weeks	Implementing Feature 3 functions and a database
UI/UX Layout Design for Feature 4 (Employee Portal)	Dela Torre	1 week	Designing UI/UX Layout for Employee Portal
Feature 4 Functionalities and Database	Camo	2 weeks	Implementing functions and creating a database for Employee Portal

UI/UX Layout Design for Feature 5 (Offboarding)	Dela Torre	1 week	Designing UI/UX Layout for Offboarding feature
Feature 5 Functionalities and Database	Camo	2 weeks	Implementing functions and a database for Offboarding
Integration of Features 3 & 4	Camo	2-3 weeks	Integrating the two system features; Onboarding Management for HR dashboard and Employee Portal
System Revision	Camo, Dela Torre	1-2 weeks	Revision of some features in the system as per feedbacks.
Feature 6 Layout and Functionalities	Camo	2-3 weeks	Layout and implementing functionalities for the added feature Freedom Wall.

Integration of the whole cluster	Camo, Dela Torre, Noora	2-3 weeks	Communicating for Integration of the whole system

Table 1. Sprint Cycle

Each sprint will end with a review and retrospective evaluation to assess progress and highlight areas for further improvement, to ensure a flexible and adaptable approach to fulfilling the demands of the client.

3.4. Scrum Artifacts

No.	To Do	In Progress	Done
1			Kick-off Meeting
2			UI Dashboard Figma Sample Designing
3			Interview with client
4			UI/UX Design Layout for Feature 1 (Online Job Posting)
5			Feature 1 Functionalities and Database
6			Developing UI/UX for Feature 2 (Applicant Tracking System)
7			Meeting with client

8			UI/UX Layout Modifications
9			Feature 2 Functionalities and Database
10			Chapter 2
11			Integration of Features 1 & 2 (Online Job Posting and ATS)
12			Chapter 3
13			Revision of the document: Chapter 1, 2, & 3
14			UI/UX Layout Design for Feature 3 (Onboarding Management)
15			Feature 3 Functionalities and Database
16			UI/UX Layout Design for Feature 4 (Employee Portal)
17			Feature 4 Functionalities and Database
18			UI/UX Layout Design for Feature 5 (Offboarding)

19			Feature 5 Functionalities and Database
20			Integration of Features 3 & 4

Table 2. Scrum Board

Product Backlog

Application Tracking System

No.	User Story	Priorities	Revised Priorities	Status
	As an HR head, I want to create new job posting so that applicants can apply for vacant job positions.		**1**	**Done**
2	As an HR head, I want to track the status of each applicant's progress so I can monitor the hiring process.	1	**1**	**Done**
3	As an HR head, I want to update the status of candidates (accepted or	1	**1**	**Done**

	rejected) so that the applicant list is always updated.			
4	As an HR head, I want to schedule interviews directly through the system, so I can manage candidate meetings efficiently.	2	**1**	**Done**
5	As a recruiter, I want to be notified when new applications are submitted, so I can review them quickly.	1	**1**	**Done**
6	As a recruiter, I want to screen submitted applications, so I can manage the recruitment process and select qualified candidates.	1	**1**	**Done**
7	As a recruiter, I want to generate a list of all qualified candidates to streamline the hiring process.	1	**1**	**Done**

8	As an applicant, I want to fill out an application form online so that I can apply for a vacant position easily.	1	**1**	**Done**
9	As an applicant, I want to be notified of my interview schedule through email so I can prepare.	2	**1**	**Done**

Table 3.1 Application Tracking System Product Backlog

Onboarding Management

No.	User Story	Priorities	Revised Priorities	Status
1	As an HR head, I want to schedule a training for new hires so that they can attend training programs.	1	**1**	**Done**
2	As a new hire, I want to access pre-employment requirements list online so	2	**1**	**Done**

	that my onboarding process smooth.			
3	As a new hire, I want to receive updates about my onboarding progress so that I can stay on track.	2	**1**	**Done**
4	As a new hire, I want to receive my employee portal login credentials so I can access my tasks and schedules.	1	**1**	**Done**

Table 3.2. Onboarding Management Product Backlog

Employee Portal

No.	User Story	Priorities	Revised Priorities	Status
1	As an employee, I want to view my current assigned tasks so that I know what I need to complete for the day.	1	**1**	**Done**

2	As an employee, I want to monitor my task progress to stay updated on what task is still pending or completed.	1	**1**	**Done**
3	As an employee, I want to view my personal information so I can make sure my records are accurate.	1	**1**	**Done**
4	As an employee, I want to update my personal contact details in the employee portal to ensure that HR management have my most up-to-date information.	1	**1**	**Done**
5	As a resigning employee, I want to be able to monitor my offboarding process in the employee portal before my account is terminated, so that I can ensure a smooth exit.	3	**1**	**Done**

6	As an employee, I want to view the schedule for my training or exit interview through the portal so that I can be prepared.	1	**1**	**Done**
7	As an employee I want to upload my completed tasks or requirements through the employee portal so that HR management can review them.	1	**1**	**Done**
8	As an employee, I want to be notified when new tasks are assigned to me in the employee portal, so that I can stay updated.	1	**1**	**Done**

9	As an employee, I want to access to an archive of my previously completed tasks and requirements through the portal so that I can reference them later when needed.	3	**1**	**Done**
10	As an employee, I want to be reminded about upcoming deadlines for task completion so that I can stay on track.	1	**1**	**Done**

Table 3.3. Employee Portal Product Backlog

Offboarding Management

No.	User Story	Priorities	Revised Priorities	Status
1	As an HR head, I want to be notified when an employee submits their resignation letter online so that I can prepare for their offboarding.	**2**	**1**	**Done**
2	As an HR head, I want to schedule the departing employee's exit interview so that the offboarding process is managed efficiently.	**2**	**1**	**Done**
3	As an employee, I want to submit my resignation letter online so that my resignation process is formalized.	**2**	**1**	**Done**
4	As an HR head, I want to be able to request directly to	**2**	**1**	**Done**

	system admin for account termination of the departing employee to maintain security.			
5	As an HR head, I want to be able to notify the financial department to process the final compensation of the departing employee.	**2**	**1**	**Done**

Table 3.4. Offboarding Management Product Backlog

Sprint Backlog

Applicant Tracking System

No.	User Story	Task Breakdown	Estimated Duration	Status
1	As an HR head, I want to create new job posting so that applicants can apply for vacant job positions.	- Design a job posting form with field such as title, description, requirements, location, schedule, salary and tags - Create an input validation to make sure all fields have all the information needed - Add authorization to make sure only HR users can create,	**1 week**	**Done**

		edit, delete job postings Implement a backend logic to store the job posts details in the database		
		Test the job posting feature to make sure the job listings are created correctly	**1 day**	**Done**
2	As an HR head, I want to track the status of each applicant's progress so I can monitor the hiring process.	- Design UI to display the progress of all applicants to HR Implement backend logic for updating candidate status	**1 week**	**Done**
		Test the applicants status tracking feature	**1 day**	**Done**

3	As an HR head, I want to update the status of candidates (accepted or rejected) so that the applicant list is always updated.	- Create database migration to add status field (Scheduled/rejected) in the applicants table. Implement backend logic to handle the scheduled/rejected status for candidates.	**1 week**	**Done**
		Test status update of applicant list for functionalities or errors	**1 day**	**Done**
4	As an HR head, I want to schedule interviews directly through the	- Create a database table for storing all applicants interviews including schedule date, interviewer,	**1 week**	**Done**

	system, so I can manage candidate meetings efficiently.	location, and applicant details - Create an interview scheduling user interface - Implement a backend logic to schedule interviews for applicants		
		Test interview scheduling functionality	**1 day**	**Done**
5	As a recruiter, I want to be notified when new applications are submitted, so I can review them quickly.	- Implement a notification system for applications submission - Create a user interface to display the total count of new applicants	**1 week**	**Done**

		Testing the notification system for functions or errors	**1 day**	**Done**
6	As a recruiter, I want to screen submitted applications, so I can manage the recruitment process and select qualified candidates.	- Create a table user interface to display all applicants list - Implement a backend logic to display the applicants profile - Implement Application Review Feature	**1 week**	**Done**
		Test the screening process for functionalities or bugs	**1 day**	**Done**
		Develop a notification system for Candidates	**1 week**	**Done**

		about their application status		
7	As a recruiter, I want to generate a list of all qualified candidates to streamline the hiring process.	- Implement backend logic to filter qualified candidates - Create a view table user interface to display the list of qualified candidates.	**1 week**	**Done**
8	As an applicant, I want to fill out an application form online so that I can apply for a vacant position easily.	- Create the job posting user interface - Design the form user interface with necessary field (applicant information)	**1 week**	**Done**

		- Implement a backend form validation for required fields. - Implement backend logic to store all the submitted applications in the database.		
		- Implement a file upload for applicants to attach their resume. - Create a storage for applicant's resume	**1 week**	**Done**
		Test the application form feature for functionality issues	**1 week**	**Done**

9	As an applicant, I want to be notified of my interview schedule through email so I can prepare.	Implement an email notification system for interview schedule	**1 week**	**Done**
		Test the notification system for functionalities and errors	**1 day**	**Done**

Table 4.1. Application Tracking System Sprint Backlog

Onboarding Management

No.	User Story	Task Breakdown	Estimated Duration	Status
1	As a new hire, I want to access pre-employment requirements	- Gather pre-employment requirements - Provide instructions for completing/	**1 week**	**Done**

	list online so that my onboarding process smooth.	submitting documents - Upload list to employee portal		
2	As a new hire, I want to receive updates about my onboarding progress so that I can stay on track.	- Develop an onboarding progress tracking system - Create a dashboard for viewing onboarding progress	**1 week**	**Done**
		- Test the onboarding progress tracking system for functionality errors	**1 eday**	**Done**
3	As a new hire, I want to receive my employee	- Provide login instructions - Implement an authorization to new	**1 week**	**Done**

	portal login credentials so I can access my tasks and schedules.	hired employee users to view their task and schedules		
		- Test access to employee portal	**3 hours**	**Done**

Table 4.2. Onboarding Management Sprint Backlog

Employee Portal

No.	User Story	Task Breakdown	Estimated Duration	Status
1	As an employee, I want to view my current assigned tasks so that I know what I need to complete for the day.	- Design taskboard UI for employee portal - Create backend to fetch and display task for employees	**1 week**	**Done**

		- Testing taskboard for functionality	**1 day**	**Done**
2	As an employee, I want to monitor my task progress to stay updated on what task is still pending or completed.	- Implement task tracking system - Display task statuses (pending/ completed) on task board	**1 week**	**Done**
		Test the task tracking system for functionalities and errors	**1 day**	**Done**
3	As an employee, I want to view my personal information so I can make sure	- Develop a personal information dashboard in the portal - Enable viewing of information	**1 week**	**Done**

	my records are accurate.	- Sync updates with HR records		
		- Test viewing information for function	**1 day**	**Done**
4	As an employee, I want to update my personal contact details in the employee portal to ensure that HR management have my most up-to-date information.	- Develop a form for updating personal information - Ensure changes are reflected in HR database	**3 days**	**Done**
		Test the edit personal information	**1 day**	**Done**

		feature for functions or errors		
5	As a resigning employee, I want to be able to monitor my offboarding process in the employee portal before my account is terminated, so that I can ensure a smooth exit.	- Set-up an offboarding progress tracker - Display the offboarding progress tracker in portal	**1 week**	**Done**
		- Test the offboarding progress tracking system	**1 day**	**Done**
6	As an employee, I want to view	- Implement schedule viewing feature in employee portal	**3 days**	**Done**

	the schedule for my training or exit interview through the portal so that I can be prepared.			
7	As an employee I want to upload my completed tasks or requirements through the employee portal so that HR management can review them.	- Create an upload function for completed tasks or requirements - Design a submission confirmation system Enable HR review	**1 week**	**Done**

		Test upload function for functionality issues	**1 day**	**Done**
8	As an employee, I want to be notified when new tasks are assigned to me in the employee portal, so that I can stay updated.	- Implement a notification system for task assignments - Define triggers for alerts(new task assignment) Display notifications in the employee dashboard	**1 week**	**Done**
9	As an employee, I want to access to an archive of my previously completed tasks and	- Create an archive for completed previous tasks - Ensure easy navigation and search functionality	**1 week**	**Done**

	requirements through the portal so that I can reference them later when needed.			
		Test the archive for functions and errors	**1 day**	**Done**
11	As an HR Head, I want to be able to assign tasks in a more efficient method	- Develop a task assignment dashboard for HR - Implement a backend logic to send the task to new hired employee	**1 week**	**Done**
		Test the task assignment for errors and bugs	**1 day**	**Done**

Table 4.3. Employee Portal Sprint Backlog

Offboarding Management

No.	User Story	Task Breakdown	Estimated Duration	Status
1	As an HR head, I want to be notified when an employee submits their resignation letter online so that I can prepare for their offboarding.	- Implement a notification system for resignation letter submission - Define triggers for alerts (new resignation letter submitted)	**1 week**	**Done**
		- Test the notification system for bugs and errors	**1 day**	**Done**
2	As an HR head, I want to schedule the departing	- Develop interview scheduling interface	**4 days**	**Done**

	employee's exit interview so that the offboarding process is managed efficiently.	- Implementing scheduling logic in the backend - Integrate calendar functionality		
		Test the interview scheduling for functionalities	**1 day**	**Done**
3	As an employee, I want to submit my resignation letter online so that my resignation process is formalized.	- Implement a document upload feature for resignation letter submission	**2 days**	**Done**

		- Test the document upload feature for errors	**1 day**	**Done**
4	As an HR head, I want to be able to request directly to system admin for account termination of the departing employee to maintain security.	- Design termination request interface - Implement admin notification system - Implement the backend process for account termination workflow - Test the HR request process	**1 week**	**Done**

Table 4.4. Offboarding Management Sprint Backlog

3.5. Micro-services Architecture

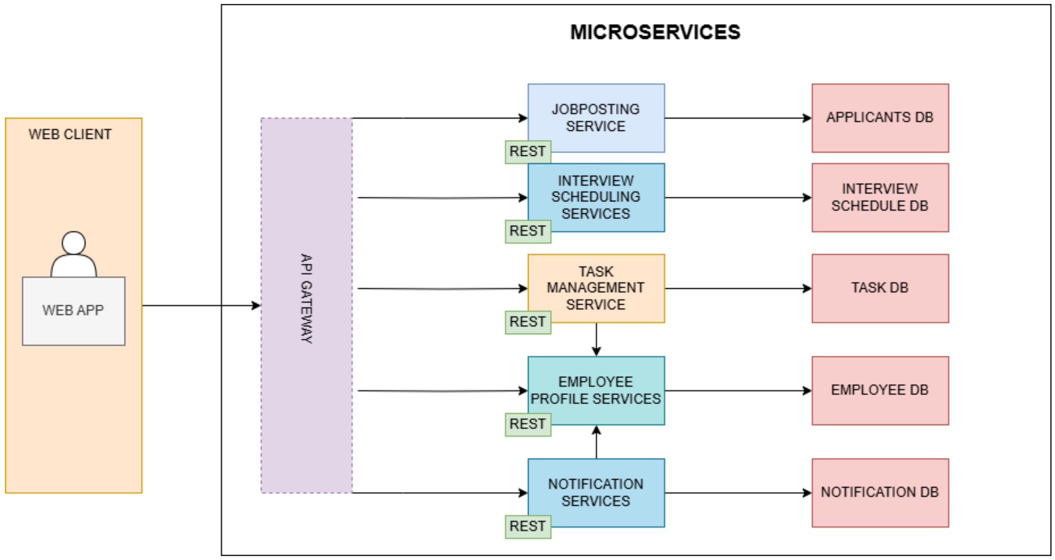

Figure 2. Microservices Architecture

This is an illustration of Microservices architecture in the system. It illustrates the system is divided into multiple services and they communicate using REST APIs.

3.6. DevOps Implementation

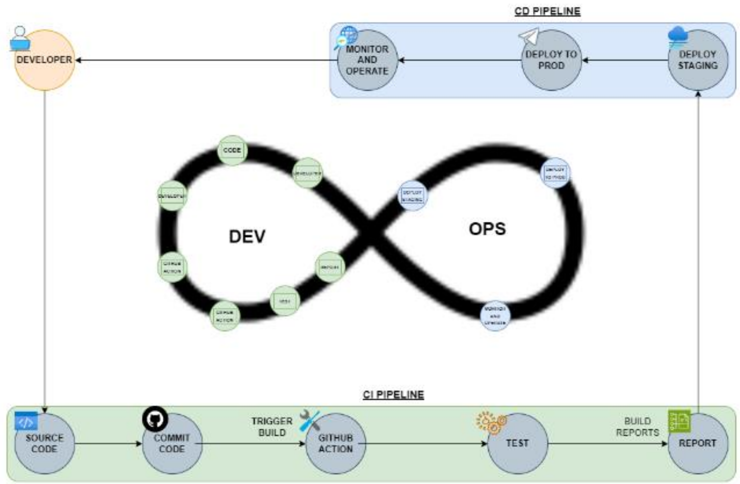

Figure 3.1. DevOps

This diagram illustrates the DevOps development cycle of the system, outlining the complete process of Continuous Integration and Continuous Delivery/Deployment

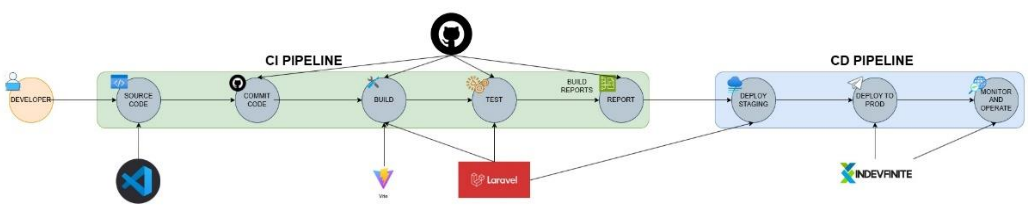

Figure 3.2. CI/CD Pipeline

3.7. Innovation Integration

This is an illustration of the business processes in the Great Wall Arts (GWA) merchandising system, serving as a blueprint for integration and connections throughout the system. The system consists of five main modules: ADMIN, HR, LOGISTICS, CORE, and FINANCE. The HR module is divided into four submodules, the LOGISTICS module into two submodules, and the CORE module into two submodules.

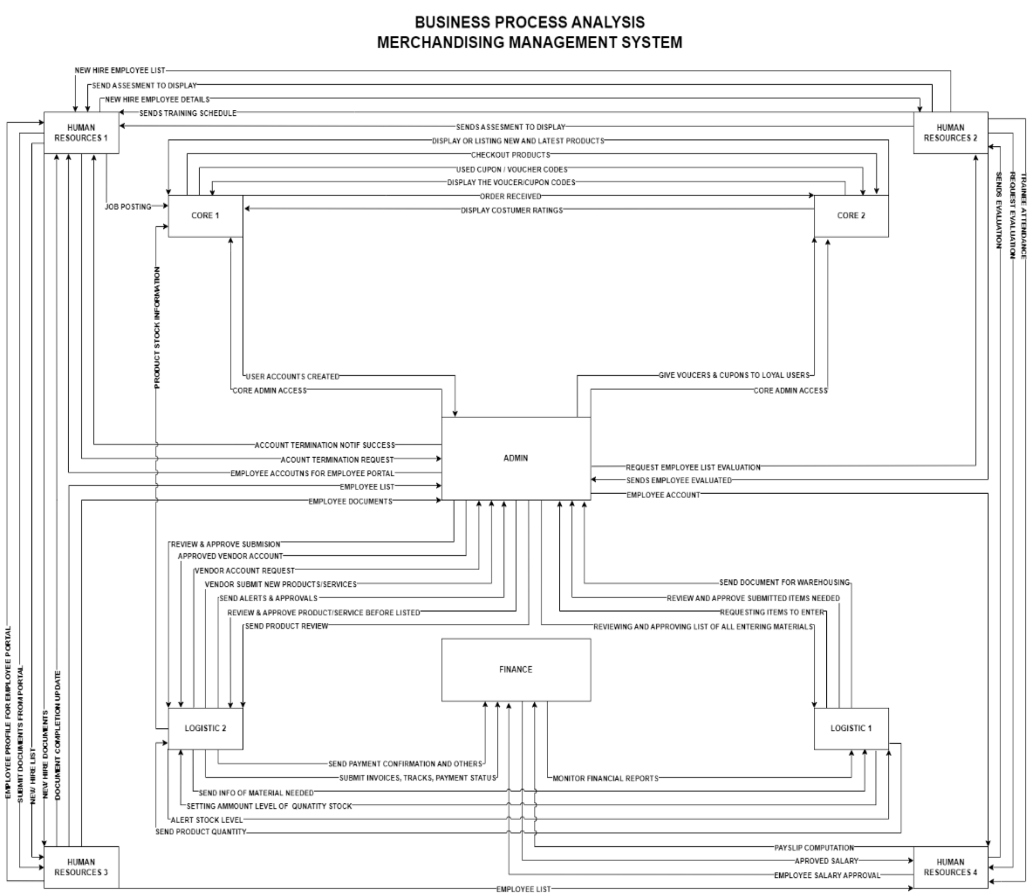

Figure 4.1 BPA Top Level 1

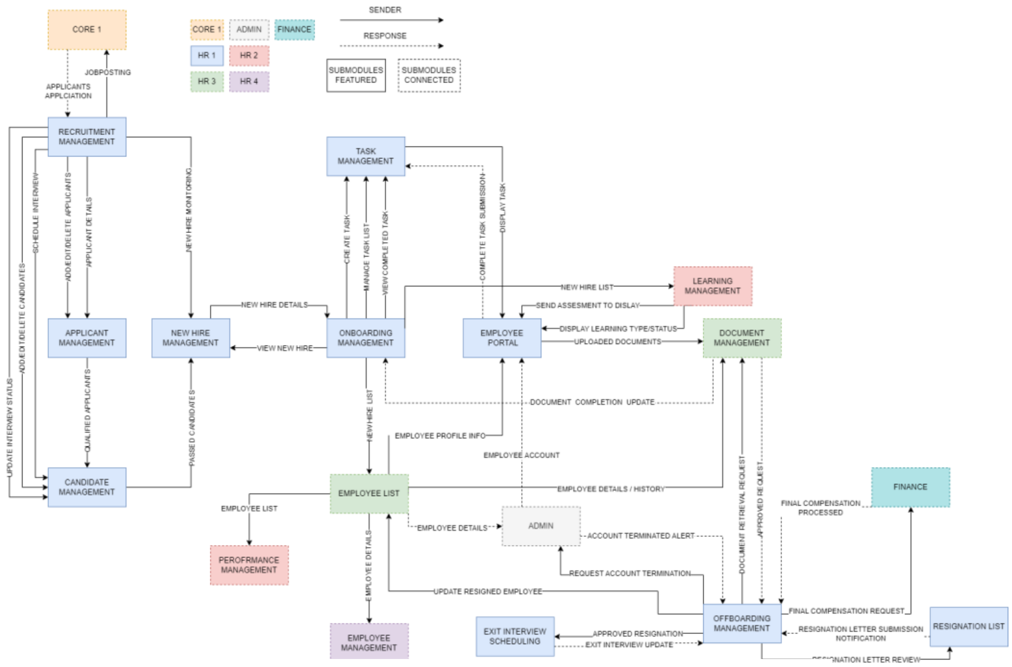

Figure 4.2 BPA Top Level 2

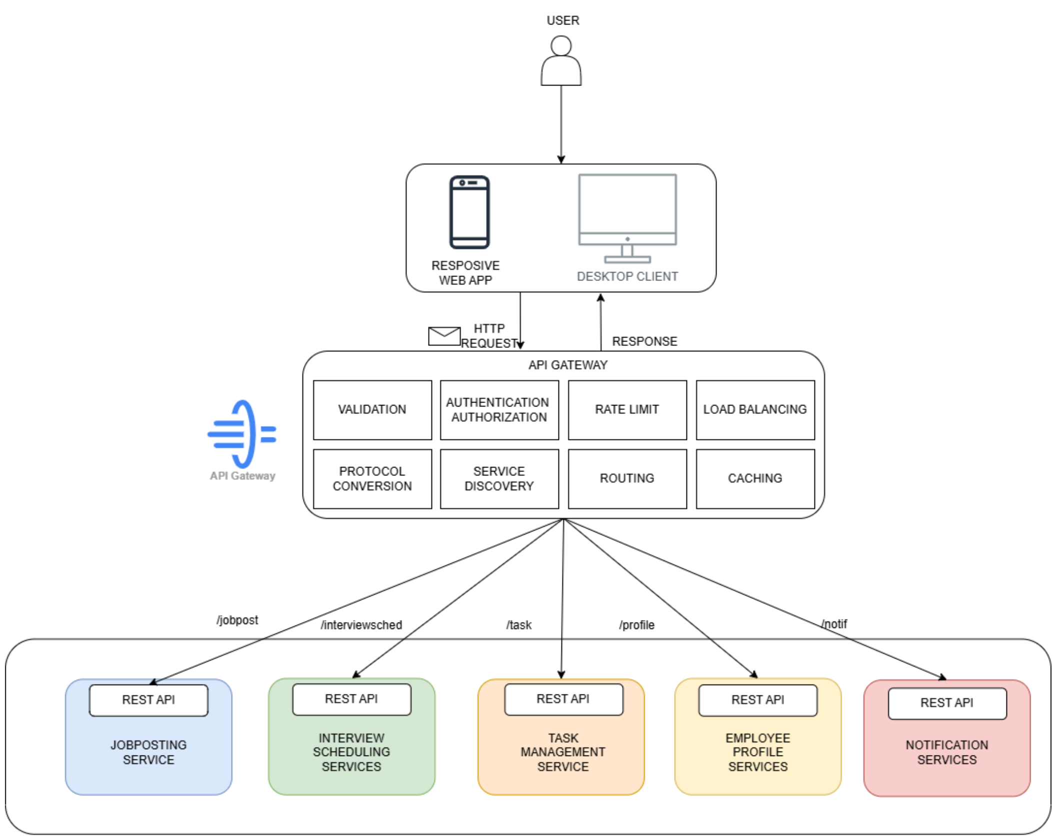

Figure 5. API Gateway

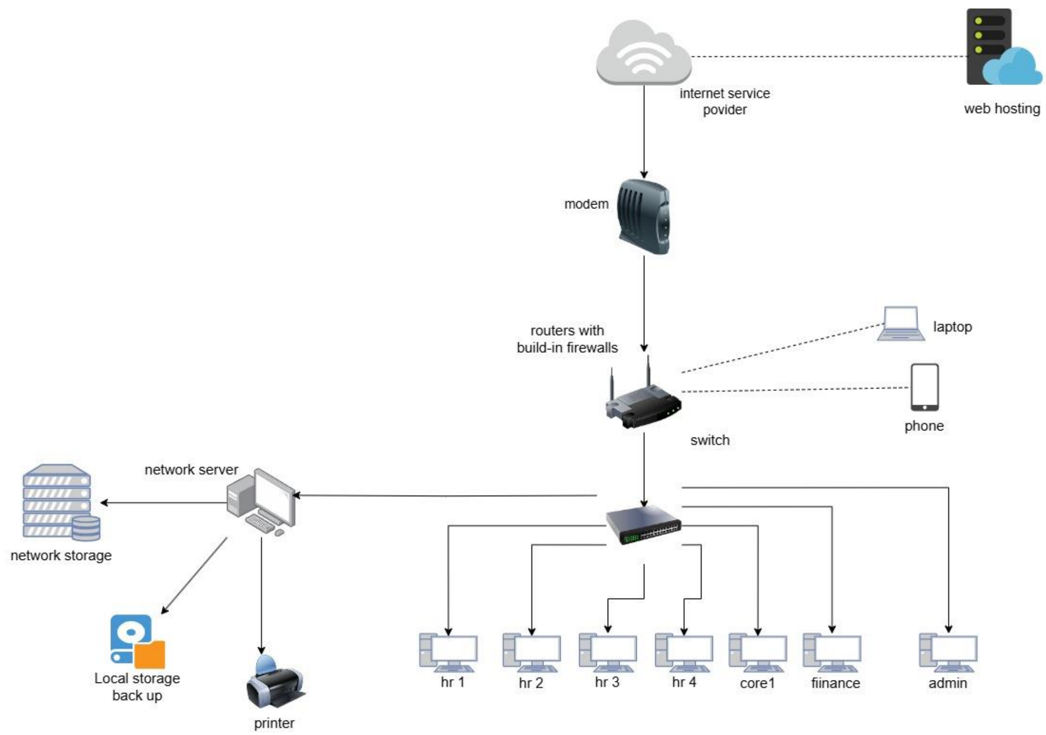

Figure 6. Network Topology

Additional Considerations

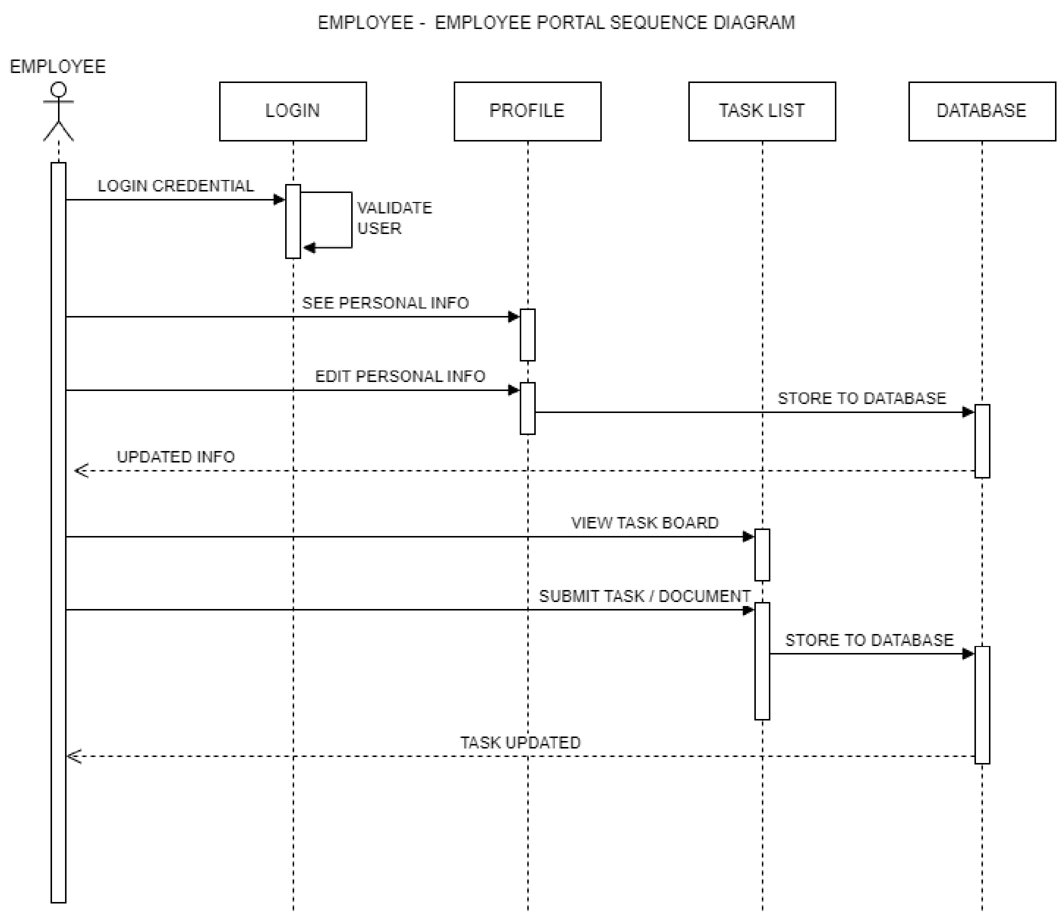

Figure 7.1. Employee - Sequence Diagram

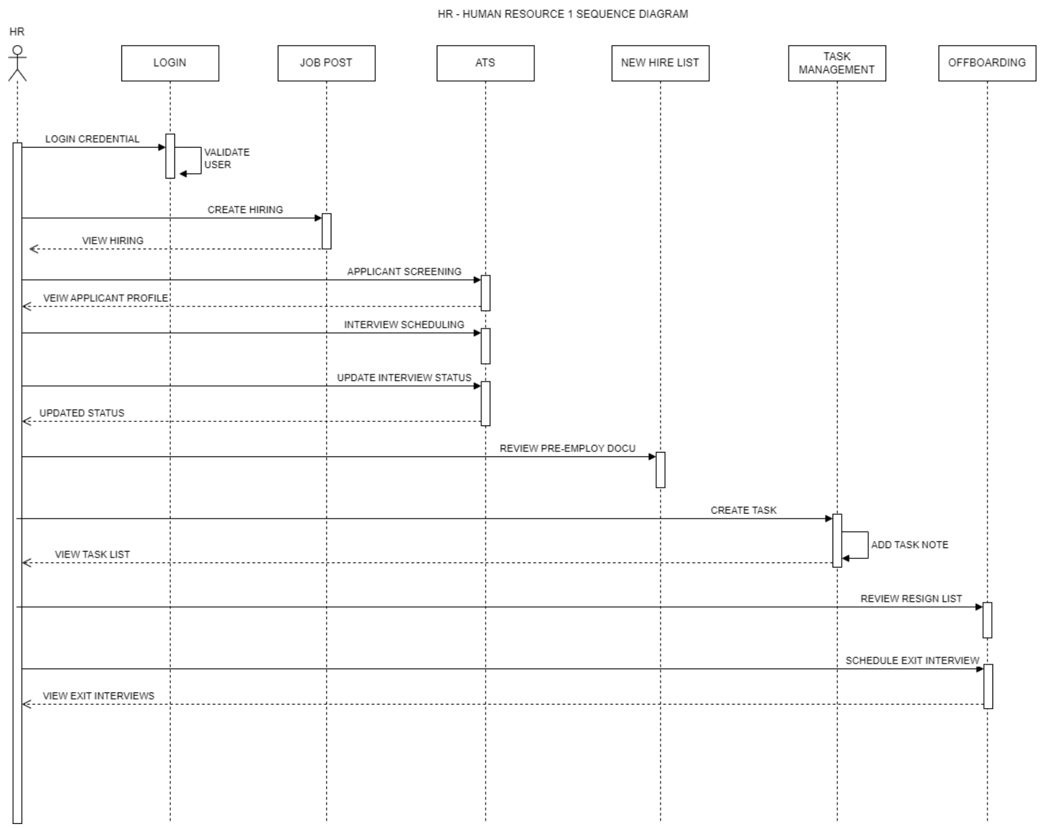

Figure 7.2. HR – Sequence Diagram

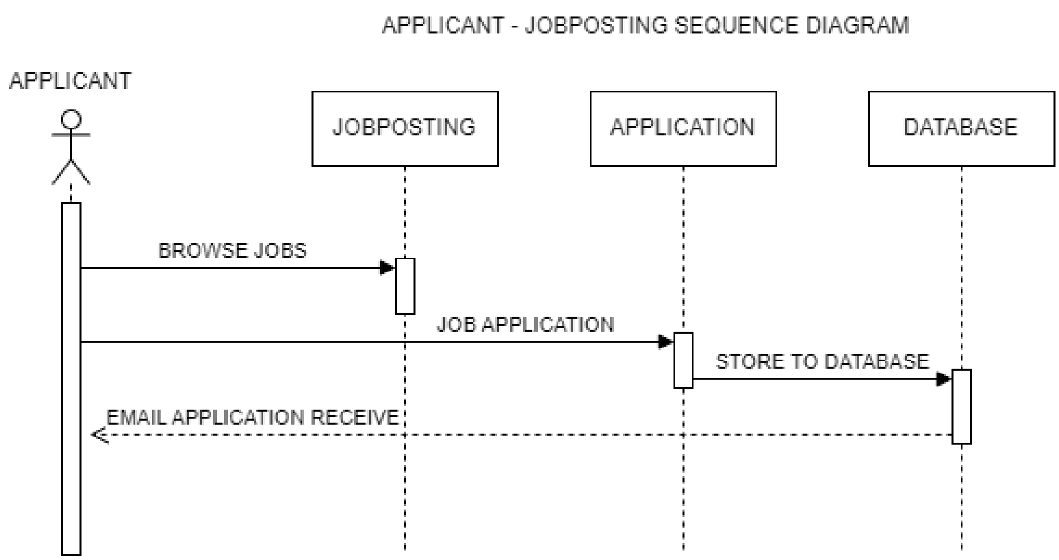

Figure 7.3. Applicant – Sequence Diagram

Chapter 4

SYSTEM DEVELOPMENT AND IMPLEMENTATION

4.1. Requirement Analysis

Stakeholder Identification

Stakeholder	Power	Interest	Impact Rating	Management Strategy
Product Owner	High	High	High	**Manage Closely:** Keep highly involved and informed
HR Personnel	High	High	High	**Manage Closely:** Keep highly involved and informed
Applicants	Low	High	Medium	**Keep Satisfied:** Provide enough info to stay content
Employee	Low	High	Medium	**Keep Satisfied:** Provide enough info to stay content
Developers	Medium	High	Medium	**Manage Closely:** Ensure they understand the project goals

Table 5. Stakeholder Identification

Requirements Gathering Techniques

Technique Used	Who was Involved	What was Collected	Challenges	Resolutions
Interview	HR Managers, Product Owner	Key features for ATS and Onboarding	Conflicting feature priorities	Prioritized critical needs.
Surveys	Employees, Applicants	Feedback on application process.	Low response rate.	Sent follow-up emails.

Table 6. Requirement Gathering Techniques

4.1.3. User Stories and Use Cases

After the requirements have been gathered, they are put together into user stories and to create use cases for the system. User stories aid in identifying how the system will be used by the stakeholders and a broad description of the system's features from the perspective of the end-user is presented by a use-case.

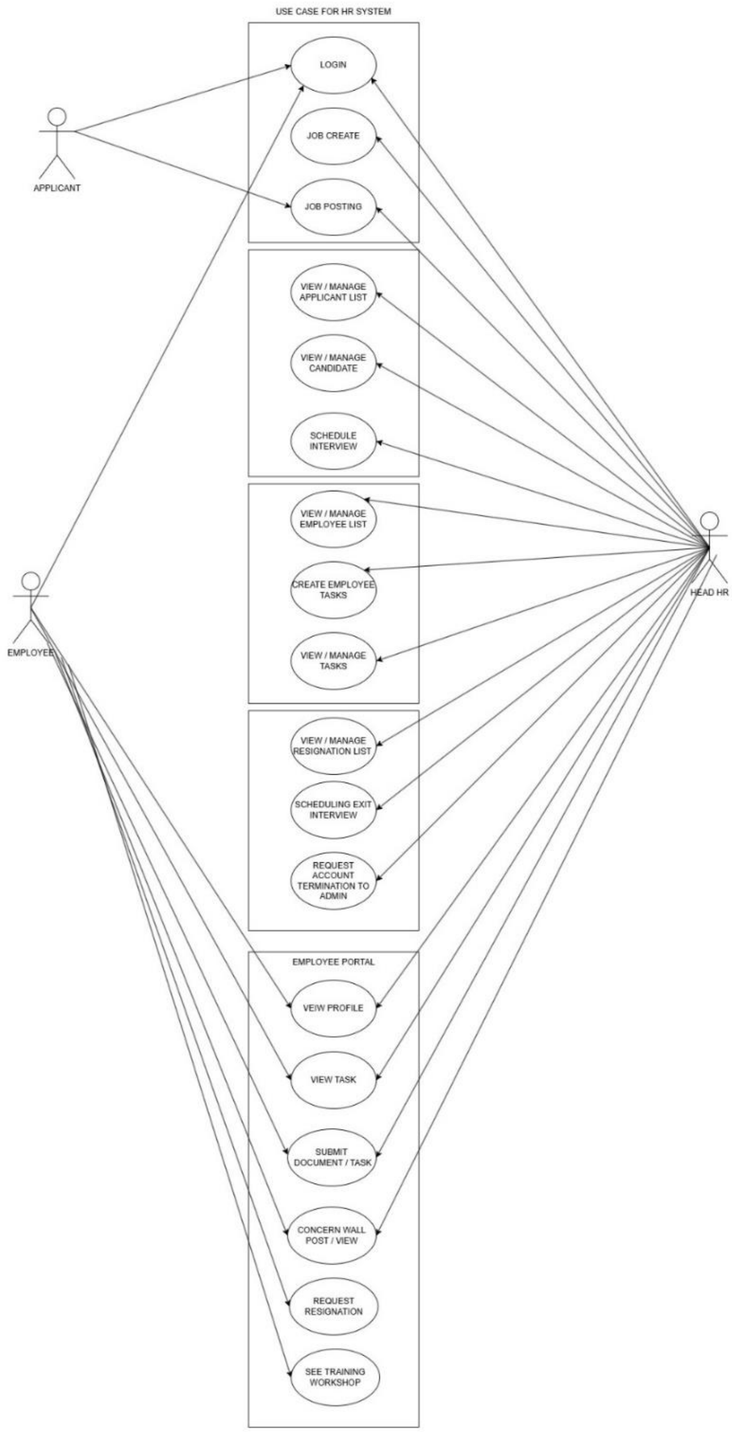

Figure 8. Use-case Diagram

Functional Requirements for Integration

The integration of various modules in the Merchandising Management System: Human Resource 1 (HR1) ensures seamless HR operations. The following are the key functional requirements:

User Authentication & Access Control

a. Users must log in using their role-based credentials (HR, Employee).
b. Role-Based Access Control (RBAC) must ensure that:
    1. HR personnel can access job posting, applicant tracking, onboarding, and offboarding features;
    2. Employees can access their Employee Portal (tasks, communication, offboarding requests); and
    3. Applicants can apply for job openings and track their application status.

Job Posting & Application Management

a. The Recruitment Management System must be notified when there is a request for a job opening, allowing HR to create a job posting.
b. Applicants must be able to apply for job openings and receive an email notification regarding their application status.
c. AI Evaluation should analyze and evaluate applicant resumes based on job requirements.
d. HR personnel should be able to schedule interviews for candidates.

e. The system must automatically send interview schedules to applicants via email (Gmail integration).

Employee Onboarding & Task Management

1. Hired applicants should receive an email confirmation of their successful application along with login credentials for the Employee Portal;
2. New hires must be able to upload pre-employment documents, and HR should be able to review and approve or request resubmissions;
3. The system should provide an interactive onboarding timeline to track pending tasks;
4. HR personnel should be able to assign and monitor onboarding tasks through the Employee Portal;
5. Employees must be able to post concerns on the Concern Wall in the Employee Portal as feedback; and
6. HR personnel must be able to review and respond to employee concerns.

Offboarding & Exit Management

1. Employees must be able to submit resignation requests online, which HR can approve or reject; and
2. Employees should receive notifications regarding the status of their resignation request.

Integration with External APIs & Services

Google Gemini AI must be used for AI-driven resume evaluation, analyzing applicant resumes based on job requirements.

Real-Time Notifications & Updates

HR personnel should receive real-time updates on recent activity, including task submissions and progress in the Task Management system.

Non-functional Requirements for Integration

1. The system must handle at least 500 concurrent users without significant latency;
2. The user interface must be mobile-responsive, designed using Tailwind CSS for an optimal experience and user friendly;
3. The system must maintain 99.9% uptime for uninterrupted HR processes;
4. Role-based access control (RBAC) should ensure only authorized users can modify or access HR data;
5. Multi-factor authentication (MFA) should be available for HR personnel with admin access; and
6. The database should store up to 100,000 applicant records while maintaining fast retrieval.

4.2. Business Process Architecture

Business Process Diagrams

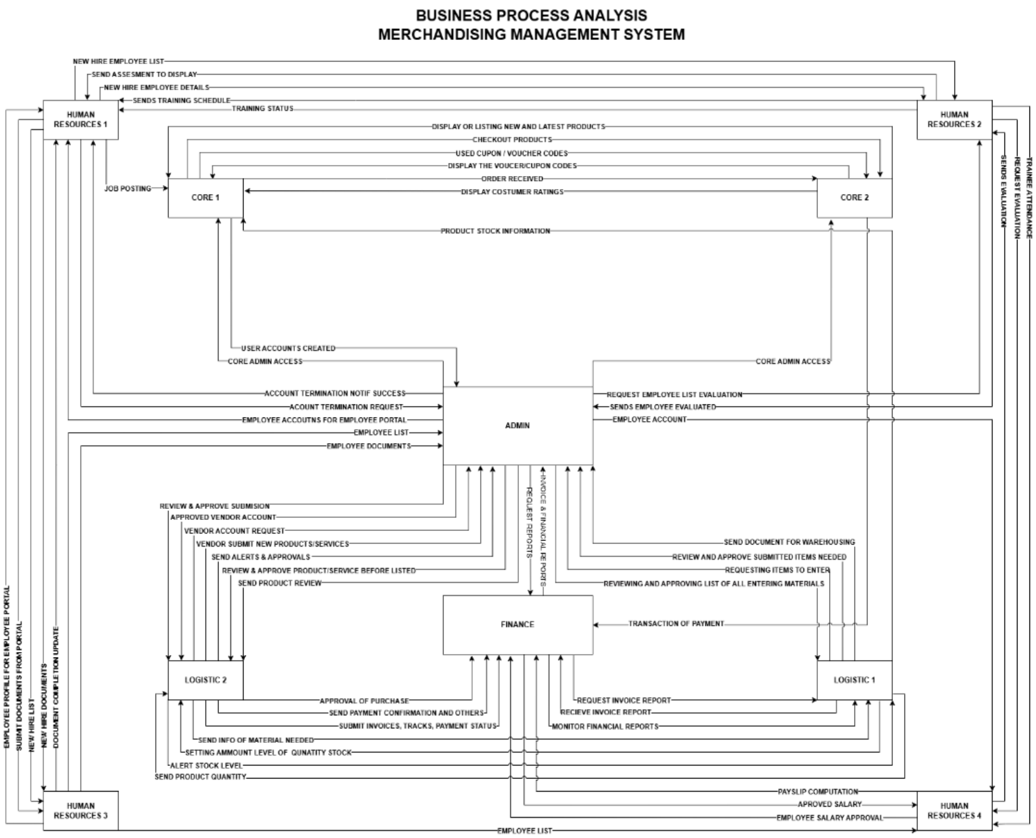

Figure 4.1. BPA Top Level 1 (pp. 94)

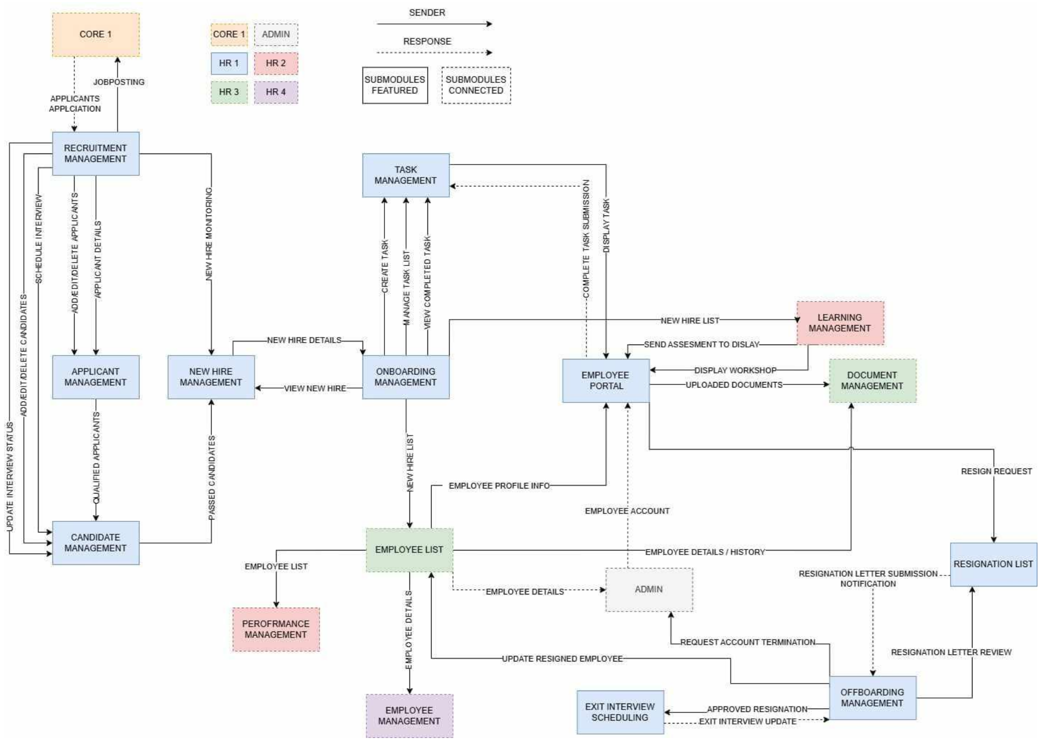

Figure 9. BPA Top Level 3

Alignment of Integrated Systems with Business Processes

The Merchandising Management System - HR1 is designed to work efficiently with various business processes, ensuring smooth operations and streamlined data flow. By aligning system modules with key functions, the platform enhances HR operations, minimizes redundancy, and improves overall productivity.

Integration with Recruitment and Onboarding Processes

1. The system simplifies recruitment by managing applicant tracking, job postings, and AI resume evaluations through the Recruitment Management module.
2. To support a structured transition for new employees, the New Hire Management feature connects with onboarding, ensuring a well-coordinated entry into the organization.
3. Additionally, the Task Management module assigns and monitors onboarding activities, helping HR personnel and new hires complete all pre-employment task and documents uploads through employee portal.

Synchronization with Employee and Learning Management

1. A centralized Employee List module serves as the main repository for employee records, maintaining accurate and up-to-date profiles.
2. The Learning Management component supports onboarding, training, and skills development, ensuring employees receive appropriate training tailored to their roles. This fosters continuous professional growth while aligning with business objectives.

Document and Process Automation

1. The Document Management module facilitates efficient handling of employee records, including onboarding paperwork, contracts, and

training materials from employee portal. This ensures secure storage and easy retrieval of critical documents.

2. Through the Employee Portal, employees can submit and essential files, while HR teams can oversee compliance and validate required submissions.

Exit and Offboarding Management

1. A structured offboarding process is supported by the Exit Interview Scheduling and Resignation List modules, ensuring proper documentation and smooth employee departures.
2. To maintain compliance with company policies, the Onboarding Management and Task Management modules track exit-related responsibilities, ensuring all necessary steps are completed before an employee leaves.

Business Process

The Business Process within HR1 involves the systematic management of HR functions from recruitment to offboarding. Traditionally, HR operations required extensive manual input, causing inefficiencies and delays in processing employee data, approvals, and compliance tasks. However, with the implementation of Business Process Automation (BPA), these processes are optimized through technology-driven solutions.

Manual Business Process (Before Automation):

1. HR staff will manually publish and unpublish job postings requested by other departments.
2. They will monitor applicants and evaluate the profiles of qualified candidates.
3. Interview schedules for shortlisted applicants are arranged, and notifications are automatically sent via email.
4. Applicant status are updated after interviews are completed.
5. HR staff will provide employee portal accounts and create tasks for newly hired applicants to upload pre-employment documents, which will then be reviewed and validated.
6. HR will process resignation requests and schedule exit interviews for resigning employees.

Automated Business Process

With the BPA implementation, the workflow has been re-engineered and automated as follows:

1. **Recruitment & New Hire Management:** Applicant data is directly stored and integrated into the New hired list when the candidates are passed from the interviews reducing redundant data entry.

2. **Onboarding & Training:** Employee records automatically sync with Learning Management, where training is assigned based on job roles.
3. **Document Management:** Employees upload documents from employee portal stored securely via an automated Document Management System.
4. **Offboarding & Exit Management:** Automated workflows assist in tracking performance, scheduling exit interviews, and managing offboarding procedures seamlessly.

4.3. Application Architecture

Components of Application Architecture

The application architecture of the Merchandising Management System is designed to ensure a seamless and efficient user experience. The system follows a microservices architecture, where different system components operate as independent services while interacting and integrated through APIs to support different system operations.

Presentation Layer (User Interface - UI)

The presentation layer is the frontend of the system that interacts with users by displaying content and capturing user input. This component is built using modern web technologies to ensure a seamless and responsive experience.

1. **HTML** – Defines the structural elements of the web-based system.
2. **Tailwind CSS** – Styles the page with a responsive, mobile-first design for a visually appealing and user-friendly experience.
3. **JavaScript** – Adds interactivity and animations to enhance the user interface and system functionality.
4. **Livewire** – Enables server-side processing with dynamic front-end components.
5. **Alpine.js** – A lightweight JavaScript framework used for functionalities like modals, action buttons, and simple UI interactions.

Application Layer

The application layer is responsible for processing user requests, enforcing business logic, and coordinating system interactions.

1. **Laravel Framework** – A PHP framework that handles business logic, data processing, and security.
2. **Microservices Architecture** – Implements modular services that communicate via REST APIs to ensure seamless interactions between system modules.
3. **Authentication and Authorization** – Security mechanisms to protect sensitive HR data.

Backend Layer (Backend System)

The backend layer manages data storage, retrieval, and business logic execution. It ensures system efficiency, security, and reliability.

1. **Laravel (PHP Framework)** – Handles backend processes, logic execution, and system security.
2. **REST APIs** – Facilitates communication between different system modules and external services.
3. **Microservices Implementation** – Ensures independent scalability and flexibility of system modules.
4. **API Gateway** – Manages request routing and authentication across microservices.

Database Management Layer

The database management layer ensures efficient data storage, retrieval, and security.

4. **MySQL Database** – A relational database that stores structured data such as applicant details, employee records, and job postings.
5. **Universally Unique Identifiers (UUIDs)** – Ensures secure and unique identification of records.
6. **Dedicated Database Instances** – Each system operates with its own database, maintaining consistency through API-based communication.

7. **Independent Hosting and Deployment** – Each system has its own domain, allowing scalability, flexibility, and fault isolation while ensuring seamless integration.

APIs and External Services

1. APIs and external services enable integration with third-party platforms and automation tools to enhance system functionality.
2. **RESTful APIs** – Enables smooth integration between system modules, ensuring data consistency and synchronization.
3. **Google Gemini AI API** – Automates resume evaluation, streamlining applicant screening and selection.
4. **API Gateway** – Facilitates secure request routing and authentication across microservices.

Application Architecture Diagram

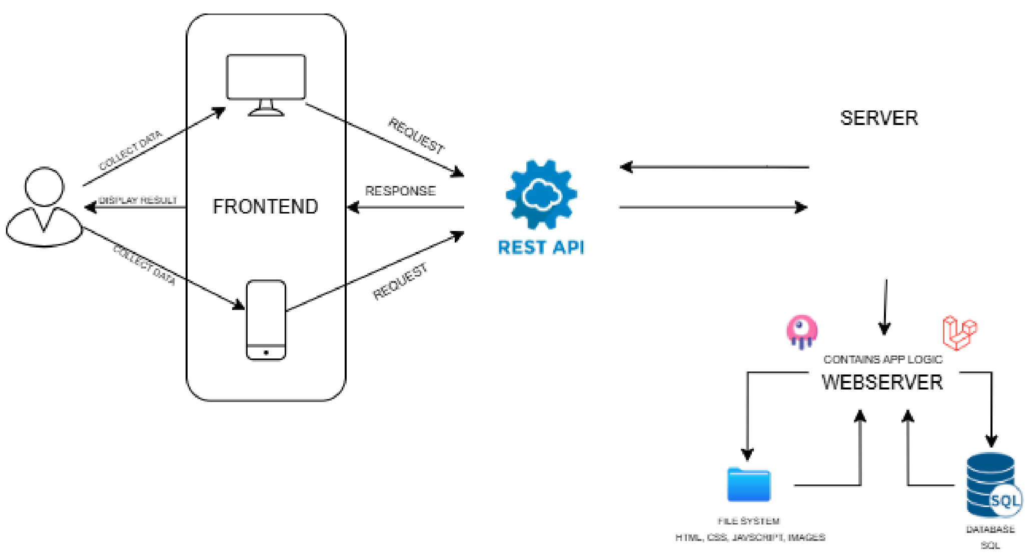

Figure 10. Application Architecture Diagram

Integration of Software Modules

Each module in the system is developed and deployed as an independent system following the rules of microservices architecture while interacting through Restful APIs:

Software Modules	Interaction Flow
Recruitment Management	- Manages recruitment, job postings, task assignments, and applicant tracking. Transfers successful candidate data to HR 2 for training and learning management
Training and Performance Management	- Manages training programs, learning modules for new hires, and performance evaluation. Receives new hire data from Recruitment Management and updates Succession Planning with employee performace reports.

Employee and Document Management	- Handles employee records, document storage, and sucession planning. Shares employee data with Payroll and Attendance Management for payroll processing and attendance tracking.
Attendance and Payroll Management	- Tracks employee attendance and processes payroll. Connected to Finance for salary reports and payslip approval.
Administration Management	- Creation, monitoring, and termination of user accounts.

Figure 11. Integration of Software Modules

Communication and Interaction Patterns

To ensure seamless and standardized interactions across the Merchandising Management System, all system components communicate using RESTful APIs. These communication patterns enable smooth data exchange and maintain consistency across all microservices. The system follows these interaction principles:

REST API Communication – Each system module interacts via REST APIs, allowing efficient data retrieval, updates, and synchronization while ensuring system flexibility and scalability.

API Gateway Management – A centralized API Gateway directs and authenticates requests, optimizing load balancing, monitoring, and security enforcement across microservices.

Asynchronous Data Processing – Certain operations, such as automated resume evaluation using Google Gemini AI API, are processed asynchronously to improve system performance and responsiveness.

Data Synchronization and Consistency – RESTful API interactions ensure that updates across Recruitment Management, Training & Performance Management, Employee & Document Management, Attendance & Payroll Management, and other modules remain synchronized in real time.

Authentication and Authorization – Laravel's built-in security mechanisms, including authentication middleware and token-based authorization, control access to different system modules, ensuring secure data transactions.

Error Handling and Logging – All API requests are logged, and system failures are tracked using monitoring tools to maintain service reliability and detect potential issues proactively.

4.4. Data Architecture

Data Sources and Types

1. Applicant Information
   a. Contains details of individuals seeking job opportunities;
   b. Includes personal information such as full name, contact details, educational background, work experience, and uploaded resumes; and
   c. Serves as the primary source of data for recruitment and hiring processes.
2. New-Hire Database
   a. Stores information about recently hired employees;
   b. Includes employment details such as job position, department, employee ID, date of hire, contract type, and salary information; and
   c. Used for onboarding processes and integration into the company's workforce.
3. Documents Database
   a. Maintains records of all relevant HR documents, including contracts, policy agreements, identification documents, and compliance forms; and
   b. Ensures secure storage and easy retrieval of employee-related paperwork for legal and regulatory purposes.

Dataflow Diagrams

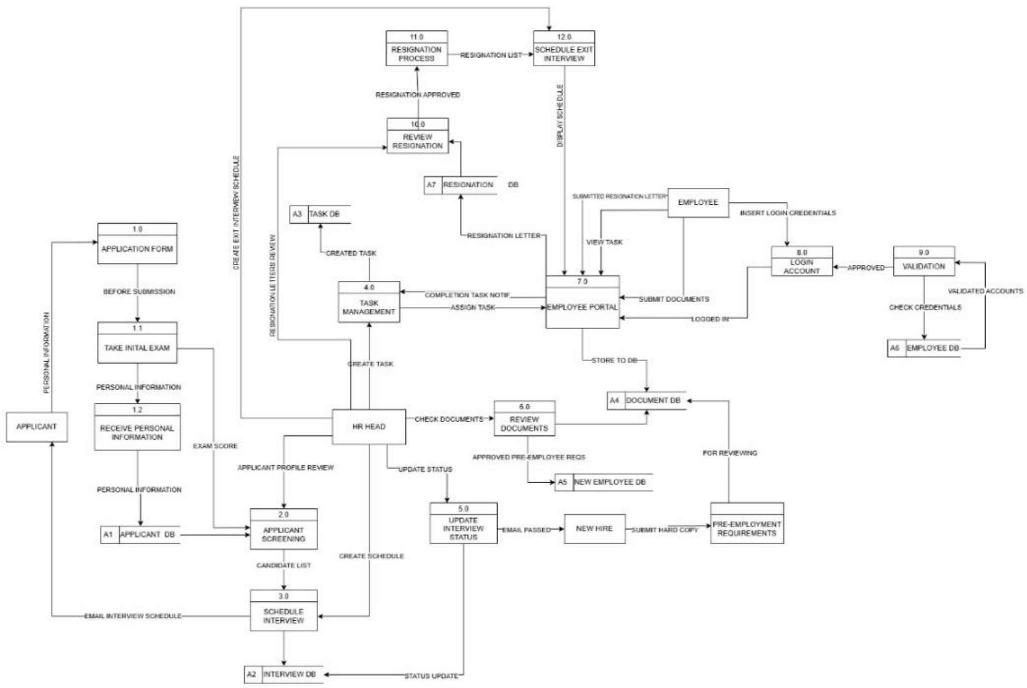

Figure 12. Dataflow Diagram

Data Storage and Management

The HR-1 Management System is designed with a structured and efficient data storage mechanism to manage various entities, including applicant information, employee records, task lists, resignation documents, and interview schedules. A relational database model is utilized to ensure data integrity, consistency, and accessibility.

Database Design

The system operates on a SQL-based relational database, incorporating the following key tables:

Applicant Table: Stores applicant details, including personal information, exam scores, and interview schedules.

Employee Table: Maintains employee records, login credentials, validation statuses, and assigned tasks.

Task Management Table: Logs created tasks, assigned employees, and task completion statuses.

Resignation Table: Contains resignation letters, approval statuses, and exit interview schedules.

Document Table: Stores employee-submitted documents and HR-approved files.

Data Storage Strategy

To maintain efficiency and security, the system implements the following storage strategies:

Normalization: The database adheres to the third normal form (3NF) to minimize redundancy and optimize retrieval speed.

Indexing: Frequently queried fields, such as employee IDs and application statuses, are indexed to enhance query performance.

Backup and Recovery: Automated daily backups and disaster recovery protocols are in place to prevent data loss.

Access Control: A role-based access control (RBAC) mechanism ensures that sensitive HR data is accessible only to authorized personnel.

Data Synchronization Across Systems

To ensure seamless operations, the HR-1 Management System incorporates real-time data synchronization across multiple functional areas. A centralized data repository model enables instant updates across all related system modules.

Synchronization Mechanism

1. **Event-Driven Updates:** Any changes in applicant status, task assignments, or resignation requests trigger automatic updates across linked modules.
2. **Database Triggers:** SQL triggers facilitate automatic updates in the Employee and Applicant tables, ensuring consistency with the Task Management and Interview Schedule modules.
3. **APIs and Webhooks:** RESTful APIs enable seamless integration with external HR tools and email scheduling services.
4. **Batch Processing:** For non-critical updates, batch synchronization occurs at scheduled intervals to optimize system performance.

Conflict Resolution

1. Timestamp-Based Merging: The most recent data entry is prioritized in case of conflicts.
2. Manual Override: HR administrators are provided with a dedicated interface to manually resolve discrepancies when necessary.
3. Audit Logs: A comprehensive audit log system records all data modifications to ensure transparency and accountability.

4.5. Technology Architecture

Technology Stack and Infrastructure

The technology architecture of this Merchandising Management System operates on an established technology stack and infrastructure. To ensure scalability, security, as well as performance, the developers used both open-source and industry-standard technologies. The technology stack includes:

Operating System: The system is deployed on Linus-based servers to ensure stability and security.

Programming Language: PHP was chosen as the primary programming language for backend development, while JavaScript was used for interactive frontend development.

Frameworks: Laravel is a PHP framework that makes web development easier and more organized. It comes with built-in features that help developers build scalable and efficient web applications faster, without the hassle of starting from scratch with livewire for dynamic web applications, while Tailwind CSS (CSS Framework) is used for front-end mobile-first web design and alpine.js a lightweight JavaScript framework that used for user interaction.

Database Management: MySQL was chosen as the relational database management system, with each microservice managing its own dedicated database instance.

Web Server: The system was hosted in Indevfinite, a cloud-based hosting, because it provides versatile system integration that focuses from website design, website maintenance, web development, and supports different tech stack that suites for microservices architecture.

Network Topology and Configuration

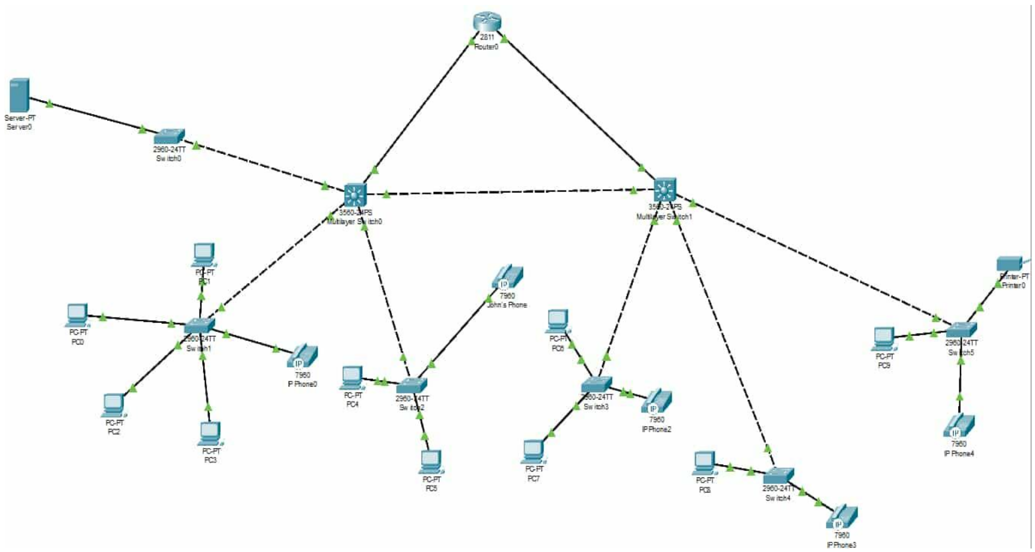

Figure 13. Network Topology and Configuration

Each of the five departments that form the organization—HR, Core, Admin, Finance, and Logistics—is connected to a distribution switch via an access switch. This distribution switch connects to the core switches, responsible for high-speed data transmission without belonging to any specific department. The modem connects to a firewall that safeguards both incoming and outgoing traffic when attempting to access the internet. Laptops and mobile devices can connect freely as a result to a router with built-in firewalls that manage internal and external communication. The network is divided into VLANs to enhance security and performance, guaranteeing that each department has regulated communication while maintaining the star topology layout.

Furthermore, a network server is included in the system for centralized data storage and backup purposes. To guarantee high availability, Link Aggregation Control Protocol (LACP) is utilized at crucial locations, like core and distribution switches, merging several physical links into one logical connection. This avoids bottlenecks and offers failover assistance, guaranteeing continuous network functionality even if a link goes down.

The star topology provides significant advantages, such as high reliability, as the failure of single devices does not affect the whole network, scalability for straightforward expansion, improved security through VLAN segmentation and firewalls, enhanced performance with load balancing using LACP, and easier troubleshooting thanks to its organized structure.

This effectively structured star topology guarantees reliable communication, security, and high availability, rendering it a strong network for the organization.

The network consists of three main layers:

Core Layer: High-speed switches handle inter-departmental data flow using Link Aggregation Control Protocol (LACP) for redundancy and load balancing.

Distribution Layer: Distributes network traffic between the core switches and access layer, ensuring efficient communication across VLANs.

Access Layer: Connects end-user devices for HR, Core, Logistics, Finance, Admin, and telephone systems using VLANs to segment traffic securely.

Network Protocols and Communication Models

Transmission Control Protocol / Internet Protocol (TCP/IP): Enables communication within the internal network and the internet.

Dynamic Host Configuration Protocol (DHCP): Assigns IP addresses dynamically to devices.

Domain Name System (DNS): Translates domain names into IP addresses.

Virtual Local Area Networks (VLANs): Segment network traffic into **logical groups** for security and optimized performance.

Link Aggregation Control Protocol (LACP): Merges multiple links into a **single logical connection** for better bandwidth and failover support.

Voice over IP (VoIP): Supports telephone communication over the network.

The network follows a client-server model, where all client devices rely on the network server for authentication, file storage, and backup management.

Security Measures

1. **Firewall Protection:** Blocks unauthorized access and enforces security policies.
2. **VLAN Segmentation:** Isolates traffic between departments for security and efficiency:

   VLAN 1: HR department

   VLAN 2: Logistics department

   VLAN 3: Core department

   VLAN 4: Finance and Admin departments

   VLAN 5: Printers and telephone systems
3. **Access Control Lists (ACLs):** Restrict unauthorized communication between VLANs.
4. **Data Encryption:** Ensures secure data transmission.
5. **Backup System:** Local storage backup for data recovery and protection.

Load Balancing and Failover Mechanisms

1. **Link Aggregation Control Protocol (LACP)**
    a. Combines multiple links to avoid bottlenecks; and
    b. Ensures redundancy if a link fails.
2. **Multiple Core Switches**
    a. Prevents a single point of failure; and
    b. Ensures continued operation if one switch fails.
3. **Server Backup System**
    a. Prevents data loss and supports business continuity.

Network Configuration

To enhance network security, performance, and reliability, the following configurations are implemented:

1. **Secure Socket Layer (SSL) Encryption**
    a. Protects data transmission between clients and the server.
    b. Ensures confidentiality, integrity, and security of sensitive information.

2. **Load Balancing**

    a. Distributes incoming network traffic across multiple server instances to prevent overloading;

    b. Improves performance, fault tolerance, and system availability; and

    c. Ensures seamless failover in case of server failure.

3. **Firewall Rules and Access Control**

    a. Implements strict firewall policies to prevent unauthorized access;

    b. Filters incoming and outgoing traffic based on security rules; and

    c. Uses Access Control Lists (ACLs) to restrict access based on roles and permissions.

By incorporating SSL encryption, load balancing, and firewall protection, our client-server network setup guarantees secure, scalable, and high-performance communication. These actions ensure robust data security, enhanced traffic management, and ongoing system dependability, facilitating smooth user experiences.

Software Technologies

The functionality of the system was enhanced using various software technologies, which includes:

Programming Languages

a. **JavaScript** – Used for front-end development, ensuring a responsive and dynamic user experience.
b. **PHP** – Handles back-end processes, such as user authentication, data retrieval, and business logic implementation.
c. **SQL** – Manages database operations, ensuring efficient data storage and retrieval.

Frameworks and Libraries

a. **Laravel** – A PHP framework that simplifies back-end development, providing robust security and efficiency.
b. **Livewire** – Enables dynamic, real-time UI interactions within Laravel applications without relying heavily on JavaScript.
c. **Alpine.js** – A lightweight JavaScript framework used for minimal front-end enhancements and interactivity.

Database Management

MySQL – A relational database used to store structured data securely, ensuring high availability and performance.

Version Control and Collaboration

GitHub – Facilitates source code management, collaboration, and version control among development teams.

Security Technologies

a. **SSL/TLS Encryption** – Protects data transmissions between users and the server.
b. **Firewall and Intrusion Detection** – Prevents unauthorized access and potential cyber threats.
c. **API Gateway and REST API Security** – Ensures secure communication between system components, managing API requests and preventing unauthorized access.

Deployment and Hosting

Indevfinite – A cloud-based web hosting server designed for scalability and large storage capacity to accommodate massive data storage and information.

Scalability and Performance Considerations

The Merchandising Management System: HR1 is designed with scalability and performance in mind, ensuring it can accommodate future growth while maintaining optimal system efficiency.

Scalability Strategies

a. **Modular Architecture**: The system follows a modular design, allowing easy expansion and integration of new features without major overhauls.

b. **Horizontal Scaling:** Supports the addition of multiple servers to distribute workload efficiently, ensuring smooth performance during peak usage.

c. **Cloud Infrastructure:** Hosted on indevfinite, providing flexible scaling options based on demand.

d. **Microservices Architecture:** Implements microservices to break down the system into independently deployable services, improving scalability, maintainability, and fault isolation.

Performance Optimization

a. Eloquent ORM Optimization: Utilizes efficient query building techniques in Laravel, such as eager loading and indexing, to enhance database performance.

b. Livewire State Management: Reduces unnecessary re-renders by efficiently managing UI state updates, improving response times and minimizing server load.

c. Alpine.js Lightweight Execution: Ensures fast front-end interactions by leveraging Alpine.js for minimal yet effective JavaScript functionalities.

d. Efficient Caching: Implements Redis for in-memory caching, reducing redundant database queries and accelerating response times.

e. Lazy Loading & Minification: Uses lazy loading techniques for images and assets, along with CSS and JavaScript minification, to enhance page load speed.

f. Asynchronous Processing: Background tasks, such as email notifications and report generation, are managed through job queues to prevent performance bottlenecks.

Load Testing and Monitoring

a. Automated Load Testing: Regular stress testing is performed to evaluate system performance under heavy usage scenarios.

b. CI/CD Testing with GitHub Actions: The system undergoes continuous integration and deployment testing using GitHub Actions, ensuring that new updates do not introduce regressions or performance issues.

c. Error Handling and Logging: Logs are maintained to identify and resolve system bottlenecks efficiently.

4.6. Development Process

The development of the Merchandising Management System followed a structured and iterative approach to ensure efficiency, maintainability, and seamless integration across system modules. Given the complexity of the system and its reliance on microservices architecture, an Agile methodology was adopted to allow flexibility in development, continuous feedback implementation, and efficient task management.

Stakeholder consultations were conducted at the beginning of the project to define the system’s core functionalities and integration requirements across modules, including Human Resource Management, Core, Logistics, Admin and Finance. These discussions helped establish clear objectives and identify the necessary functional and non-functional requirements. A product backlog was created to outline all system features, which were then prioritized for development.

Planning & Requirement Gathering

Stakeholder meetings were held to identify gaps in existing processes and define system goals. The overall workflow was mapped, detailing how different system modules interact within the Merchandising Management System. A structured product backlog was developed, prioritizing core features of every system module that aligned with the stakeholder's expectations.

System Design & Architecture

A Microservices Architecture was selected to ensure modular development, independent deployment, and scalability across system modules.

Entity-Relationship Diagrams (ERD), Data Flow Diagrams (DFD), and API endpoints were designed to establish seamless communication between microservices.

The technology stack was finalized including:

Technology Stack	Tools & Frameworks
Backend	Laravel (PHP Framework)
Frontend	Tailwind CSS, JavaScript, Livewire, Alpine.js
Database	MySQL with UUIDs for unique record identification
External APIs	Google Gemini AI (resume evaluation), Gmail API (email notifications)

Figure 14. Tools and Frameworks

4.7. Implementation

The Merchandising Management System was implemented using a microservices architecture, where each module operates independently while interacting through RESTful APIs. The implementation process ensured scalability, maintainability, and seamless integration across HR, Core, Logistics, Finance, and Admin modules.

Technical Implementation Details

The Merchandising system was developed in a modular and independent microservice that interacts with other subsystems, including HR1, HR2, HR3, HR4, CORE, LOGISTICS, ADMIN, and FINANCE. The microservices architecture allowed each module to operate autonomously while maintaining seamless communication through RESTful APIs. Laravel was used as the primary backend framework, enabling structured data handling, secure authentication, and efficient API management.

Each system was hosted and deployed independently, ensuring high availability and fault isolation. API endpoints were structured following RESTful principles, allowing efficient data exchange and synchronization. Data storage was managed using MySQL, with each service operating its own dedicated database instance to enhance security and maintain data integrity.

To maintain data integrity, MySQL was chosen as the relational database management system, with each microservice managing its own dedicated database instance. This design choice enhances security, reduces inter-service dependencies, and ensures consistent data synchronization across all system components through well-defined API interactions.

The implementation was structured with the following key principles:

1. **Microservices Architecture** – Each subsystem operates independently, with its own logic and database.
2. **RESTful APIs** – RESTful is implemented for each modules and submodules to communicate over HTTP requests.
3. **Stateless Interactions** – Each API request contains all necessary information, avoiding the need for session tracking.

Tools and Technologies Used

The development and implementation of the Merchandising Management System were supported by a selection of tools and technologies that ensured efficiency, scalability, and security. Each tool was carefully chosen to optimize performance and support seamless integration between microservices. The key tools and technologies used include:

Tools and Technologies	**Definition**
Integrated Development Environment (IDE)	Visual Studio Code (VS Code) was utilized for writing, debugging, and managing the system's source code efficiently.
Backend Framework	Laravel, a PHP-based framework, was selected for its MVC architecture, built-in authentication, and REST API support.
Frontend Development	Tailwind CSS was used for styling, providing a responsive and mobile-friendly user interface, while Livewire and Alpine.js were implemented to handle interactive components dynamically.
Database Management System (DBMS)	MySQL was chosen due to its reliability, scalability, and compatibility

	with Laravel for structured data storage and retrieval.
API Communication	RESTful APIs were implemented using Laravel's built-in API functionalities to facilitate seamless data exchange across HR, Core, Logistics, Admin, and Finance subsystems.
Version Control and Collaboration	Git and GitHub were utilized for source code management, version control, and team collaboration, ensuring an organized and efficient development workflow.
Continuous Integration and Deployment (CI/CD)	GitHub Actions was configured to automate the testing and deployment process, enhancing system stability and reducing manual errors.
Dependency Management	Node Package Manager (NPM) was used for managing JavaScript

	dependencies, while Composer handled PHP dependencies, ensuring that all required packages were properly maintained.
Hosting and Deployment	INDEVFINITE was used as it provide versatile system integration that focuses from website design, website maintenance, web development and supports different Tech stack that suites for microservices architecture.

Figure 15. Tools and Technologies Used

Each of these technologies played a crucial role in the successful development and deployment of the system, ensuring that it met the functional and technical requirements outlined for the Merchandising System.

Code Integration and Interoperability

Integration between HR1 and other system components was achieved through Laravel's RESTful API implementation. API endpoints were documented and standardized to facilitate seamless interaction between microservices. Authentication and authorization mechanisms were implemented to ensure that only authorized requests were processed.

To maintain interoperability, strict API contracts were followed, and payload structures were standardized across all system modules. JSON format was used for data exchange, ensuring compatibility with all integrated services. Error-handling mechanisms were incorporated to prevent data conflicts and maintain system stability.

Version control practices were enforced using Git, with branch management strategies such as feature branching and pull request reviews to maintain a clean and structured codebase. Continuous integration (CI) pipelines were established to automate testing and deployment processes, minimizing the risk of errors during integration.

Integration Testing and Debugging

Integration testing and debugging were carried out using Laravel's built-in testing framework and GitHub's Laravel Workflow CI/CD Pipeline, ensuring continuous validation of system functionality. The testing process followed a structured approach to verify seamless communication between microservices and maintain data integrity.

Automated test scripts and continuous monitoring mechanisms were implemented to streamline validation processes and detect anomalies in real time. By leveraging the Laravel CI/CD Pipeline, testing efficiency was improved, ensuring that each deployment-maintained system stability and performance.

Comprehensive testing strategies were employed to validate system functionality and interoperability. The testing process included:

1. **Unit Testing:** Individual components and services were tested to ensure that each function operated as expected.
2. **Integration Testing:** RESTful API calls were tested between HR1 and other microservices to verify seamless data exchange.
3. **End-to-End Testing:** The entire workflow, from HR to other System Modules was validated to ensure system cohesion.
4. **Bug Tracking and Debugging:** Issues identified during testing were logged, categorized, and resolved using debugging tools integrated with the development environment.

4.8. Testing and Quality Assurance

To ensure the Merchandising Management System – HR1 runs smoothly and meets high-quality standards, a thorough testing and quality assurance process was carried out. Different testing methodologies were used to verify that each component functions correctly and integrates seamlessly with other modules.

Testing and Quality Assurance

A mix of manual and automated testing was performed to identify and fix any issues before deployment. The key testing strategies included:

Unit Testing: PHP-Unit was used to test individual functions and components of the Laravel-based system to ensure they work as expected.

Integration Testing: Data flow and communication between HR1 and other modules (HR2, HR3, HR4, ADMIN, and FINANCE) were tested through RESTful API calls to verify smooth interactions.

End-to-End (E2E) Testing: Every module and submodule was tested to ensure data consistency and interoperability, allowing seamless data exchange throughout the system.

Regression Testing: Automated test scripts, such as those in a GitHub CI/CD pipeline, were executed whenever new features were added to ensure that existing functionality remained intact

Performance Testing: The Merchandising system – HR1 was evaluated under different workloads to measure response time, scalability, and overall efficiency.

Security Testing: Laravel's built-in security features were tested to detect vulnerabilities related to authentication, encryption, and access control.

Test Cases and Test Data

To ensure all system functionalities were properly tested, structured test cases were developed, covering key areas such as authentication, data exchange and security.

Here are the highlighted major cases:

Test Case	Description	Expected Result	Status
User Authentication	Validate login and role-based access	User is authenticated successfully	**Passed**
API Data Exchange	Verify data transfer between HR1 and HR2,HR3,HR4	Data is synchronized accurately	**Passed**
Job Posting Module	Test job creation and applicant submission	Jobs are posted and applications received	**Passed**

AI Resume Evaluation	Evaluate Resume based on Job post	Evaluated Accurately with score	**Passed**
New Hire List Module	Webhooks Realtime data transfer	Hr 2 and Hr 3 will have the new hired employee	**Passed**
Document management	Document Upload API	File uploaded from employee portal to document management	**Passed**
Leave Management	Leave Request from portal	Request sent and updated status	**Passed**
Security Validation	Check system vulnerabilities	No unauthorized access allowed	**Passed**

Table 7. Test Cases and Test Data

Real-world HR scenarios were used to generate test data, including employee records, job applications, details, and system transactions.

Testing Strategies and Methodologies

Once all test cases were executed, the system's performance and stability were analyzed. Any detected issues were logged and categorized based on severity:

Critical Bugs: Issues that caused major system failures, such as authentication errors or data loss.

High-Priority Bugs: Problems affecting essential workflows, High-volume of applicants caused of Server Overload.

Medium-Priority Bugs: UI/UX inconsistencies and minor logic errors that did not disrupt core functionality.

Low-Priority Bugs: Responsive issues, minor layout misalignments, and usability improvements based on the device.

All critical and high-priority bugs were fixed before deployment, while minor improvements were scheduled for future updates.

Quality Assurance Measures

To maintain system reliability and prevent defects, several quality assurance measures were implemented:

Code Reviews: Each piece of code was reviewed by other developers before being merged to ensure quality and consistency.

Continuous Integration and Deployment (CI/CD): GitHub’s CI/CD pipeline automatically ran tests before deploying new updates, reducing the risk of errors.

User Acceptance Testing (UAT): HR personnel and stakeholders tested the system in real-world conditions to validate its usability and functionality before final deployment.

Post-Deployment Monitoring: The system was continuously monitored using logs and performance metrics to quickly identify and resolve any emerging issues.

Through these structured testing and quality assurance processes, the Merchandising Management System: HR1 is designed to be reliable, secure, and optimized for smooth HR operations.

4.9. Results and Evaluation

Project Outcomes and Deliverables

The Merchandising Management System has produced a range of outcomes and deliverables that impact the efficiency and management in human resource operations. Key project deliverables include the successful implementation of an applicant tracking system, employee task assignment, and employee onboarding and offboarding. These features has streamlined the hiring, onboarding, and offboarding processes, enhancing HR efficiency and improving employee engagement.

A technical documentation, such as user manuals and training guides, to ensure effective system management and maintenance.

Alignment with Project Objectives

The project's objectives were focused on addressing the inefficiencies in manual HR processes, particularly in recruitment management, onboarding, and offboarding.

The following key objectives were met:

1. Streamlining hiring and applicant tracking, where the HR personel manage job postings and track applicant progress seamlessly reducing the need for paper-based submissions, improving efficiency;
2. Enhancing employee onboarding, where the system provided an organized task board that ensured a structured training and orientation process;
3. Employees could submit concerns and feedback through the provided Freedom Wall which helps in improving HR decision-making and problem-solving; and
4. Employees could submit resignation letter digitally, making the offboarding process more efficient.

Stakeholder and User Feedback

Stakeholder feedback is gathered based on their experience while using and testing the system. The evaluation follows a client acceptance grading form. According to the feedback, the only aspect that requires modification is the color scheme, which should be aligned with the client's business branding for the employee portal.

All other components and functionalities were reported to be functioning well, with no additional feedback or issues raised by the stakeholders.

Lessons Learned:

The team gained valuable insights into how the merchandising management system operates, particularly in how the company manages its human resources and handles mass applicants and employees. The researchers learned about the specific processes the stakeholders follow and the overall flow of their business operations.

Through this experience, the team acquired invaluable knowledge that can be applied to future projects and professional work. The researchers developed a deeper understanding of the challenges they faced, identified areas that need improvements, and discovered opportunities for innovation to enhance their business processes.

As a team, the researchers also learned how to integrate individual modules into a cohesive system, as well as how to work with different systems simultaneously. We realized that not everything goes according to plan, flexibility and readiness to adjust are essential. Following deadlines and adhering to project timelines are crucial to completing the project successfully.

From the planning phase to the development phase, we learned that using and following an Agile development approach can significantly improve project flow and enhance overall progress. The iterative nature of Agile allowed us to adapt quickly, respond to changes, and continuously improve our system based on feedback.

Regular meetings played a key role in our progress, allowing us to continuously exchange information, align on objectives, and gather critical data from stakeholders throughout the development process.

Chapter 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1. Key Takeaways and Summary

The integration of the Applicant Tracking System (ATS) with Onboarding Management within the Merchandising Management System: Human Resource 1 has successfully streamlined the hiring and onboarding processes. The project addressed key challenges faced by Great Wall Arts, including inefficient manual tracking, delays in onboarding, and lack of centralized data management. By automating job postings, application tracking, and task assignments, the system enhanced HR efficiency and improved the experience for both applicants and employees. The project demonstrated the value of digital transformation in HR management, highlighting how automation and structured workflows contribute to operational efficiency and employee engagement. Stakeholder feedback played a crucial role in refining system functionalities, ensuring alignment with organizational needs and expectations.

5.2. Project Achievements and Contributions

The project achieved significant milestones in optimizing the recruitment and onboarding workflows for Great Wall Arts. The development of an online job posting feature enabled applicants to submit their applications seamlessly, while the ATS ensured systematic tracking of candidate progress. The onboarding module provided a structured process for new hires, allowing them to complete tasks, upload necessary documents, and access training materials through an interactive portal. HR personnel benefited from real-time notifications and automated document management, reducing administrative workload and expediting decision-making. The inclusion of an offboarding system further enhanced employee lifecycle management by providing a clear, efficient resignation process. This project not only improved HR operations but also contributed to the broader goal of enhancing employee experience and engagement in a merchandising-focused environment.

5.3. Future Work and Enhancements

While the system successfully addressed key challenges, there are opportunities for future enhancements. One area for improvement is the implementation of a mobile-responsive version to increase accessibility for applicants and employees who prefer to access the platform via mobile devices. Enhancing real-time notifications, such as interview reminders and onboarding task deadlines, would further improve user engagement. Additionally, integrating advanced analytics and reporting tools would provide HR personnel with deeper insights into recruitment trends, onboarding effectiveness, and employee retention patterns. Future iterations could also incorporate AI-driven recommendations for job applicants and personalized onboarding experiences tailored to specific roles. By continuously refining the system based on user feedback and emerging industry trends, the project can further optimize HR processes and drive long-term efficiency in workforce management.

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APPENDICES

Appendix A: Process Diagrams

a. Applicant Process

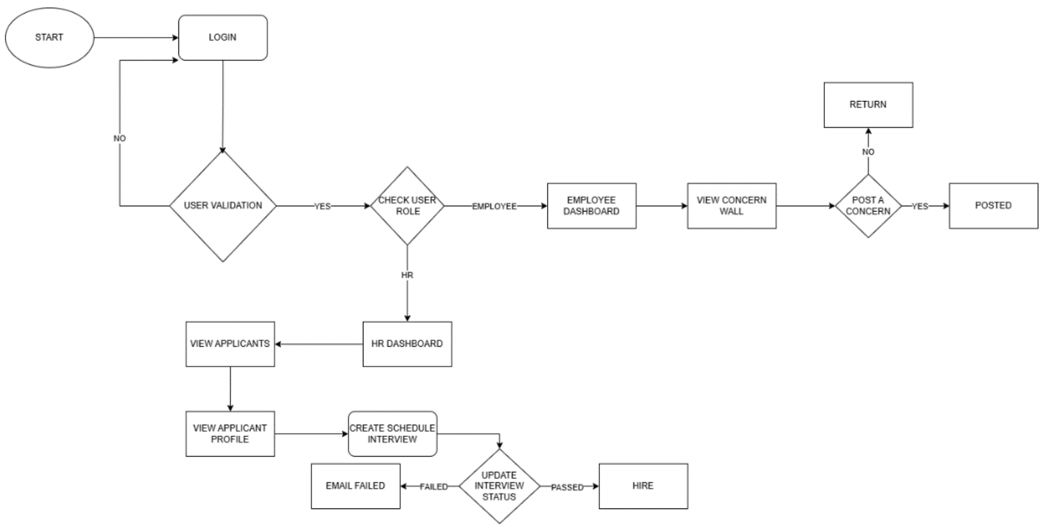


Figure 16.1. Applicant Process Diagram

b. Employee Process

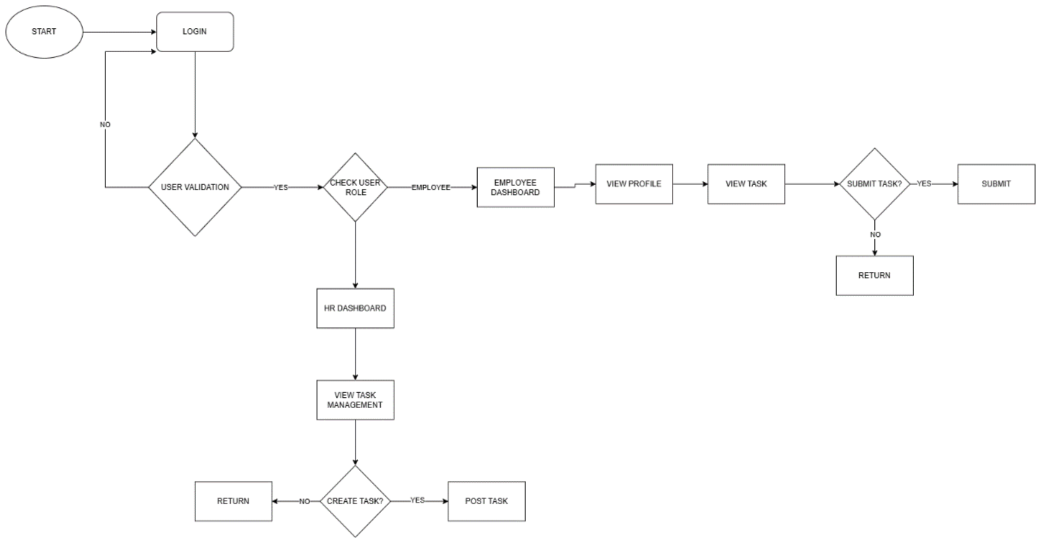


Figure 16.2. Employee Process Diagram

c. Resignation Process

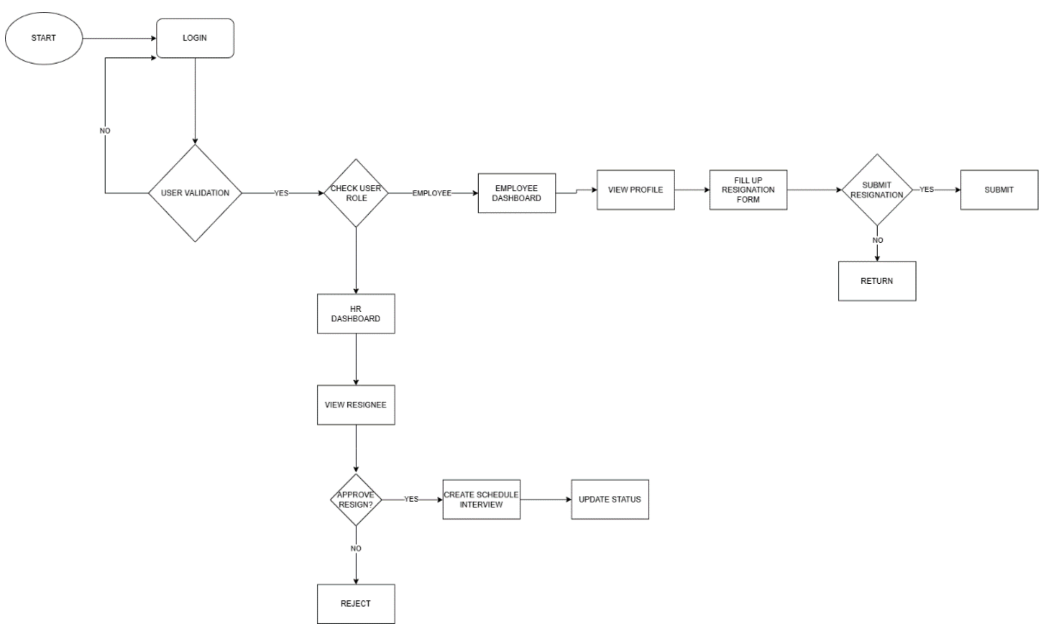


Figure 16.3. Resignation Process Diagram

Appendix B: User Surveys

Survey Questionnaire: The questionnaires were divided into 3 sections; for HR, for Employees, and for Applicants.

For HR

1. Did you logged in the system as exactly as your role?
2. How effective is the AI resume evaluation in shortlisting qualified candidates?
3. Do you think the concern wall inspired by X (formerly Twitter) feature, helped in employee engagement, and will be of help in resolving the issues faster?
4. Do you think the system reduced the time you spend on applicant tracking?
5. Do you think the taskboard will be able to help you monitor the onboarding progress more efficiently?
6. Are the resignation and offboarding features easy to manage and track?
7. How helpful are the real-time notifications and automated updates in managing applicants and employees?
8. What feature do you find most useful?

For Employees

1. Did you logged in the system as exactly as your role?
2. Do you feel that the system helps you stay informed about your tasks and progress?
3. Do you feel safe when using the anonymous feedback wall (concern wall) when voicing out your issues with your fellow employee or with the company?
4. How useful do you find the anonymous Feedback Wall in voicing concerns or suggestions?
5. How easy was it to understand and complete your onboarding tasks through the portal?
6. What feature do you find most useful?

For Applicants

1. How easy was it to find and apply for a job through the system?
2. Did you receive timely updates about the status of your application?
3. Was the system accessible and easy to use on your device (e.g., mobile or desktop)?
4. Did the application form cover all necessary information without being too long?
5. Can you still use your email address to apply on other job position in the site when you already used it?

For all types of roles (HR, Employee, Applicant)

1. As a user, do you feel that your personal information is safe?
2. Is the system user-friendly?
3. On a scale of 1–5, how satisfied are you with the system's performance?
4. What improvements or features in the system would you like to see added in the future?

Link to Survey: https://forms.gle/G2ibth9a8GrqULPd6

Survey Results:

https://docs.google.com/spreadsheets/d/1cwdVqUG0ti0TTnFO506J-L2ktPwpGeRh1o1lPDffbPE/edit?usp=sharing

Appendix C: User Stories and Use Cases

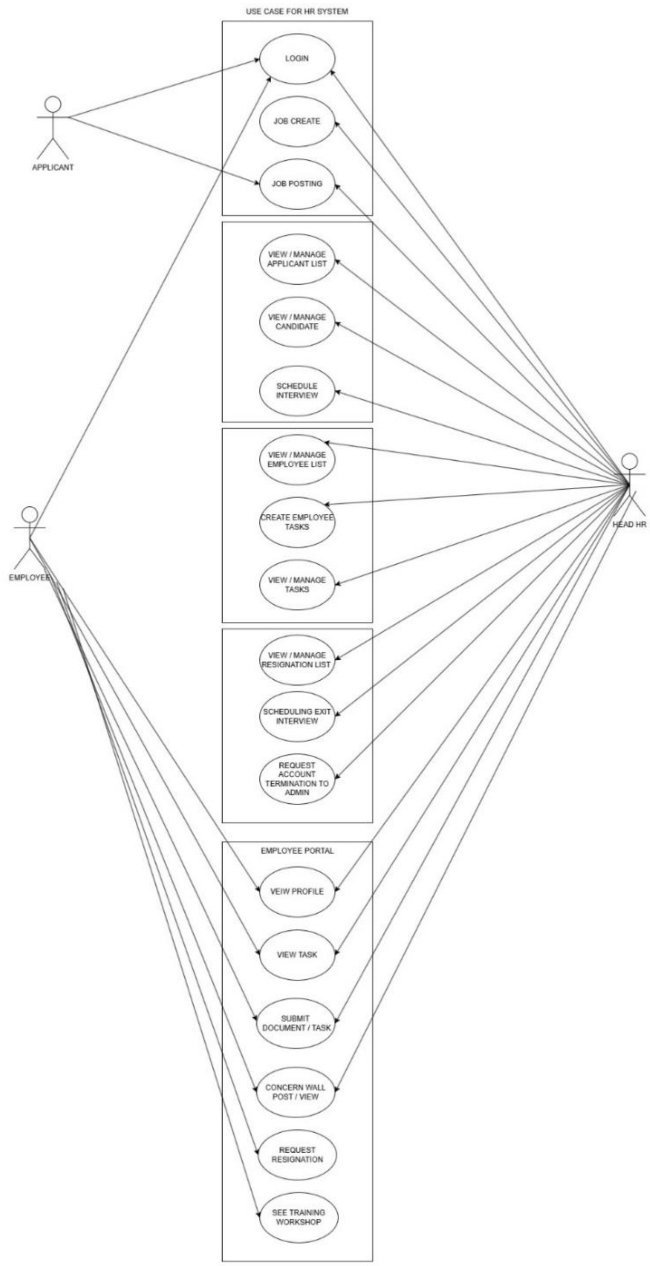


Figure 8. Use-case Diagram (pp. 101)

Appendix D: Sample Data

Below is a sample dataset that includes application forms inspired by real-life biodata templates and job postings, along with resignation letters and task submissions. The dataset also features supporting files and images, reflecting common practices in professional environments.

1. Application Form
    1) **First Name:** Juan
    2) **Middle Name (optional):** B.
    3) **Last Name:** Tamad
    4) **Address:** #3 JuanTamad St. San Juan City.
    5) **Email:** juantamad69@gmail.com
    6) **Nationality**: Filipino
    7) **Religion:** Catholic
    8) **Sex:** Male
    9) **Birthday:** 9/12/1980 (dd/mm/yy)
    10) **Contact:** 09459788091
    11) **Civil Status:** Single
    12) **Resume:** Resume/CV PDF
    13) **Referred by:** Barabas
2. Task Creation
    1) **Department :** Sales

2) **Title :** SEND SSS NUMBER AND SOFT COPY

3) **Start Date:** 9/4/2025 (dd/mm/yy)

4) **Deadline:** 13/4/2025 (dd/mm/yy)

5) **Task Description:** MUST SEND SSS NUMBER BY IMAGE AND A SOFT COPY OF SSS FILE

3. Announcements

   1) **Title:** URGENT THINGS ARE URGENT

   2) **Description:** SOME SAMPLE TEXT OF ANNOUNCEMENT

4. Task Submissions

   1) Image/PDF File

5. Resignation Form

   1) **Job Position:** Sales

   2) **Email**: juantamad69@gmail.com

   3) **Contact Number:** 09459788091

   4) **Reason**: New job opportunity at another field.

   5) **Resignation Letter:** PDF file of letter

Appendix E: Stakeholder Interview Transcripts

Interviewee: Ms. R. L.

Role: Company Owner

Date: April 27, 2024

Purpose: Understand current challenges in human resource operations.

Questions and Responses:

Q1: Can you describe how you currently handle job applications?

A: We handle it in a traditional way, manually. We would very like to have an online page in our website for job application.

Q2: What features you would like to see in an online application page?

A: We need an on-page job application system on the website where applicants can see open positions and submit their applications directly. Applicants should be able to submit their requirements online as part of the application process. We would also like if the applicants can receive an information through email regarding their application.

Q3: What tools do you use for keeping your employee records?

A: We don't have a proper database, we only use Excel or Google Sheet for keeping records.

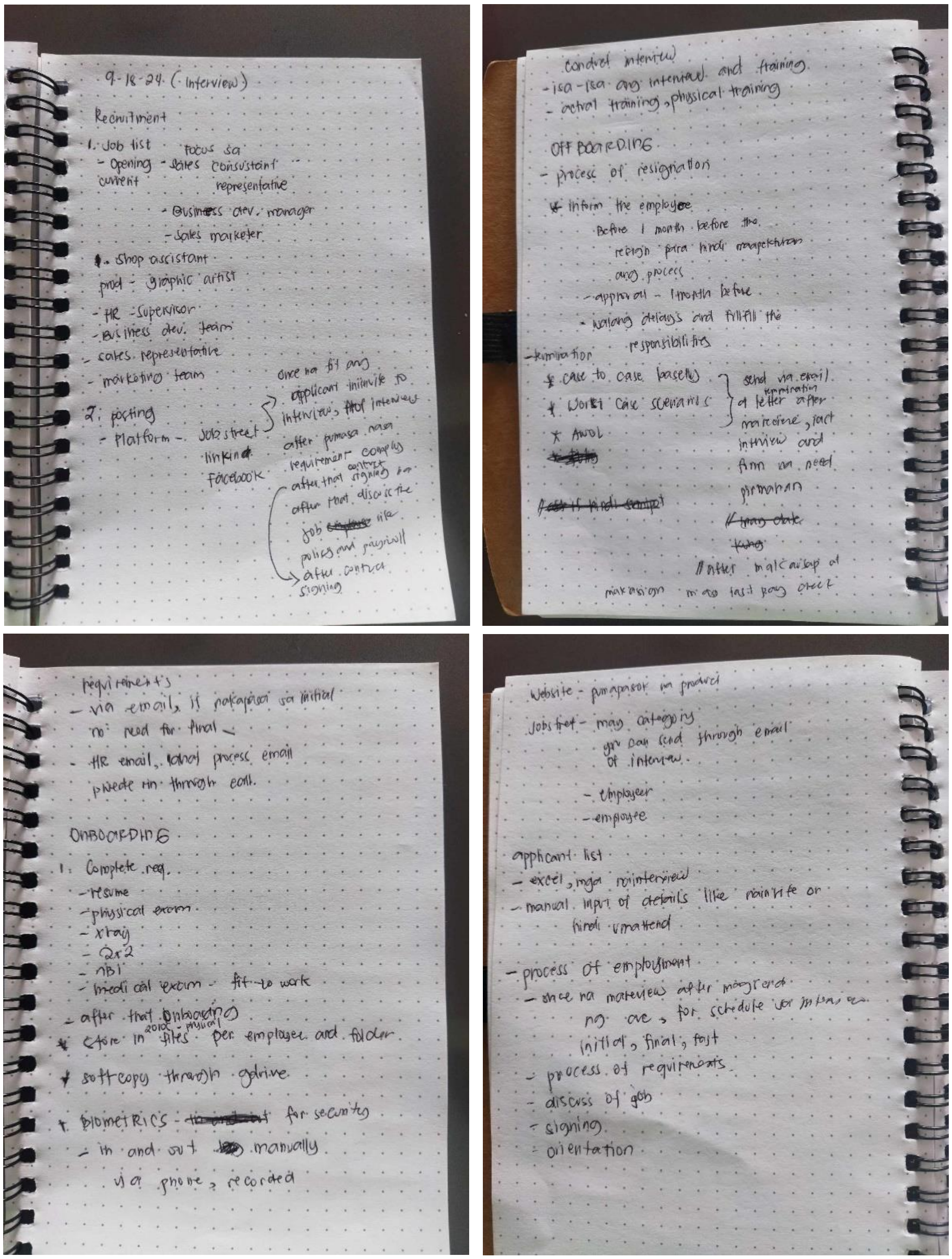


Figure 17. Stakeholder Interview Transcripts

Appendix F: System Performance Test Results

This appendix presents the detailed results of the performance testing carried out on the Merchandising Management System – Human Resource 1, focusing on response times, load handling, and scalability assessments.

Test Scenario	Expected Result	Actual Result	Status
Application Submission (50 applicants simultaneously)	Submission time ≤ 2 seconds	1.8 seconds	Passed
AI Resume Evaluation (50 applicants simultaneously)	Evaluation time ≤ 5 seconds	4.0 seconds	Passed
Publish/Unpublish (10 jobs simultaneously)	Update time ≤ 2 seconds	1.3 seconds	Passed
Emailing Interview schedule (10 applicants)	Emailing schedule time ≤ 3 seconds	2 seconds	Passed
Task Creation (10 simultaneously)	Posted time ≤ 2 seconds	1.8 seconds	Passed

Task submission (10 users simultaneously)	Submission time ≤ 3 seconds	2.5 seconds	Passed
Users Login (10 users simultaneously)	Submission time ≤ 2 seconds	1.5 seconds	Passed

Table 8. System Performance Test Result

Appendix G: Data Security Measures

This appendix is the real measures of data security measures of the system the table below is used to be showed the detailed and summarized view of measures.

Category	Description
Access Control	Role-Based Access Control (RBAC): Access is granted based on user roles (HR Staff, Employee, Applicants). Session Management: Automatic session timeout after 15 minutes of inactivity.
Data Encryption	Encryption in Transit: All data transmitted between clients and servers is encrypted using SSL/TLS protocols via HTTPS, which has been deployed indefinitely to ensure secure communication. Encryption at Rest: Sensitive data is encrypted using AES-256 encryption.

API Security	**API Gateway**: Manages and secures all API traffic, ensuring only authorized requests are processed. **API Tokens:** Each API request requires a secure API token for authentication and authorization. Tokens are time-bound and encrypted to prevent unauthorized access.
Data Backup & Recovery	Daily Automated Backups: Regular daily backups are stored in encrypted formats. Disaster Recovery Plan: Data recovery procedures ensure system restoration within 4 hours in the event of data loss or failure.
Data Privacy Compliance	User Consent: Employees are informed about data collection, processing, and storage practices. Consent is obtained before any data handling. Data Retention Policy: Personal data is

	retained only as necessary and securely deleted after the retention period.
HTTPS Deployment	Secure Communication: The system is fully deployed over HTTPS indefinitely, ensuring all data in transit is encrypted and protected from eavesdropping and man-in-the-middle attacks.
Security Awareness & Training	Regular security training sessions are conducted for HR staff and system administrators, focusing on data protection protocols, recognizing social engineering attacks, and securing access credentials.

Figure 18. Data Security Measures

Appendix H: Adviser Acceptance

Name of Group Members/Individual:

1	CAMO, JOBERT H.	4	NOORA, CHRISTIAN A.
2	CULLO, JAY F.	5	ROSARIO,ROSELYN A.
3	DELA TORRE , JULIUS ERVIN B.	6	

Title of the Project: EIS: ***Merchandising Management System (Human Resource 1): Onboarding and Offboarding, Task Management, and Online Job Posting with AI Resume Evaluation***

	Technical Proofreader	Grammarian
Name	Mr. Alejandro B. Adovas	Ms. Ma. Nica P. Bersola
Designation	Professor, College of Computer Studies	Professor, College of Computer Studies Secretary to the Vice President for Academic Affairs
Work Place Address	Bestlink College of the Philippines Millionaire's Village, Novaliches Quezon City	Bestlink College of the Philippines Millionaire's Village, Novaliches, Quezon City
Academic/ Professional Qualifications / Membership	• Master of Science in Information Technology (ongoing) • Bachelor of Science in Computer Science • Tesda Certified Level 1 Trainer (TM1) • Instructor I, Bestlink College of the Philippines	• Bachelor of Secondary Education, Major in English • Licensed Professional Teacher
Email Address	andy.adovas@gmail.com	nicsbersola20@gmail.com
Telephone	09054622478	09612383153
Signature	*I hereby confirm that I have undertaken to supervise the project mentioned above and I do certify that I am not a member of the Project Examination Board (PEC)*	
	Mr. Alejandro B. Adovas *Date: May 20, 2025*	Ms. Ma. Nica P. Bersola, LPT *Date: May 20, 2025*

Bestlink College of the Philippines

1071 Brgy Kaligayahan, Quirino Highway, Novaliches,Quezon City

Project Acceptance Form

Bestlink College of the Philippines
Bachelor of Science in Information Technology
Lot 762 Topaz and Sapphire St. Millionaires Village Barangay San Agustin, Novaliches, Quezon City

Project Details

- **Project Title:** Merchandising Management System (Human Resource 1): Onboarding and Offboarding, Task Management, and Online Job Posting with AI Resume Evaluation
- **Project Description:** The Merchandising Management System: Human Resource 1 is a web-based system designed to optimize and enhance important HR processes like task management, applicant tracking, onboarding, and offboarding. The solution incorporates features like AI-powered resume evaluation, real-time e-mail notifications, and onboarding procedures. It also features a unique anonymous feedback wall that allows employees to voice out their concerns in confidence not being known, while HR monitors them and responds as needed. For companies that place a high priority on merchandising, the system provides an organized and user-friendly method of managing the whole employee lifetime. It was developed to increase HR performance and employee satisfaction.
- **Developed By:**

Camo, Jobert H.	(21121161)
Cullo, Jay F.	(20016896)
Dela Torre, Julius Ervin B.	(21017037)
Noora, Christian Jay A.	(21017516)
Rosario, Roselyn A.	(21121089)

- **Academic Program:** Bachelor of Science in Information Technology (BSIT)
- **Capstone Adviser:** Mr. Paulo B. Tolentino
- **Supervising Professor:** Mr. Ronald G. Roldan, Jr.
- **Company Name:** Great Wall Arts
- **Company Representative:** Ms. Russeline Laceda
- **Date of Project Submission:** April 16, 2025

Acceptance and Agreement

- The undersigned, representing **Great Wall Arts**, acknowledges the completion of the project titled **Merchandising Management System (Human Resource 1): Onboarding and Offboarding, Task Management, and Online Job Posting with AI Resume Evaluation** developed by the students of **Bestlink College of the Philippines**. By signing this form, the company agrees to the following:

1. **Acceptance of the Project:**
   The company accepts the project as developed by the students and acknowledges that it meets the agreed-upon requirements and specifications.
2. **Use of the Project:**
   The company agrees to use the project for its intended purpose as outlined in the scope. The company may use, modify, or distribute the project as needed, subject to agreed-upon terms.

3. **Feedback and Evaluation:**
   The company agrees to provide feedback on the project's performance, usability, and any areas for improvement. This feedback will be used for academic evaluation and potential future enhancements.
4. **Confidentiality and Intellectual Property:**
   The company acknowledges that the project may contain confidential information and agrees to handle it responsibly. The agreement between the university and the company will govern any intellectual property rights.
5. **No Warranty:**
   The project is provided "as is" without any warranties, express or implied. The university and the students are not liable for any issues arising from using the project.

Signatures

Company Representative:
I, **Ms. Russeline Laceda,** representing **Great Wall Arts,** acknowledge the project's completion and agree to the terms outlined above.
Signature: ______
Name: SARAH COSMIANO / JOHN LLOYD DE FIESTA
Designation: HR / IT
Date: May 21, 2025

University Representative:
I, **Mr. Ronald G. Roldan, Jr**., representing **Bestlink College of the Philippines**, confirm that the students have completed the project as part of their academic requirements.
Signature: ______
Name: Ronald Roldan Jr.
Designation: ______
Date: 05-02-25

RATING/GRADE:	Maximum Points	Score Given
1. Functionality:	**20**	20
- Completeness of features	10	10
- Accuracy and reliability	10	10
2. Usability	**15**	14
- User interface design	8	7
- User experience (UX)	7	7
3. Technical Design and Architecture	**15**	13
- System architecture and modularity	8	7

- Code quality and organization	7	6
4. Performance	**15**	13
- Speed and responsiveness	8	7
- Load handling and scalability	7	6
5. Security	**10**	9
- Data protection and privacy	6	5
- System vulnerabilities	4	4
6. Testing and Validation	**10**	9
- Unit testing and bug fixing	5	4
- User acceptance testing (UAT)	5	5
7. Documentation	**10**	8
- User manuals and guides	5	4
- Technical documentation	5	4
8. Innovation and Creativity	**5**	4
- Novelty of the solution	5	4

Feedback Section (Optional)

The company is encouraged to provide feedback on the project to help improve its quality and usability.

1. **What aspects of the project do you find most useful?**

2. **What improvements or additional features would you suggest?**

3. **Overall satisfaction with the project (Rate from 1 to 5):**
   ☐ 1 (Poor) ☐ 2 (Fair) ☐ 3 (Good) ☑ 4 (Very Good) ☐ 5 (Excellent)

Additional Comments:

COLOR SHOULD BASE IN BRAND GUIDELINE

Appendix I: Panel Evaluation and Signature

Figure 19.1. Pre-Oral Defense

Figure 19.2. Oral Defense

Appendix J: Capacity Planning

1. Overview

Capacity planning ensures that the Merchandising Management System – HR1 can support business growth over the next five years while maintaining performance, scalability, and reliability. This process involves analyzing resource allocation, system demand, and future expansion requirements.

2. System Resource Requirements

The system is designed to operate efficiently while accommodating increased usage over five years. A backend server with a minimum of 4 vCPUs and 8GB RAM, along with SSD storage, ensures optimized API performance. The database server, powered by MySQL, requires 6 vCPUs and 16GB RAM to handle the growing volume of transactions and data storage. Cloud-based frontend hosting with auto-scaling capabilities allows the system to adapt to rising user traffic. Storage requirements start at 500GB, with expansion capacity based on document and employee record growth. Additionally, a 200 Mbps network bandwidth supports increasing concurrent users and data interactions.

3. Scalability Considerations

To ensure long-term performance and adaptability, the system employs several strategies. Auto-scaling mechanisms within the cloud infrastructure adjust resources dynamically as demand fluctuates. Database optimization techniques, including indexing, query caching, and partitioning, enhance data retrieval efficiency. The microservices architecture enables independent scaling of system components based on workload requirements, ensuring that no single service becomes a bottleneck.

4. Performance Benchmarking

Regular performance testing maintains system efficiency and supports expected growth. The API response time remains below 300ms, ensuring smooth data exchange. The system is designed to handle 500+ concurrent users without performance degradation. Database queries execute within 100ms, optimizing data retrieval speed. With a target uptime of 99.9%, high availability is ensured through proactive monitoring and system redundancies.

5. Future Expansion Strategy

The system is built to accommodate business growth over the next five years through strategic expansion measures. Cloud-based scaling provisions additional computing resources dynamically, allowing flexibility as demand increases. Load balancing techniques distribute incoming requests efficiently across multiple servers, preventing system overloads. Data archiving mechanisms ensure older records are offloaded to minimize database load while preserving accessibility. To maintain security and compliance, regular updates and monitoring are implemented, aligning with evolving data protection regulations.

By following these capacity planning strategies, the Merchandising Management System – HR1 remains scalable, efficient, and ready to support long-term operational growth.

Appendix K: Pilot Companies' Background

Great Wall Arts was identified as the pilot company for the HR1: Interactive Onboarding and Offboarding Task Management and Online Job Posting System. Great Wall Arts, in Pasig, Metro Manila, sells handcrafted Filipino products that utilize modern engraving and printing techniques. Great Wall Arts, a small-to-medium enterprise (SME), was established in 2018 by a Christian couple and has grown consistently by combining traditional craftsmanship with modern-day production processes.

Great Wall Arts was chosen to be a pilot site because of its growing HR recruitment practices and the management was willing to consider digital transformation and structured onboarding workflows to pilot the HR1 functionalities.

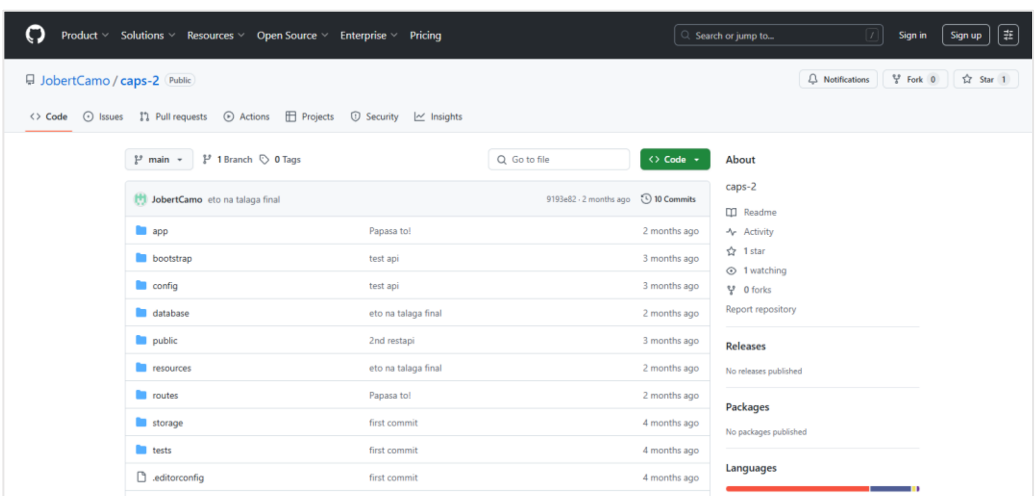
The site focus was to streamline friction in employee onboarding and offboarding, automate repetitive HR functions, and improve internal communication. During the pilot site period, Great Wall Arts provided valuable feedback on the usability of the HR1 system as it related to their current practices which improve the usability and effectiveness of the HR1 system.

Figure 20. Meeting with Client

Appendix L: Cloud Copy of the Codes (1 year validity)

The source code for the Merchandising Management System – HR1 is securely stored in a cloud-based repository, ensuring accessibility for development, maintenance, and future enhancements. The repository is hosted on GitHub and remains valid for one year, allowing authorized team members to access, review, and update the system as needed.

Repository Link: GitHub - CAPS-2 https://github.com/JobertCamo/caps-2?fbclid=IwY2xjawI2CoNleHRuA2FlbQIxMAABHVMROSbtZH0NytGIswqSey9rrpnU2z0AFfnKMI6V7f5349j5aaacxlLqoQ_aem_XEGXFnHciU5-Igx5KCv08A



Appendix M: IMRAD Format Summary

Introduction

In the era of business expansion and digitalization, human resource management has emerged as the backbone of organizational efficiency, especially for companies that heavily rely on effective recruitment and onboarding. The recruitment and onboarding processes are delayed, thereby resulting in more errors and overall reduced productivity by traditional recruitment processes and manual paper work. With the vision to address the issues, this project integrates critical human resource functionalities, such as an AI-based Resume Evaluation, Online Job Posting, and Task Management in a Merchandising Management System: Human Resource 1. This system is designed to optimize job posting, candidate tracking, and employee task management by offering an end-to-end, technology-enabled solution, thus enabling seamless transition from hiring to onboarding and organizational effectiveness and employee engagement.

Methods

The agile scrum technique used in this project enables incremental system modifications based on input from stakeholders. Surveys and interviews are used by the researchers to collect information on the efficacy and usability of the system. The Applicant Tracking System (ATS), Onboarding Management, and Employee Portal were all integrated into the system, which was created in a modular fashion.

Results

The Merchandising Management System has significantly improved human resource operations. The company decreased the amount of manual labor needed for applicant screening by improving the posting of job hiring and application tracking. The use of AI-based resume review sped up the recruiting process by finding a qualified candidates based on a predetermined criteria. By managing tasks efficiently, the employee portal ensured that new recruits completed the required onboarding tasks. Additionally, the offboarding module made sure that exit procedures were professional and well-documented. The solution increased overall employee engagement, reduced processing time, and improved HR efficiency.

Discussion

Compared to conventional human resource (HR) practices, utilization of AI-based recruitment and online task management yielded several benefits. The capability of the system to automate basic human resource (HR) processes resulted in less human error and workload. The addition of an interactive onboarding and offboarding system also facilitated an organized and employee-oriented system. However, issues of user

adaptation and data security problems had to be resolved, including additional streamlining such as AI-based learning programs and upgraded information security protocols.

Keywords: Merchandising Management System, Applicant Tracking System, AI Resume Evaluation, HR Automation, Onboarding

Appendix N: Comparison of the EIS to existing EIS's (5 pages)

The Merchandising Management System: Human Resource 1 is an integrated system designed to enhance the hiring and onboarding processes within the company Great Wall Arts. This system enhances efficiency by automating job postings, applicant tracking, onboarding and offboarding workflows, reducing manual administrative tasks for HR team. By digitizing key HR functions, it ensures a seamless transition from recruitment to employment, allowing new hires to complete requirements and training schedules within a structured and interactive system.

Comparing this system with existing Employee Information Systems (EIS) is essential to highlight its unique capabilities and improvements over traditional HR management methods. While traditional systems primarily focus on employee data storage, the Merchandising Management System: Human Resource 1 incorporates automated workflows, applicant tracking, and onboarding management, making it a comprehensive solution for modern HR operations. This comparison will demonstrate how the system addresses existing challenges, enhances user experience, and provides a scalable approach to managing workforce data efficiently.

Unique Features and Key Advantages:

1. **AI-Powered Resume Evaluation**

   Traditional EISs rely heavily on human screening of resumes, which can be time-consuming and prone to bias. The AI module in this system evaluates resumes based on relevance, skills, and experience against job requirements, speeding up the shortlisting process while promoting unbiased hiring decisions.

2. **Automated and Structured Workflows**

   While most EISs provide basic functionality for storing and retrieving employee data, this system offers structured workflows for each HR process. For instance, onboarding involves a step-by-step progression with predefined tasks and document submissions, visible to both the employee and HR.

3. **Interactive Employee Portal**

   Employees can access a centralized portal where they can view pending tasks, submit documents, receive onboarding materials, and track the status of their application or resignation. Existing EISs often lack such interactive features, which limits employee engagement and self-service.

4. **Custom Offboarding Process**

   Many traditional systems neglect the offboarding phase, which often leads to inconsistencies in resignations, exit interviews, and clearance. The proposed system provides an automated offboarding module that includes resignation submission, clearance checklists, and feedback forms, ensuring a smooth and traceable exit process.

5. **Anonymous Feedback Wall (Concern Wall)**

   A distinct feature of this system is the Feedback Wall, inspired by platforms like X (formerly Twitter). This wall enables employees to post concerns, suggestions, or feedback anonymously, fostering a safe space for open communication. Employees can voice issues related to coworkers, management, or company culture without fear of retaliation, increasing psychological safety and transparency. While these posts appear anonymous to fellow employees, the HR department has visibility into the identity of the poster, allowing them to follow up appropriately if the concern requires attention. Traditional EISs rarely include tools that promote real-time, anonymous communication, making this feature a unique channel for improving employee well-being and HR responsiveness.

Comparison Table: New vs. Existing EIS features

Feature	Traditional EIS	Merchandising Management System: HR1
Data Management	Employee records only	Employee records + AI resume insights

Table 9. Comparison Table – New vs. Existing EIS

Feature	Traditional EIS	Merchandising Management System: HR1
Recruitment	Manual screening	AI resume filtering + automated posting
Onboarding	Limited/Manual	Structured, task-based onboarding
Offboarding	Often absent or manual	Dedicated module with traceable steps
Task Management	Not typically included	Centralized taskboard for new hires
Applicant Tracking	Basic data logging	Real-time tracking + status updates
Employee Interaction	Minimal	Interactive employee portal
Concern Wall	Feedback is usually formal and traceable	Allows anonymous employee posts visible only to HR

Table 9. Comparison Table – New vs. Existing EIS

Impacts on HR Efficiency

In comparison to traditional and existing systems, the Merchandising Management System: Human Resource 1 has a greater impact on HR efficiency, accuracy, and employee satisfaction. The integration of advanced features reduces unnecessary administrative tasks, allowing HR staff to focus on more high-level responsibilities. The enhanced user interface, employee portal with an integrated anonymous feedback wall features also contribute to increasing employee engagement and quicker resolution of issues.

Furthermore, the system’s capacity to generate real-time reports, send automated notifications, and track individual task completion increases transparency and accountability within the organization. This digital development reduces the risk for human error, guarantees policy compliance, and creates a more professional experience for both applicants and current employees.

DETAILED TECHNICAL DOCUMENTATION

6.1. System Architecture

The system architecture of the Human Resource Management System (HRMS) is modular and integrates multiple subsystems to support the full employee lifecycle—from recruitment and onboarding to offboarding and succession planning. It is designed to streamline HR processes, ensure data consistency, and facilitate seamless communication between different departments.

The architecture uses an integrated system approach, where each module communicates through defined data flows and integration points. Core modules include Recruitment Management (ATS), Onboarding Management, Task & Document Management, Employee Portal, Learning Management, Offboarding Management, and Succession Planning.

Detailed Diagrams Illustrating System Components and Interactions

Referencing the BPA Level 3 Diagram, the following components and their interactions are highlighted:

Component	Description
Recruitment Management (ATS)	Handles job postings, applicant screening, evaluations, and interview scheduling. Integrates

	with Succession Planning for hiring strategy alignment.
Onboarding Management	Manages new hire onboarding processes, coordinates document collection, and assigns onboarding tasks from employee portal.
Task Management	Facilitates the creation, assignment, and monitoring of tasks during onboarding and offboarding processes.
Document Management	Manages uploading, reviewing, and validating employee documents. Integrated with Task Management and HR notifications.
Employee Portal	Central hub for employees to view tasks, submit documents, manage profiles, apply for resignations, and access learning materials, pre-employment document submissions.

Offboarding Management	Manages resignation applications, HR approvals, exit interviews, and coordination of the offboarding process.
Learning Management	Tracks training schedules and updates employee learning records. Displays training workshops in the Employee Portal.
Succession Planning	Provides candidate lists for job requisitions, monitors candidate progress, and feeds into the recruitment pipeline.
Administration Management	Oversees account termination requests and ensures role-based access control.

Figure 21. System Components and Interactions

6.2. Information Systems Integration

This section describes the integration of various information systems within the Human Resource Management System (HRMS) to enhance operational efficiency and data consistency. The integration ensures seamless communication and data sharing between different modules and systems, reducing manual work and improving accuracy.

This is the sample diagram that shows the data flow in every module that are integrated:

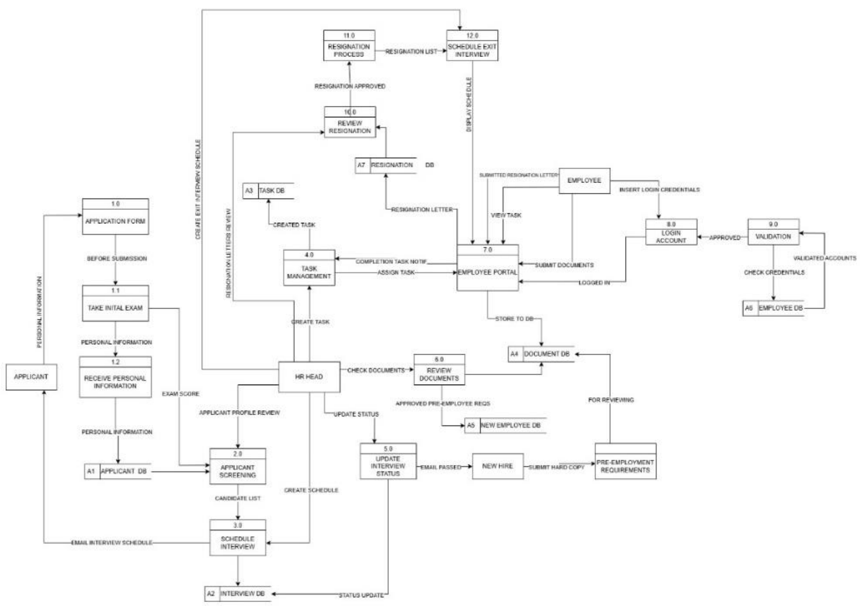


Figure 12. Dataflow Diagram (pp.118)

The Human Resource Management System (HRMS) is integrated with the following systems to streamline HR processes and improve productivity:

1. **Training Management System:** Integrated to manage new hired training schedules, assign training/trainee and track new hired training progress.
2. **Document Management System:** Integrated with the Employee Portal to allow employees to submit required pre-

employment documents directly, which are then routed to HR for review of documents.

3. **Employee Management System (Employee List):** Centralized management of employee profiles and records, ensuring all integrated modules have updated and accurate employee data.
4. **Succession Planning Module:** Integrated to identify and manage potential candidates for key roles, utilizing data from performance evaluations and employee records, identifying job roles to that are open to be posted for hiring in job posting.
5. **Admin Dashboard/Module:** Provides administrative users with access to all integrated systems for monitoring, validation, and management of HR operations.

These integrations eliminate redundant data entry, ensure data consistency across systems, and improve decision-making by providing real-time access to employee-related data.

Data transformation and mapping processes

The HR1 environment accomplishes structured data transformation and mapping so that consistency, quality and communication across these different modules—ATS, Onboarding, and Task Management—becomes automatic. When personal information, resumes, and testing scores are submitted through online job applications, these attributes are transformed into pre-mapped, standard structures for each of the onboarding modules. These completed transformation mappings are auto-populated directly into a record that is mapped to the employee record upon hiring, which gets the system pre-loaded with important information such as assigned job title(s) and task assignments.

Moving through onboarding, completed submitted requirements and mapped tasks data reporting allow HR to follow and monitor required complaisance and task completion through this user profile. Similarly, any employee data request changes through submitted options for changes such as personal data information changes or resignation requests, are dynamically mapped within the system so related modules will have instant access. These mapped data transformations happen with no or little manual encoding and eliminate repetitiveness will maintain quality data across all HR instances. The HR1 system had clean, modular data transformation pipelines which allows for scalability in the future with other enterprise tools integrations.

6.3. Application Design and Development

This section documents how input data from external APIs and frontend modules is validated, transformed, and mapped into the backend database models across different controllers used for API integration and interaction with other modules in the application.

EmployeeController:

Endpoint: update(Employee $employee, Request $request)

Transformation & Mapping Steps:

1. **Validation:** Checks format and required presence of fields (email, contact, job_position, salary, department).
2. **Mapping:** Maps validated form data to the Employee model's attributes.

Example Mapping:

1. email (frontend) → $employee->email (database)
2. job_position (form input) → $employee->job_position (DB)

JobPostController:

Endpoint: store(Request $request)

Transformation & Mapping Steps:

Validation: Ensures required job fields are correctly formatted.

Transformation:

Converts tags from comma-separated string to lowercase and trims whitespace.

Mapping:

Creates a Job record excluding tags, which are handled separately.

Endpoint: update($job, Request $request)

Validation: Ensures status is present.

Mapping: Updates only the status field of the job.

Endpoint: update($job, Request $request)

Validation: Ensures status is present.

Mapping: Updates only the status field of the job.

NotificationController

Endpoint: show($notification)

Mapping Logic:

Maps the id in the request to the correct Notification model entry.

No transformation is applied in the current version.

6.4. Database Schema and Data Management

Relationship Table

a. One-to-Many Relationships:

From Table (One Side)	To Table (Many Side)	Foreign Key in Many Side Table
User	Jobs	user_id
User	Task	user_id
User	Resignations	user_id
User	Post	user_id
User	Comments	user_id
Post	Comments	post_id
Task	Completed_Task	task_id
Task	Failed_Task	task_id
Applicants	Interviews	applicant_id
Department	User	department_id

Table 10.1. One-to-Many Relationship Table

b. Many-to-Many Relationship:

Tables Involved	Pivot Table	Foreign Keys in Pivot Table
Jobs ↔ Tags	job_tag	job_id, tag_id

Table 10.2. Many-to-Many Relationship Table

c. Independent Tables

Table Name	Purpose
Announcement	For site announcements
Notification	For user notifications
New_Hired	Stores new employee details

Table 10.3. Independent Table

Database Table Documentation

Table Name	Purpose	Key Fields
User	Stores all system users: Admin, HR, Employees	id (PK), department_id (FK)
Department	Stores department names	id (PK)
Jobs	Job postings created by Users	id (PK), user_id (FK)
Job_Tag	Many-to-Many connector between Jobs & Tags	id (PK), job_id (FK), tag_id (FK)
Tags	Tags/Labels for Jobs	id (PK)
Applicants	Stores applicants' information for hiring	id (PK)
Interviews	Schedule & details of applicant interviews	id (PK), applicant_id (FK)
New_Hired	Stores details of successfully hired applicants	id (PK)
Task	Task management for users	id (PK), user_id (FK)
Completed_Task	Stores completed task info	id (PK), task_id (FK)
Failed_Task	Stores failed task info	id (PK), task_id (FK)

Resignations	Resignation requests from users	id (PK), user_id (FK)
Post	User posts (for social/news feed feature)	id (PK), user_id (FK)
Comments	Comments on posts	id (PK), user_id (FK), post_id (FK)
Announcement	Public announcements for all users	id (PK)
Notification	User-specific notifications/messages	id (PK)

Table 11. Database Table Documentation

Summary of Relationships:

Relationship Type	From	To
One-to-Many	User → Jobs	user_id FK in Jobs
One-to-Many	User → Task	user_id FK in Task
One-to-Many	User → Resignations	user_id FK in Resignations
One-to-Many	User → Post	user_id FK in Post
One-to-Many	Post → Comments	post_id FK in Comments
One-to-Many	User → Comments	user_id FK in Comments
One-to-Many	Task → Completed_Task	task_id FK in Completed_Task
One-to-Many	Task → Failed_Task	task_id FK in Failed_Task
One-to-Many	Applicants → Interviews	applicant_id FK in Interviews
One-to-Many	Department → User	department_id FK in User

Table 12. Relationships Summary Table

Entity-relationship diagrams (ERDs)

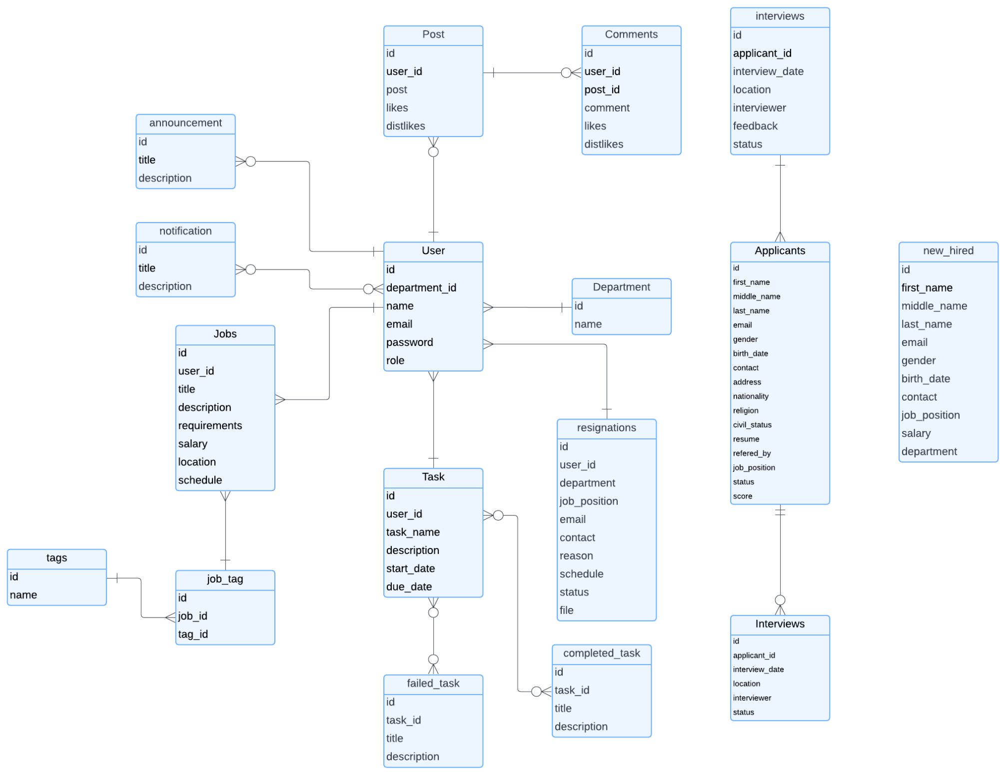


Figure 22. Entity-relationship Diagrams

Data modeling and normalization

The data visualization for Merchandising Management System Human Resource 1, an Entity-Relationship Diagram (ERD) was created to represent all necessary tables, their attributes, and the relationships between them. This modeling helps ensure the database is efficient, organized, and free from data redundancy.

Database Schema Design

Entities and Attributes

a. **User Table**

Field Name	Data Type	Description
id	INT	Primary Key
department_id	INT	Foreign Key from Department table
name	VARCHAR	Full Name
email	VARCHAR	Unique Email
password	VARCHAR	Password
role	VARCHAR	User Role (Admin, HR, Staff etc.)

Table 13.1. User Table

b. **Department Table**

Task	Output
Laravel Migration	PHP Migration Files
SQL Scripts	Create Table SQL
CRUD Functions	Laravel Controller Code

Table 13.2. Department Table

c. **Applicants Table**

Field Name	Data Type	Description
id	INT	Primary Key
first_name	VARCHAR	First Name
middle_name	VARCHAR	Middle Name
last_name	VARCHAR	Last Name
email	VARCHAR	Email
gender	VARCHAR	Gender
birth_date	DATE	Birth Date
contact	VARCHAR	Contact No.
address	TEXT	Address
nationality	VARCHAR	Nationality
religion	VARCHAR	Religion
civil_status	VARCHAR	Civil Status
resume	TEXT	Resume File
referred_by	VARCHAR	Referred By
job_position	VARCHAR	Applied Position
status	VARCHAR	Application Status

score	INT	Exam Score

Table 13.3. Applicant Table

d. Jobs Table

Field Name	Data Type	Description
id	INT	Primary Key
user_id	INT	Foreign Key (User)
title	VARCHAR	Job Title
description	TEXT	Job Description
requirements	TEXT	Job Requirements
salary	DECIMAL	Offered Salary
location	VARCHAR	Location

Table 13.4. Jobs Table

e. Task Table

Field Name	Data Type	Description
id	INT	Primary Key
user_id	INT	Assigned User ID
task_name	VARCHAR	Task Name
description	TEXT	Task Description
start_date	DATE	Start Date
due_date	DATE	Deadline Date

Table 13.5. Task Table

f. Interview Table

Field Name	Data Type	Description
id	INT	Primary Key
applicant_id	INT	Foreign Key (Applicant)
interview_date	DATE	Interview Date
location	VARCHAR	Interview Location
interviewer	VARCHAR	Interviewer Name
feedback	TEXT	Feedback from Interview

status	VARCHAR	Interview Result

Table 13.6. Inteview Table

g. Resignation Table

Field Name	Data Type	Description
id	INT	Primary Key
user_id	INT	Foreign Key (User)
department	VARCHAR	Department Name
job_position	VARCHAR	Job Position
email	VARCHAR	User Email
contact	VARCHAR	User Contact
reason	TEXT	Reason for Resigning

Table 13.7. Resignation Table

h. Completed Task Table

Field Name	Data Type	Description
id	INT	Primary Key (Auto Increment)
task_id	INT	Foreign Key (Reference Task ID)
title	VARCHAR	Title of Completed Task
description	TEXT	Task Description

Table 13.8. Completed Task Table

i. Failed Task Table

Field Name	Data Type	Description
id	INT	Primary Key (Auto Increment)
task_id	INT	Foreign Key (Reference Task ID)
title	VARCHAR	Title of Failed Task
description	TEXT	Task Description

Table 13.9. Failed Task Table

j. Announcements Table

Field Name	Data Type	Description

id	INT	Primary Key (Auto Increment)
title	VARCHAR	Announcement Title
description	TEXT	Announcement Content

Table 13.10. Announcements Table

k. Notifications Table

Field Name	Data Type	Description
id	INT	Primary Key (Auto Increment)
title	VARCHAR	Notification Title
description	TEXT	Notification Content

Table 13.11. Notifications Table

l. Tags Table

Field Name	Data Type	Description
id	INT	Primary Key (Auto Increment)
name	VARCHAR	Tag Name

Table 13.12. Tags Table

Normalization Table

The Merchandising Management in Human Resource 1 system was created to help manage human resource activities such as posting jobs, tracking applicants, handling interviews, assigning tasks, and managing employee resignations. To make sure the data is clean, organized, and easy to manage, normalization was used during the database design process.

The table shown checks each table in the system to see if it follows the rules of First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). All tables passed 1NF and 2NF, meaning that the data is in simple, atomic format and that all fields depend on the table's main key.

However, some tables—like Applicants, new_hired, and resignations—did not pass 3NF. These tables include extra or repeated information such as job_position, department, or email, which can be pulled from other tables instead. This can lead to problems like duplicated data or inconsistencies.

The rest of the tables, including User, Jobs, Tasks, Post, Comments, and others, are already normalized up to 3NF. These are well-organized, without extra data, and follow proper relationships between tables. This helps the system run better and makes the database easie

Table	1NF	2NF	3NF	Notes
applicants	✓	✓	✗	Transitive dependency: job_position → department, salary
new_hired	✓	✓	✗	Repeats job info (same issue as applicants)
resignations	✓	✓	✗	Redundant info from user_id (e.g., email, department)
user	✓	✓	✓	Fully normalized
jobs	✓	✓	✓	Fully normalized
task	✓	✓	✓	No transitive dependencies
post	✓	✓	✓	No transitive dependencies
comments	✓	✓	✓	FKs in place, atomic fields
interviews	✓	✓	✓	Relational, no dependency issues
department	✓	✓	✓	Atomic, independent table
announcements	✓	✓	✓	Simple structure
notifications	✓	✓	✓	Clean one-to-many with users

completed_task	✓	✓	✓	Dependent only on task_id
failed_task	✓	✓	✓	Same as above
tags	✓	✓	✓	Lookup table, atomic
job_tag	✓	✓	✓	Pivot table (many-to-many)

Table 14. Normalization Table

CRUD (Create, Read, Update, Delete) operations

The following code snippets represent the CRUD operations of the system, listed below:

1. task-creation.blade.php

```
new class extends Component {

public $department;
public $title;
public $date_start;
public $deadline;
public $description;
public function submit()
{
   $users = User::where('department', $this->department)->get();
      $validated = $this->validate([
      'title' => ['required'],
      'description' => ['required'],
      'department' => ['required'],
      'date_start' => ['required'],
      'deadline' => ['required'],
   ]);
```

```
    $tasks = $users->map(function ($user) use ($validated) {
        return [
            'user_id' => $user->id,
            'title' => $validated['title'],
            'description' => $validated['description'],
            'date_start' => $validated['date_start'],
            'deadline' => $validated['deadline'],
        ];
    });

    Task::insert($tasks->toArray());
    $this->redirect('/hr-task');
}

}; ?>
```

2. task-preview.blade.php

```
new class extends Component {
    public function tasks()
    {
        return Auth::user()->tasks->take(5);
    }
    public function with()
    {
        return [
            'tasks' => $this->tasks(),
```

```
        ];
    }
}; ?>
```

3. task-submit.blade.php

```
new class extends Component { use WithFileUploads;
public $task;
public $file;
public $date_completed;

public function submit()
{
    $this->validate([
        'date_completed' => ['required'],
        'file' => ['required', File::types(['pdf','png','jpeg','jpg'])],
    ]);
    $file = $this->file->store('tasks', 'public');
    $filePath = url('/') . '/' . 'storage' . '/' . $file;

    UserTask::create([
        'user_id' => $this->task->user_id,
        'name' => Auth::user()->name,
        'title' => $this->task->title,
```

```
            'task_description' => $this->task->description,
            'date_completed' => $this->date_completed,
            'file' => $filePath,
        ]);
        $this->task->delete();
        $this->redirect('/task-list');
    }
}; ?>
```

4. create-job.blade.php

```
new class extends Component {

    public $title;
    public $requirements;
    public $salary;
    public $location;
    public $schedule;
    public $description;
    public $tags;
    public $department;

    public function submit()
    {
        $data = $this->validate([
            'title' => ['required', 'string'],
            'description' => ['required','string', 'max:255'],
            'requirements' => ['required', 'string', 'max:255'],
```

```
            'salary' => ['required', 'string'],
            'location' => ['required', 'string'],
            'schedule' => ['required'],
            'tags' => ['required'],
            'department' => ['required'],
        ]);

        $this->authorize('create', Job::class);
        $job = Auth::user()->jobs()->create(Arr::except($data, 'tags'));
        // $job = Job::create(Arr::except($data, 'tags'));

        if(!empty($data['tags']))
        {
            foreach (explode(',', strtolower($data['tags'])) as $requirement) {
                $job->tag(trim($requirement));
            }
        }
        // $this->reset();
        $this->dispatch('success-notif');
    }
}; ?>
```

5. job-list.blade.php

```
new class extends Component {
     public $q;
    public $alphabetical;
    public function getJobs()
    {
        return Job::query()
```

```
            ->when($this->q, fn($query) => $query->where('title', 'LIKE', '%' .
$this->q . '%'))
            ->when($this->alphabetical, fn($query) => $query->orderBy('title' ,
'asc'))
            ->latest()
            ->paginate(8);
    }
    public function publish(Job $job)
 {
        // dd($job);
        $job->update([
            'status' => 'published'
        ]);
    }
    public function unpublish(Job $job)
    {
        // dd($job);
        $job->update([
            'status' => 'private'
        ]);
    }

    public function delete(Job $job)
    {
        $job->delete();
        $this->dispatch('delete-notif');

    }
```

```
    public function with(): array
    {
        return [
            'jobs' => $this->getJobs(),
        ];
    }
}; ?>
```

6. job-edit.blade.php

```
class extends Component {

    public $id;
    public $title;
    public $salary;
    public $location;
    public $schedule;
    public $description;
    public $requirements;
    public function mount(Job $job) {

        $this->id = $job->id;
        $this->title = $job->title;
        $this->salary = $job->salary;
        $this->location = $job->location;
        $this->schedule = $job->schedule;
        $this->description = $job->description;
        $this->requirements = $job->requirements;
    }
```

```
    public function saveJob()
    {
        $validated = $this->validate([
            'title' => ['required'],
            'salary' => ['required'],
            'location' => ['required'],
            'schedule' => ['required'],
            'description' => ['required'],
            'requirements' => ['required'],
        ]);
        $this->authorize('create', Job::class);
        $job = Job::findOrFail($this->id);

        $job->update($validated);
        $this->dispatch('update-notif');
        sleep(1);
        $this->redirect('/create-job');
    }
}; ?>
```

6.5. Network Configuration

Network topology and configuration details

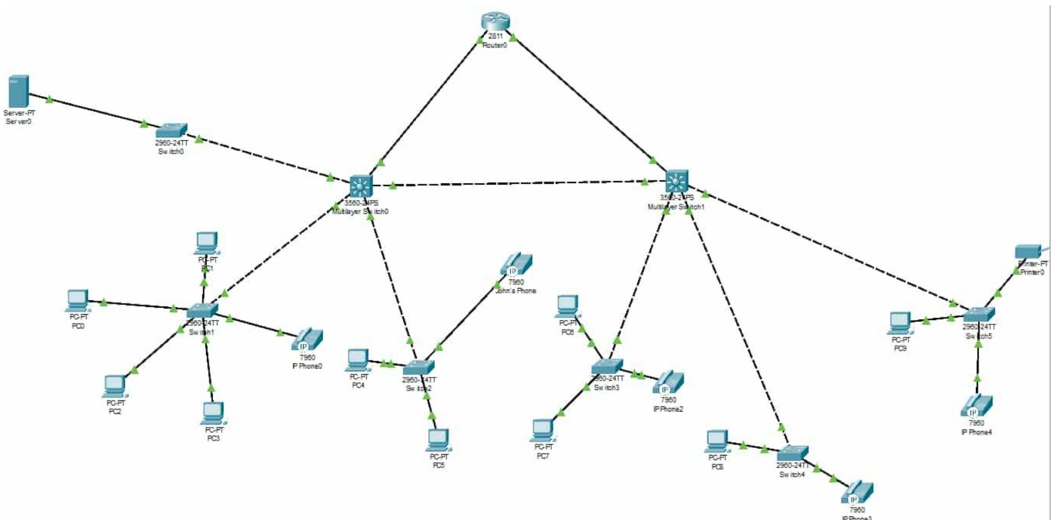


Figure 13. Network Topology and Configuration (pp.123)

The star topology provides significant advantages, such as high reliability, as the failure of single devices does not affect the whole network, scalability for straightforward expansion, improved security through VLAN segmentation and firewalls, enhanced performance with load balancing using LACP, and easier troubleshooting thanks to its organized structure.

This effectively structured star topology guarantees reliable communication, security, and high availability, rendering it a strong network for the organization.

Network Configuration

1. VLAN Configuration

Devices Involved: All switches (Core and Access)

Purpose: Segregate network traffic for different departments or device types.

```
vlan 10      name DATA10
vlan 20      name DATA20
vlan 30      name DATA30
vlan 40      name VOICE
vlan 50      name PRINTER
vlan 60      name SERVER
```

2. Access Port Configuration

a. Example for PC connected to VLAN 10:

```
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 10
```

b. For IP Phones (with Voice VLAN):

```
interface FastEthernet0/2
 switchport mode access
 switchport voice vlan 40
 switchport access vlan 10
```

3. Trunk Port Configuration (Core to Access Switches)

```
interface GigabitEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60
```

Repeat for all trunk links between Core and Access switches.

4. Router-on-a-Stick (Inter-VLAN Routing)

Example on Router interface GigabitEthernet0/0:

interface GigabitEthernet0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

---

interface GigabitEthernet0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

---

interface GigabitEthernet0/0.30

encapsulation dot1Q 30

ip address 192.168.30.1 255.255.255.0

```
interface GigabitEthernet0/0.40

encapsulation dot1Q 40

ip address 192.168.40.1 255.255.255.0
```

```
interface GigabitEthernet0/0.50

encapsulation dot1Q 50

ip address 192.168.50.1 255.255.255.0
```

```
interface GigabitEthernet0/0.60

encapsulation dot1Q 60

ip address 192.168.60.1  255.255.255.0
```

5. Server Port Configuration (Example)

```
interface FastEthernet0/24

 switchport mode access

switchport access vlan 60
```

6. Spanning Tree Protocol (STP) Overview

Purpose: Prevent loops in the Layer 2 network by electing a Root Bridge and defining port roles.

Root Port – Best path toward the root bridge.

Designated Port – Forwarding port on the segment.

Blocked Port – Backup path; placed in blocking state to prevent loops.

Use the following command to verify:

show spanning-tree

7. Verification Commands

These commands are used to verify the configuration and ensure proper VLAN and trunk setups:

show vlan brief

show interfaces trunk

show running-config

show ip interface brief

show spanning-tree

8. VLAN Assignment Table

Device	Interface	VLAN Assigned	Purpose
Switch0	Fa0/1	VLAN 10	PC Connection
Switch1	Fa0/2	VLAN 20	IP Phone
Switch2	Fa0/3	VLAN 30	Printer
Switch3	Fa0/4	VLAN 40	Voice

Switch4	Fa0/5	VLAN 50	Printer
Core Switch	Gi0/1	Trunk	To Router
Core Switch	Gi0/2	Trunk	To Access Switch

Table 15. VLAN Assignment Table

Note:

Always ensure that both ends of trunk links are configured identically and allowed VLANs match. Also verify that spanning-tree roles are healthy and not blocking necessary ports.

Security measures (e.g., firewalls, encryption)

1. Access Control
2. Data Encryption
3. API Security
4. Data Backup & Recovery
5. Data Privacy Compliance
6. HTTPs Deployment
7. Security Awareness & Training

Protocols and communication methods used

The HR1 system follows a microservices architecture where services communicate over secure HTTP (HTTPS) protocols using RESTful APIs. JSON is used as the primary data interchange format to maintain lightweight and structured communication. For internal service-to-service communication, message brokers such as RabbitMQ or Kafka can be integrated (if applicable) to ensure asynchronous communication and improve system responsiveness. Authentication is handled via OAuth2 tokens and session-based validation for secure API access.

Load balancing and failover mechanisms

The Merchandising Management System HR1 employs load balancing across multiple cloud instances to evenly distribute incoming traffic and ensure high availability. Failover mechanisms are in place to automatically redirect traffic to standby servers in the event of a system failure, minimizing downtime and ensuring continuous service delivery.

6.6. Deployment and Infrastructure

Deployment strategies (e.g., cloud, on-premises)

The Merchandising Management System - HR1 is deployed using a cloud-based infrastructure to ensure scalability, accessibility, and efficient system maintenance. The deployment strategy leverages GitHub as the version control system, allowing continuous integration and deployment through automated pipelines. Cloud hosting provides flexibility in resource allocation, supporting increased user demand while maintaining system stability.

Hardware specifications and server configurations

The system runs on a cloud-hosted server configured with a minimum of 4 vCPUs, 8GB RAM, and 250GB SSD storage, ensuring optimized performance for HR processes. The network bandwidth is set at 100 Mbps to facilitate smooth data exchange between microservices and external systems. The operating system is Ubuntu 22.04 LTS, providing stability and security for backend operations. Database management is handled by MySQL, with efficient indexing and query optimization to enhance data retrieval speeds.

Configuration management (e.g., scripts, automation tools)

Configuration management is automated using GitHub Actions, which streamlines code deployment and system updates through continuous integration and deployment (CI/CD) pipelines. Environment variables are managed using `.env` files, securely storing sensitive information such as database credentials and API keys. Server configurations and system maintenance tasks, including database migrations and log management, are automated using shell scripts to reduce manual intervention. This structured deployment approach ensures system reliability, simplifies updates, and maintains operational efficiency across all HR modules.

Scalability and resource optimization

The Merchandising Management System: Human Resource 1 is built for scalability with no loss of performance. It may accommodate increasing numbers of users, job applicants, and other human resources records. It has a modular design enabling individual new modules to be added, or current modules to be incorporated into new human resources processes such as benefit management or payroll, with minimal costly disruption to ongoing processes. The use of cloud-ready technologies will allow for a vertical or horizontal scaling of the system resource use based on demand, maximizing infrastructure usage, and limiting operational costs. Additionally, the system streamlines resource-hungry tasks like AI resume evaluation and document processing through asynchronous processing and queue-based task handling, thereby lowering server load, and providing an efficient user experience for all modules while the system is able to dynamically allocate resources to the tasks processing these processes.

6.7. Security Measures

Ensuring the confidentiality, integrity, and availability of data is a fundamental aspect of the HR1 Management System. To protect sensitive information, prevent unauthorized access, and maintain system reliability, several security protocols have been implemented.

Authentication and authorization mechanisms:

Role-Based Access Control (RBAC) is implemented:

1. HR personnel can access job posting, applicant tracking, onboarding, and offboarding features.
2. Employees can access their portal with features like task tracking and offboarding.
3. Applicants can apply for job openings and track application status.
4. User Authentication is required for system access via role-based credentials.
5. Multi-factor Authentication (MFA) is available specifically for HR personnel with admin access

Encryption and data protection

1. The system uses a centralized MySQL database, and UUIDs are used to ensure safe and unique data handling, especially for the job posting section.
2. Network topology incorporates security measures:
3. Router with built-in firewalls
4. Data redundancy via local backup and network storage
5. Domain hosting with support for load balancing and anti-DDoS protection is mentioned, which adds an additional layer of security and reliability

Incident response and mitigation procedures

1. The architecture and modular design (e.g., microservices) indirectly support fault isolation, which is helpful for minimizing the impact of incidents.
2. The CI/CD pipeline and DevOps approach imply regular testing and automated deployment, which aids in quickly deploying patches and updates if vulnerabilities are found

6.8. Testing and Quality Assurance

Detailed test cases and scenarios

Scenario Description	Steps	Actor	Expected Outcome	Status
User Login	1. Navigate to login 2. Enter User Credentials 3. Click login	HR / EMPLOYEE	**Valid Credentials:** HR is directed to HR system Employee is Directed to Employee portal **Invalid Credentials Error message:** Email or password do not match! / The email /password field is required / The email field must be a valid email address.	**Passed**
Post a Jobs	1. Navigate to "Post a job" 2. Check for job listed 3. Publish / unpublish job or delete	HR	**Publish:** Job is posted on job posting **Unpublish:** The posted job is removed from job posting **Delete:** Job is deleted from list and job posting	**Passed**

Job Posting	1. Applicant Search for jobs 2. Click jobs 3. Click Apply now 4. Fill-up Application Form 5. Check Terms and Condition 6. Submit Application	Applicant	**Success Message:** Application Submitted, Goodluck! :D **Error Message:** Specific Field is Required to be filled / Terms and Condition Required / Application Failed	**Passed**
AI Resume Evaluation	1. Appliant Submit PDF CV/RESUME 2. CV / RESUME is evaluated 3. Evaluation Feedback is Noted	Google Gemeni AI	**All applicants:** is listed on All appliccants **Passed CV/RESUME:** Applicant is listed on Priority Applicants **AI Feedback:** The Feedback message evaluated CV/ RESUME can be seen	**Passed**
Application Emailing	1. Applicant Submit Application Form 2. Email applicants Gmail Account after Submission	System	**Success:** Email sent to applicants Gmail Account **Failed:** No Email Sent, notify HR Gmail for Failed response	**Passed**
Add Applicants	1. Click Add applicants 2. HR Fillup add form	HR	**Success Message:** Applicant Added **Error Message:** Input Filed is Required	**Passed**

Interview Scheduling	1. View Applicant Profile 2. Click “Create Schedule” 3. Fill up Modal form for Interview Schedule	HR	**Success Message:** Interview Schedule Success Listed in Scheduled interview **Error Message:** Input Field is Required	**Passed**
Interview Emailing	1. Schedule Submit 2. Sent email of interview schedule to applicants gmail after submission	System	**Success:** Interview Schedule sent to applicants Gmail Account **Failed:** No interview schedule Email Sent, notify HR Gmail for Failed response	**Passed**
Interview Status Update	1. Click Applicant Scheduled 2. Status update of scheduled interview applicants 3. Choose Remarks 4. Fill a Feedback 5. Submit	HR	**Success Message:** Status Updated **Error Message:** Remark and Feedback Field is required	**Passed**
Interview Status update Emailing	1. Click “Interview Status” 2. Click “Hire” or “ Reject”	System	**Success:** Sent Email Hired or Rejected Message in Applicants Gmail Account **Failed:** No Hired or Failed Email Sent, notify HR Gmail for Failed response	**Passed**

			Hired Applicants: Hired is Listed to newhired **Rejected:** Rejected applicants is listed on failed table	
Task creation	1. Click “Add Task” 2. Fill-up Form 3. Submit Task	HR	**Success Message:** Task created Successfully Posted in employee portal and listed in Task List **Error Message:** Input Filed is Required / Task Creation Failed	**Passed**
Post Announcement	1. Click “Create announcement” 2. Fill-up Announcement Form 3. Submit	HR	**Success Message:** Announcement is Posted Announcement is posted on employee portal and listed in Announcement List **Error Message:** Failed to Post an announcement	**Passed**
Employee Task Submission	1. Click “Task List” 2. Click “Task” 3. Submit File or Image 4. Input Date submission 5. Click “ Mark as Done”	EMPLOYEE	**Success Message:** Task Submitted The submission of Task is listed in HR system of Recent Activities **Error Message:** Date submission Required, Proof file/ image is Required	**Passed**
Resignation form	1. Click “Profile” 2. Click “Resignation Form”	EMPLOYEE	**Success Message:** Resignation Submitted	**Passed**

	3. Fill-up Form 4. Submit Resignation		The resignation request is listed on Offboarding table **Error Message:** Input Field is Required	
Schedule Exit interview	1. Click " Resignee More Details" 2. Fill-up Exit interview Schedule	HR	**Success Message:** Exit interview scheduled successfully The scheduled interview is listed in inline-Exit interview **Error Message:** Schedule Field is Required	**Passed**
Exit Interview email	1. Submit Schedule 2. Email Exit interview to Employees gmail account	System	**Success:** Sent Email Exit interview schedule Message in employees Gmail Account **Failed:** No Exit interview Email Sent, notify HR Gmail for Failed response	**Passed**
Resignation Approval	1. Click "Scheduled exit interview" 2. Approve Resignation	HR	**Success Message:** Resignation approved **Error Message:** Resignation status approval failed	**Passed**
API Notifications / List Passing	1. New Job request Notifications 2. Newhired employee list passed to admin and training	SYSTEM	**Output:** Notify for new job request, Newhired list is passed to admin and training management, Account termination notifications to admin	**Passed**

	management 3. Account Termination notifications to admin			
Concern Wall Posting	1. Open Employee Portal 2. Publish a post 3. Fill-up form 4. Submit Concern 5. Comment on posts 6. React like and dislike concern posts	EMPLOYEE/HR	**Success Message:** Concern is posted Successfully **Error Message:** Post Something is Required **Reacts:** Like and Dislike is adds up **Comments:** Comments is listed inside a post	**Passed**
Change Password	1. Click "View Profile" 2. Change Password 3. Fill-up New Password 4. Submit	EMPLOYEE/HR	**Success Message:** Password is updated Error Message: Password is already taken, Password is invalid, Confirm-password is required, Confirm password not match	**Passed**
Logout	1. Click "Profile picture" 2. Click "Logout"	EMPLOYEE/HR	**Success:** Directed to logout **Failed:** Logout failed please try again	**Passed**

Table 16. Detailed Test Cases and Scenarios Table

Test data and expected outcomes

User Login

Field	Test Data (Example)	Expected Outcome
Email	**Valid:** catherinesaminan9@gmail.com	**Valid Credentials**: Redirected to dashboard **Invalid Credentials:** Email or password do not match! / The email /password field is required / The email field
	Invalid: Null / adad.com	
Password	**Valid:** #saminanGWA	
	Invalid: NULL / admin' --	

Table 17.1. User Login – Test DataTable

Request Job Post

Field	Test Data (Example)	Expected Outcome
Job title	**Valid:** Sales	**Valid Credentials :** "Job posted successfully!" and listed to "Post a Job List" **Invalid Credentials / Null:**
	Invalid: Null	
Salary	**Valid:**500	
	Invalid: Null / Characters /Numbers	
Schedule	**Valid:** Options WFH, Full-time, Part-time	
	Invalid: Null / Select option	
Location	**Valid:** Options Quezon City, Pasig/ Null	

	Invalid: Null / Select option	Please Fill-up the input field, Invalid input, Select Options.
Tags	**Valid:** #Salestalk etc.	
	Invalid: Null	
Department	**Valid:** Options Sales etc.	
	Invalid: Null / Select option	
Description	**Valid:** Words for something	
	Invalid: Null	
Requirements	**Valid:** Skills etc.	
	Invalid: Null	

Table 17.2. Request Job Post – Test DataTable

Application Form

Field	Test Data (Example)	Expected Outcome
First Name	**Valid:** Julius	**Valid Credentials:** **-** Analyzing Load before submission - Submitted to HR Applicant Lists - Applicant receive an email of Application Received - CV/RESUME Analyzed by the AI - Application Succesfull Notification **Invalid Credentials:** - Application failed notification
	Invalid: Null	
Middle Name (optional)	**Valid:** B	
	Invalid: ASD. / More than 2 letters	
Last Name	**Valid:** Dela Torre	
	Invalid: Null	
Address:	**Valid:** #3 Cassanova Drive, Culiat Q.C.	
	Invalid: Null	
Email:	**Valid:** delatorrejuliuservin9@gmail.com	
	Invalid: Null / no @ / Ex.delatorrejuliuservin.com	
Nationality	**Valid:** Filipino, American, Chinese etc.	
	Invalid: Null	
Religion	**Valid:** Catholic, Muslim, Born Again	

	Invalid: Null / Not Selected	- File Must be PDF - Select an Option - Specific Input field is required (Firstname, Lastname Birth Date, Etc.)
Sex	**Valid:** Selected – Male, Female	
	Invalid: Null / Not Selected	
Birthday	**Valid:** 20/08/2003	
	Invalid: Null / Not Selected	
Contact	**Valid:** 09459788091	
	Invalid: Null / Characters / Invalid format 12345678901	
Civil Status	**Valid: Selected –** Single, Married, Divorced	
	Invalid: Null / Not Selected	
Resume:	**Valid:** CV pdf., Resume pdf. 5gb file - size	
	Invalid: Null / Non-PDF/ Big size	
Referred by (Optional)	**Valid:** Camo	
	Invalid: 1231124	
Terms and Condition	**Valid:** Checked / Accepted	
	Invalid: Null / Not Accepted or Declined	

Table 17.3. Application Form – Test DataTable

Interview Schedule

Field	Test Data (Example)	Expected Outcome
	Valid: 17/04/2025	

<table>
<tr><td>Appointment Date</td><td>Invalid: Null / Not Selected / ABSCD...</td><td rowspan="5">Valid Credentials: Interview Schedule Successful.

Email of Interview Schedule is Sent to applicants Gmail

Invalid Credentials: Schedule Failed to Create

Specific Input field is required (Date, Interviewer, Location)

No Email Sent to applicants Gmail

Calendar of Interview is Highlighted</td></tr>
<tr><td rowspan="2">Interviewer</td><td>Valid: Camo</td></tr>
<tr><td>Invalid: Null / 123131</td></tr>
<tr><td rowspan="2">Location</td><td>Valid: Pasig, Quezon City</td></tr>
<tr><td>Invalid: Null / Not Selected / 123131</td></tr>
</table>

Table 17.4. Interview Schedule – Test DataTable

Add Task

<table>
<tr><th>Field</th><th>Test Data (Example)</th><th>Expected Outcome</th></tr>
<tr><td rowspan="2">Department</td><td>Valid: Sales Dept</td><td rowspan="5">Valid Data: Task Successfully Posted

Created Task is listed and Posted on Employee Portal Announcement</td></tr>
<tr><td>Invalid: Null / Not Selected / Numbers</td></tr>
<tr><td rowspan="2">Title</td><td>Valid: Send SSS Number</td></tr>
<tr><td>Invalid: Null / 12312313131</td></tr>
<tr><td rowspan="2">Start Date</td><td>Valid:18/04/2025</td></tr>
<tr><td>Invalid: Null / Not Selected / ABCD...</td><td rowspan="2">Invalid Data: Taks Failed to Post</td></tr>
<tr><td>Deadline</td><td>Valid: 17/04/2025</td></tr>
</table>

	Invalid: Null / Not Selected / ABCD...	No Task is Listed and Posted Error Message Shows
Task Description	**Valid:** Any words that can describe a Task	
	Invalid: Null / Numbers / No commas	

Table 17.5. Add Task – Test DataTable

Post Announcement

Field	Test Data (Example)	Expected Outcome
Title	**Valid:** Urgent to pass	**Valid Credentials:** Announcement already posted in employee portal and listed to announcement **Invalid Credentials:** Not announcement was listed and no posted
	Invalid: Null	
Announcement Details	**Valid:** Something like announce of nothing	
	Invalid: Null	

Table 17.6. Post Announcement – Test DataTable

Document Task Submission

Field	Test Data (Example)	Expected Outcome
Date Completed	**Valid:** 18/04/2025	**Valid Credentials:** Document Uploads passed to Document Management and Notify in Recent Activities, Listed and Checked as complete Task **Invalid Credentials:** No Submitted, Pending.
	Invalid: Null / ABCD...	
Proof upload	**Valid:** IMG / PDF File	
	Invalid: Null / txt	

Table 17.7. Document Task Submission – Test DataTable

Resignation Form

Field	Test Data (Example)	Expected Outcome
Job Position	**Valid:** Sales	**Valid Credentials:** Resignation Submitted and listed to Offboarding Management
	Invalid: Null / 1231231313	
Email	**Valid:** delatorrejuliuservin8@gmail.com	
	Invalid: Null / no "@"	
Contact Number	**Valid:** 09459788091	
	Invalid: Null / NaN / #Format	
Reason	**Valid:** Some Text any words	**Invalid Credentials:** No Resignation submitted
	Invalid: Null	
Resignation Letter	**Valid: PDF File**	
	Invalid: Null / non Pdf File	

Table 17.8. Resignation Form – Test DataTable

Test results and validation against requirements

User Login

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid login (HR)	catherinesaminan9@gmail.com / #saminanGWA	Redirected to dashboard	Redirected to dashboard	Matches login success flow	Pass

Invalid email format	adad.com / #saminanGWA	Error: Must be valid email	Validation error shown	Proper format validation implemented	Pass
Blank password	catherinesaminan9@gmail.com / (Null)	Error: Password required	Password error shown	Required field validated	Pass

Table 18.1. User Login – Test Result Table

Request Job Post

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Complete form submission	Sales / 500 / WFH / Quezon City / #Salestalk / Sales / Description text / Skills	Job posted successfully	Confirmation message shown, job listed	All required fields validated and accepted	Pass
Missing schedule	Sales / 500 / (Null) / Quezon	Error: Schedule required	Validation error shown	Required field check	Pass

	City / #Salestalk / Sales / Description / Skills			implemented	
Invalid salary	Sales / ABC / WFH / Pasig / #Salestalk / Sales / Description / Skills	Error: Invalid salary input	Salary input rejected	Numeric validation enforced	Pass

Table 18.2. Request Job Post – Test Result Table

Application form

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid application	Julius / B / Dela Torre / Q.C. / delatorrejuliuservin9@gmail.com / Filipino / Catholic / Male / 20/08/2003 / 09459788091 / Single /	Submitted, Email received, AI analysis done	All validations passed and confirmation shown	Meets submission flow, analysis, and email logic	Pass

	PDF Resume / Terms checked				
T&C not checked	All valid except Terms not accepted	Error: Terms required	Form rejected	Required acceptance verified	Pass
Invalid email	Julius / B / Dela Torre / Q.C. / delatorre.com	Error: Must be valid email	Email validation error shown	Fails on format check	Pass

Table 18.3. Application Form – Test Result Table

Interview Schedule

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Complete form	17/04/2025 / Camo / Quezon City	Schedule created, email sent	Interview listed, email sent	Functional flow verified	Pass
Missing interviewer	17/04/2025 / (Null) / Pasig	Error: Interviewer required	Validation error shown	Required field check in place	Pass

Table 18.4. Interview Schedule – Test Result Table

Add Task

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid task	Sales Dept / Send SSS Number / 18/04/2025 / 20/04/2025 / Task description	Task posted successfully	Task appears in portal	Listed in system and displayed	Pass
Missing deadline	Sales Dept / Title / 18/04/2025 / (Null) / Description	Error: Deadline required	Validation error displayed	Date field validation applied	Pass

Table 18.5. Add Task – Test Result Table

Post Announcement

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid task	Sales Dept / Send SSS Number / 18/04/2025 / 20/04/2025 / Task description	Task posted successfully	Task appears in portal	Listed in system and displayed	Pass
Missing deadline	Sales Dept / Title / 18/04/2025 / (Null) / Description	Error: Deadline required	Validation error displayed	Date field validation applied	Pass

Table 18.6. Post Announcement – Test Result Table

Document Submission

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid proof	18/04/2025 / PDF	Uploaded and marked complete	File listed and status updated	Document flow validated	Pass
Missing proof	18/04/2025 / (Null)	Error: File required	Upload rejected	File requirement validation applied	Pass

Table 18.7. Document Submission – Test Result Table

Resignation Form

Test Case	Test Data	Expected Outcome	Actual Result	Validation Against Requirement	Status
Valid resignation	Sales / delatorrejuliuservin8@gmail.com / 094597880	Submitted to Offboarding	Confirmation shown	Submission flow works correctly	Pass

	91 / Reason text / PDF letter				
Invalid email	Sales / delatorrejuliuservin8gmail.com / 09459788091 / Reason / PDF	Error: Invalid email format	Validation triggered	Email format validation enforced	Pass

Table 18.8. Resignation Form – Test Result Table

Performance testing and optimization effort

Test Scenario	Test Objective	Expected Outcome	Actual Result	Optimization Effort	Status
Concurrent User Login	Ensure system can handle 100+ users logging in simultaneously	No timeout, all users authenticated within 2–3 seconds	All users authenticated in < 2.5s	Cached login assets, DB index added to user table	Pass
Job Posting	Validate performance when 50+	No slowdowns or	Job posted in <	Added job table indexing and backend	Pass

Load Test	jobs are posted within a short time frame	database lags	1.5s consistently	queue for inserts	
Resume Upload & AI Evaluation	Test PDF upload and AI feedback response time	AI response in < 5s	Average: 4.3s	Optimized file validation and AI call response handling	Pass
Email Notification Dispatch	Check how many emails system can dispatch within 1 minute	Minimum 40 emails/min	Sent: 52 emails/min	Queued and batched email system using Laravel mail queue	Pass
Application Form Load Time	Measure form load and submit duration under stress	Form loads < 2s, submits < 3s	Load: 1.4s, Submit: 2.1s	Minified JS/Tailwind, optimized backend response	Pass
Interview Schedule Batch Creation	Schedule 20+ interviews in one session	Each scheduled in < 2s	Avg: 1.2s per schedule	DB optimization for scheduling	Pass
Task Submission File	Handle file uploads from 30	All files uploaded	Success: No crashe	Increased max upload size,	Pass

Handling	employees simultaneously	successfully in < 5s window	s or failed uploads	asynchronous file saving	
Concern Wall Reacts and Comments	React and comment stress test (100+ interactions/ min)	All interactions recorded with no lag	No delay, all feedback logged	Used debounce for input and lazy-load reactions	Pass

Table 19. Performance Testing and Optimization Effort

6.9. System Monitoring and Maintenance:

To ensure the system's long-term stability, reliability, and efficiency, proper monitoring and maintenance strategies are essential. This section outlines the tools, methods, and contingency procedures integrated into the system for its smooth and secure operation.

Tools and techniques for monitoring system health

The system employs several monitoring mechanisms to observe the server environment, application processes, and user interactions. Monitoring is performed in real-time to quickly detect anomalies or performance issues.

Logging and error handling mechanisms

1. **Server Performance Tools**: Monitoring of CPU usage, memory load, and disk space is managed through hosting provider dashboards (Indevfinite, Server7)
2. **Application Monitoring**: Laravel's native logging capabilities track backend activity, while optional tools like **Laravel Telescope** provide visibility into requests, database queries, and job queue performance.
3. **Activity Logs**: User actions, including login attempts and key API interactions, are recorded and can be reviewed by administrators.

Scheduled maintenance and updates

Regular maintenance ensures that the system remains secure, up-to-date, and stable. To automate quality control and ensure that updates do not disrupt core functionality, the system utilizes **GitHub CI/CD pipelines** for continuous testing and deployment.

1. **Version Control & CI/CD**: The system's source code is managed via **GitHub**, where pull requests trigger **automated test suites** using **GitHub Actions**. This ensures that all changes pass validation before merging.
2. **Automated Testing**: PHPUnit and feature tests are run during the CI pipeline to verify API responses, database operations, and user flows.

3. **Deployment Process**: After successful test runs, the system is deployed manually or via pipeline to a staging or production environment.
4. **Maintenance Mode**: Laravel's php artisan down command is used during critical updates to prevent user activity and data corruption.

Disaster recovery and backup procedures

To safeguard data and minimize loss during unexpected failures, a robust backup and recovery strategy is implemented.

1. **Automated Backups**: Daily backups of the MySQL database are scheduled using cron jobs or Laravel’s backup packages. Backups are stored securely on remote storage or cloud platforms.
2. **Manual Restore Capability**: Admins can restore the latest backup manually using phpMyAdmin or command-line tools.
3. **Disaster Response Plan**: In case of data loss, a rollback to the most recent stable backup is performed, and affected services are restored step by step.

6.10. APIs and Integration Points

Documentation of APIs used for integration

The Merchandising System in Human Resource Management uses APIs to integrate and exchange data with other systems by utilizing RESTful architecture. These APIs are used not only for internal communication but also for external modules such as:

Training Management System – to fetch data on newly hired employees

Document Management System – to retrieve documents uploaded via the Employee Portal

Administrative System – to fetch new hire lists and receive account termination notifications

The APIs follow the RESTful architecture, with data exchanged in JSON format over HTTP.

API endpoints, methods, and data formats

New Hire Employee API — Shared with Training System and Admin Panel

Endpoint	Method	Description	Used By
/api/employees	GET	Get all employees	Training System
/api/employees/{id}	GET	Get employee details	Training System
/api/employees/{id}	PUT	Update employee info	Admin Panel
/api/employees/{id}	DELETE	Remove employee	Admin Panel

Table 20.1. New Hire Employee API

Task API — Integrated with Document Management System

Endpoint	Method	Description	Used By
/api/tasks	GET	Fetch all employee tasks	Document Management
/api/tasks/{id}	GET	Get specific task	Document Management
/api/tasks	POST (planned)	Submit task with documents	Employee Portal → Document System

Table 20.2. Task API

Notification API — For Account Termination Alerts

Endpoint	Method	Description	Used By
/api/notifications	GET	Get all system notifications	Admin Panel
/api/notifications/{id }	GET	Get a specific notification	Admin Panel
[Planned] /api/notifications	POST	Create/send notification to admin (e.g., termination alert)	Internal — triggered on resignation

Table 20.3. Notification API

Job Post API - External Job Request Integration

This API is designed to **receive job posting requests** from external system **Succession planning System** and submitted to **Post a job List.**

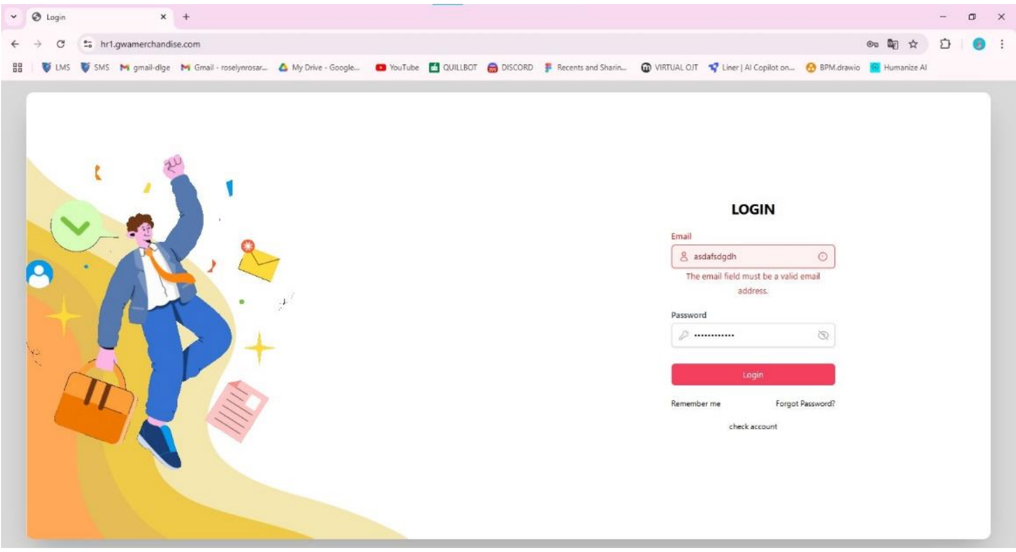
Endpoint	Method	Description	Used By
/api/jobs	**GET**	Fetch all job requests submitted by external system	**Recruitment Management**
/api/jobs	**POST**	Receive a new job request (create a job reuqest post)	**Succession Planning System**
/api/jobs/{id}	**GET**	Get specific job request details	**HR Reviewers**
/api/jobs/{id}	**PUT**	Update the status of the job post (e.g., Publish, Unpublish)	**Recruitment Management**

Table 20.4. Job Post API

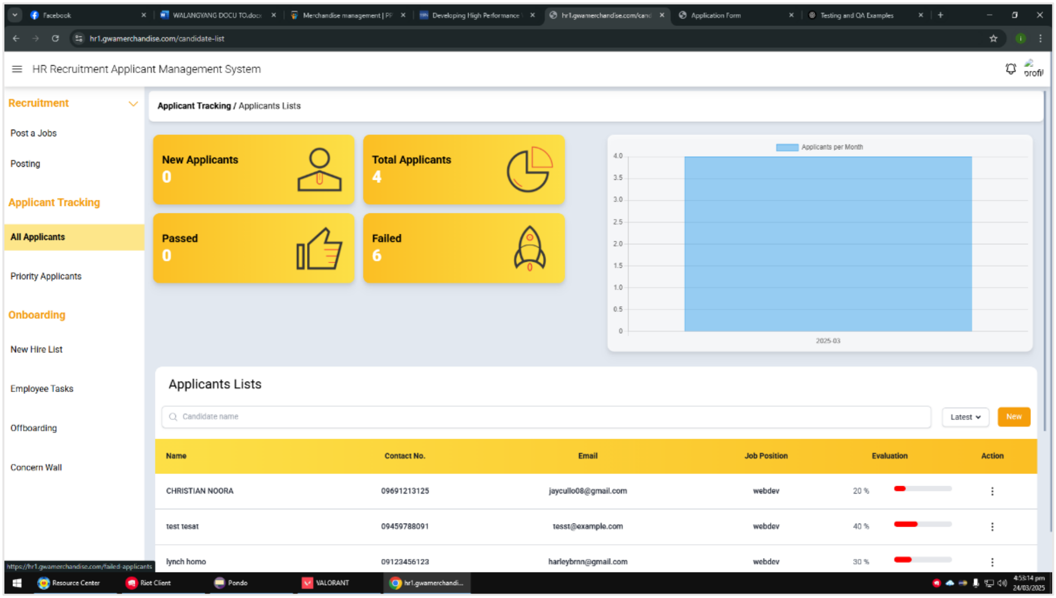
6.11. User Documentation

Instructions for system users

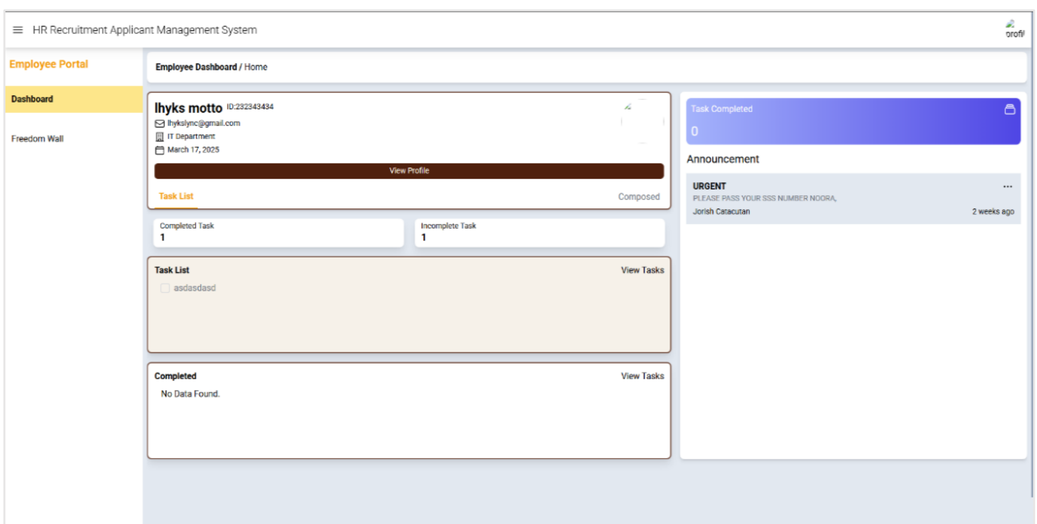
Login: The user must login their credentials. If the entered credential is not valid, the form will show as an error. The accounts that will be used to login the system is provided by the admin.



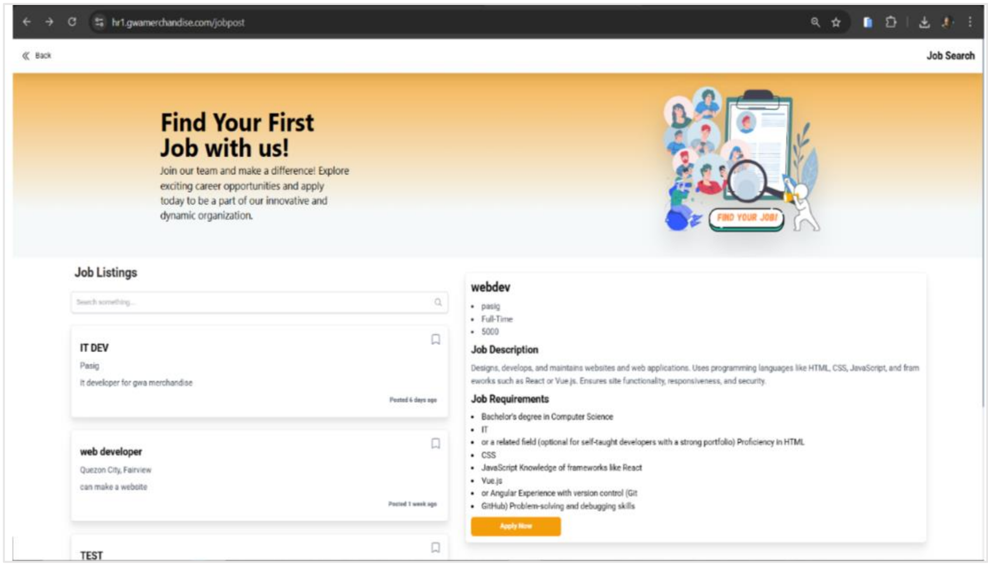
Expected Result: The user will be directed to HR dashboard if the credentials entered in the login is for an HR account**.**



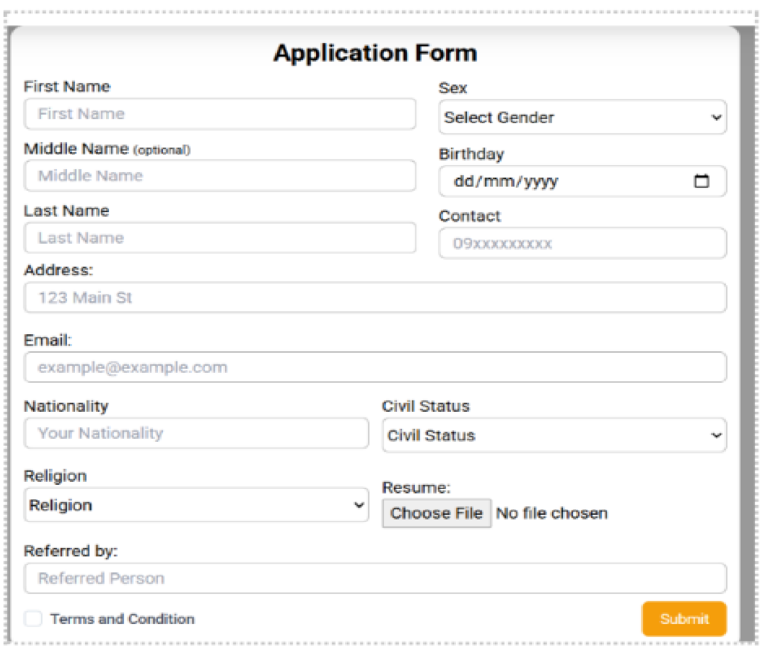
If the user entered a credential for an Employee account, they will be directed to employee portal.



JOB APPLICATION (for Applicants):

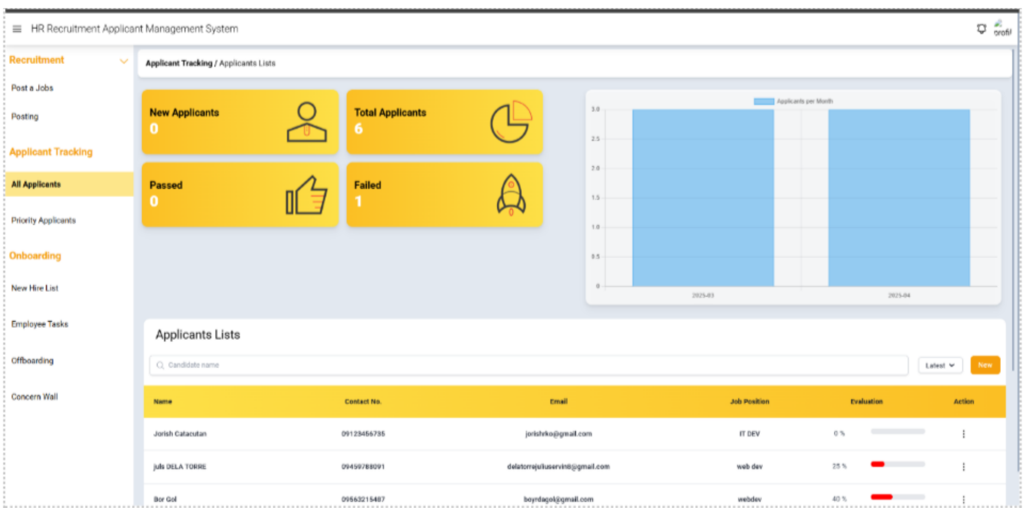


To apply for a job, applicants can browse the available listings and click “Apply Now” on a desired position.



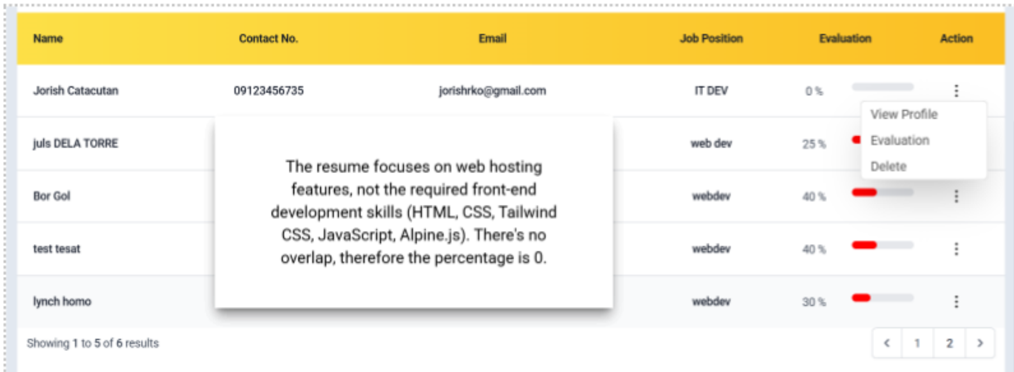
They will be directed to an application form where they provide basic details and upload their resume. After submission, they will receive a confirmation email that their application has been received by HR. The system will also use AI to evaluate their resume based on the job requirements, offering a compatibility score and feedback to help improve their chances of being hired.

APPLICANT TRACKING (for HR):

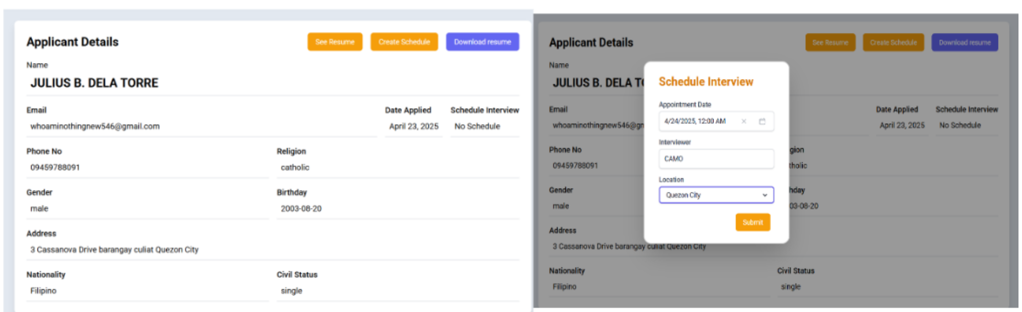


This section of the HR Recruitment Applicant Management System allows HR to track and manage applicants in one interface. After an application is submitted, the applicant's details appear in a list with key info like name, contact, email, job applied for, and evaluation progress. Summary cards and a monthly bar graph provide quick insights into applicant status and trends. HR can search, filter, and take actions on each applicant, the score bar is for score based on the AI evaluation

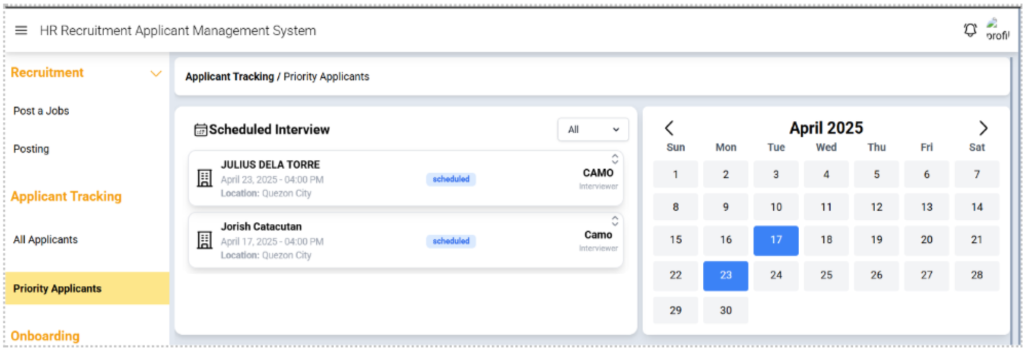
To see the evaluation click the three button



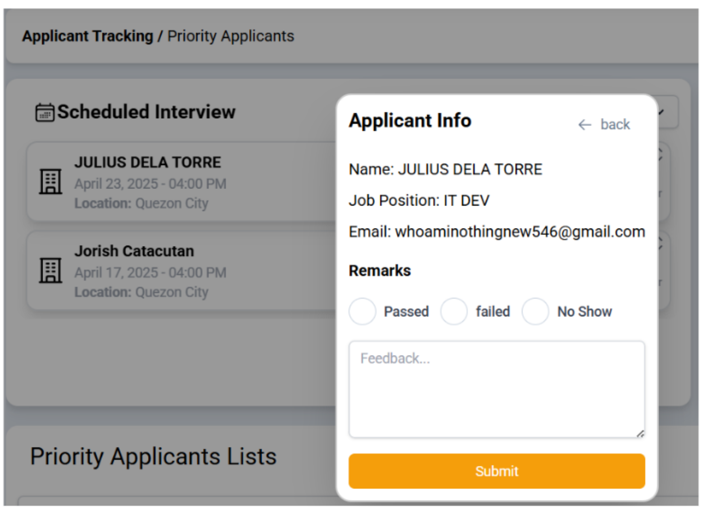
If the applicant is Passed based on AI evaluation the user navigates to Priority applicants, then click the three button to see the applicants profile and create a schedule for an interview



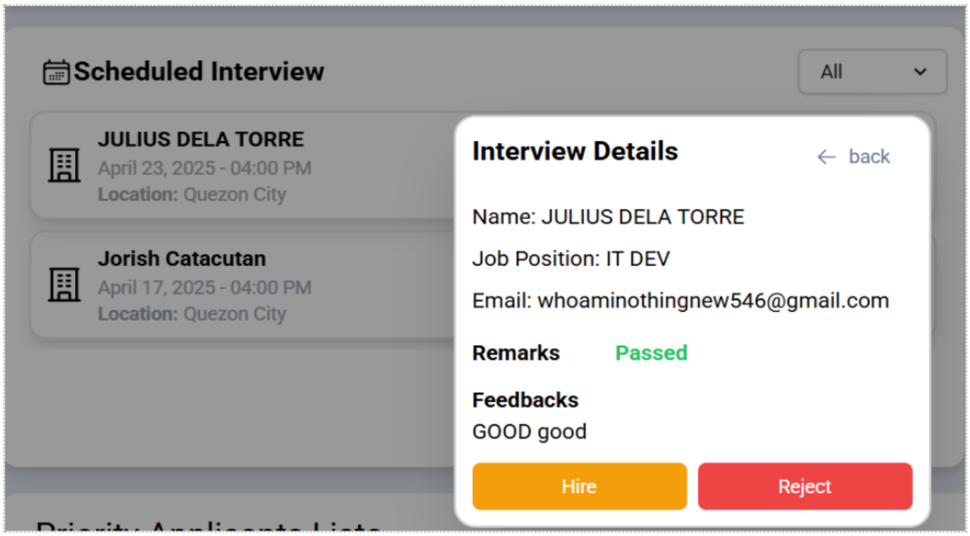
To see the scheduled applicant simply click the priority applicants in sidebar and see the scheduled interview queue. The calendar on the right side highlights specific dates with scheduled interviews, making them easy to notice.



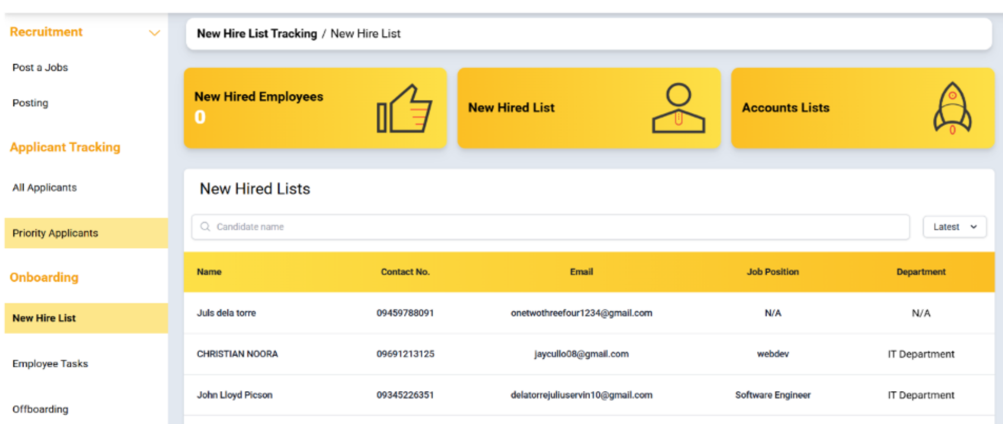
To update the status of the interview simply click the name of the applicant that is scheduled. The HR must click the Remarks and put a feedback before submitting.



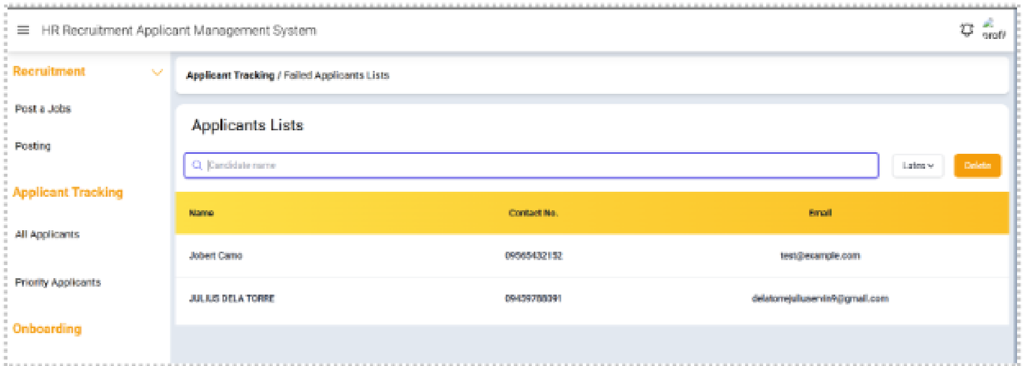
After the submission, the HR should check the interview details by clicking the up and down arrows above the interviewer's name to view the details. If the HR clicks 'Hire,' an email will be sent to the applicant's email, indicating they passed. If rejected, the email will indicate failure.



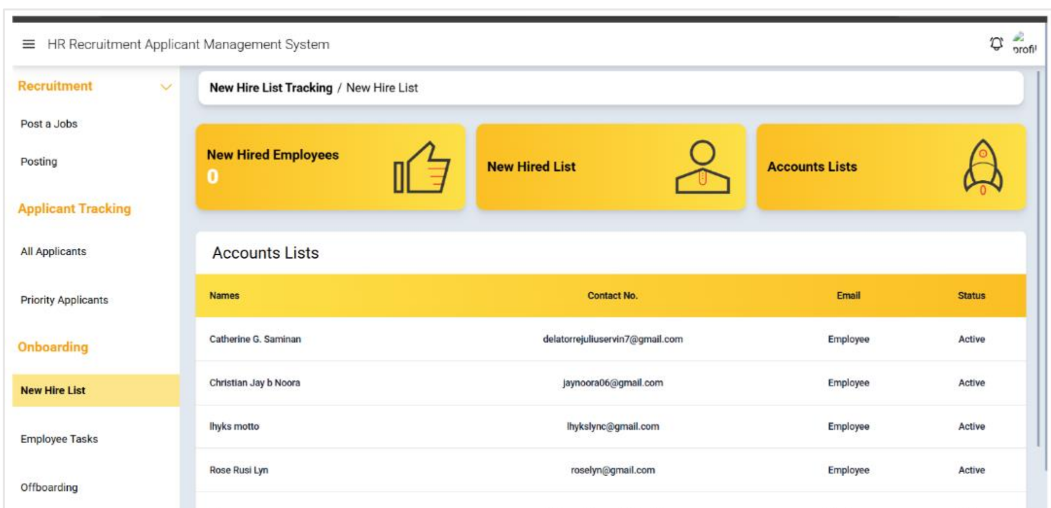
If the passed applicant is **hired**, the HR can now check the new hired list by navigating in sidebar



To see the **failed applicants**, navigate on all applicants by clicking in the sidebar and click the card Failed, the Delete button is for deleting the failed applicants or wait after 3 months, so they can apply again using the same used gmail.

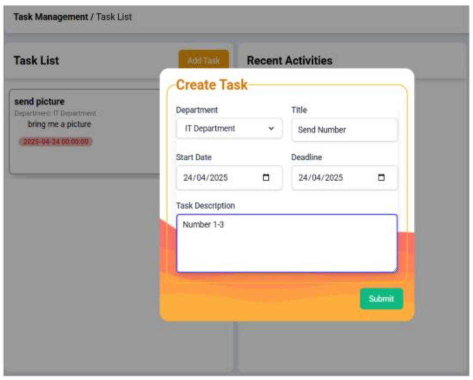


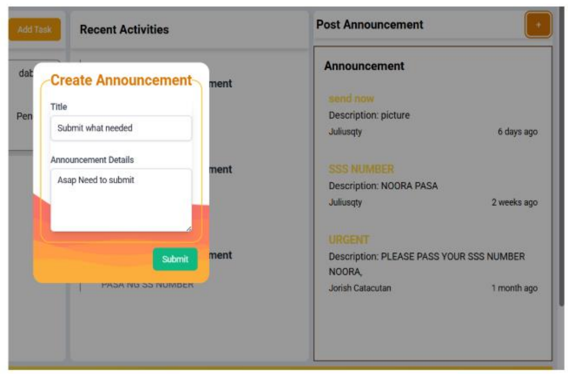
To see accounts list that created by Admin after they receive the new hired list to create an employee accounts for employee portal, click the Accounts Lists



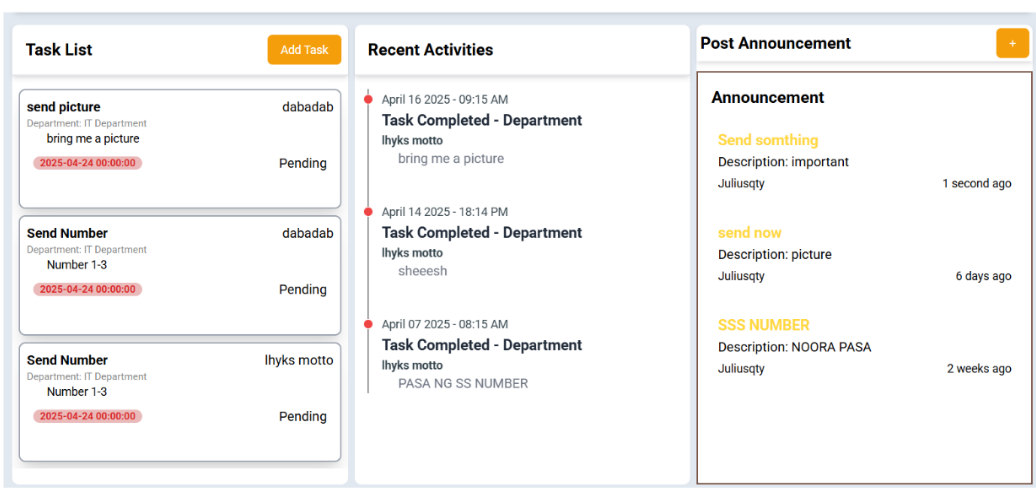
HR – TASK MANAGEMENT:

In Employee Tasks to Create a Task and Announcement Click the Add Task or the “+” Button to Post a Announcement



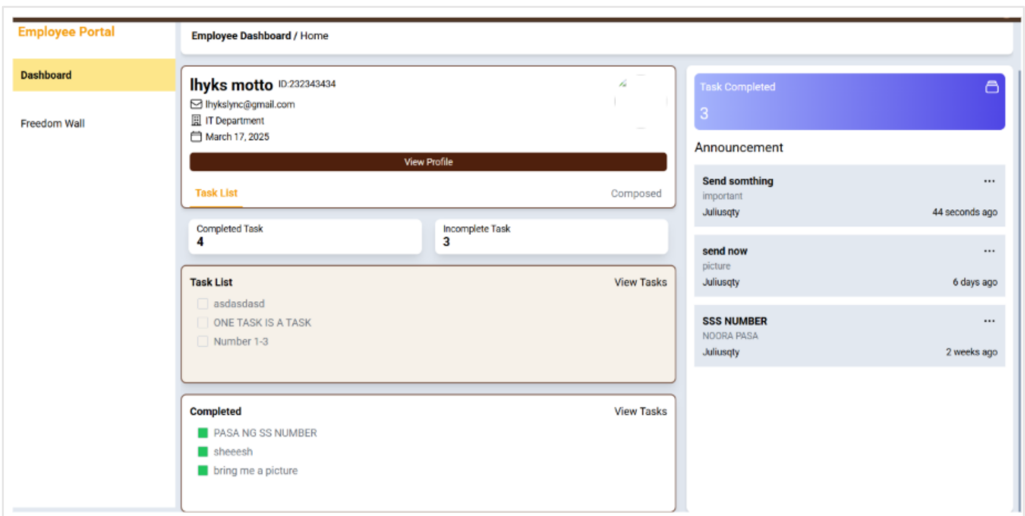


After the form is filled out and submitted, the task and announcement are posted on the employee portal and listed in the Employee Task section of the HR system.

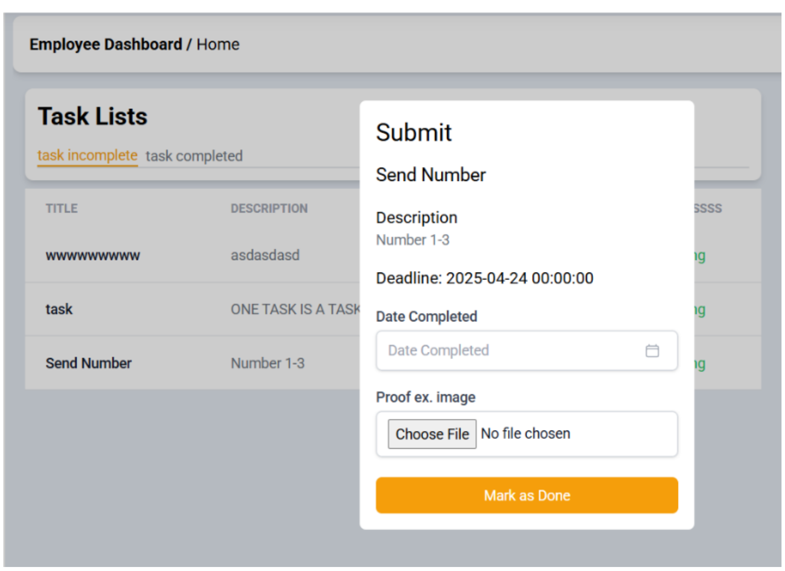


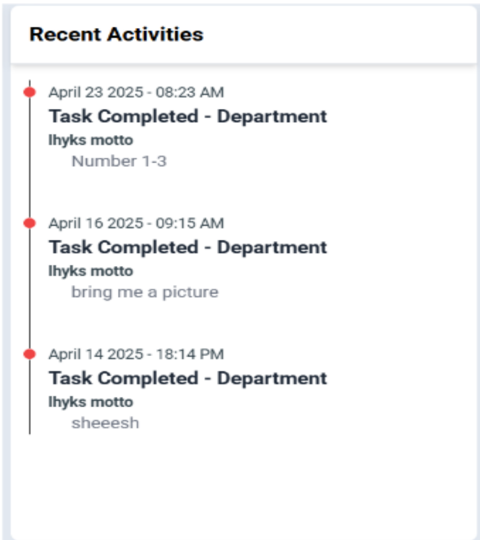
EMPLOYEE – TASK SUBMISSION:

To submit a completed task in the **employee portal**, the employee navigates to the dashboard and clicks on the task in the task list. The employee can also see the announcement on the right side of the dashboard.



Fill out the form and upload an image or Pdf file for proof.

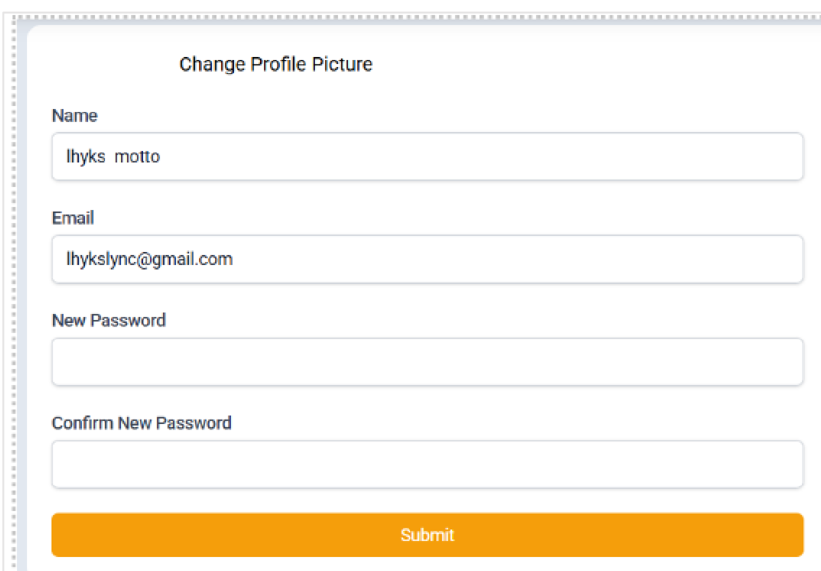




After the task is submitted, the HR can see the recent activities in Task management who submitted and complete a task

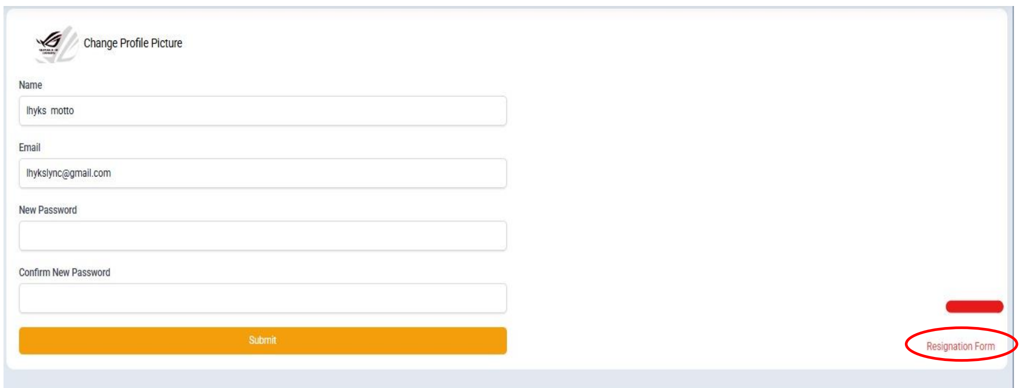
CHANGE PASSWORD (for Employee):

To change password or the profile picture the user employee click the View profile and fill-up form for change password, for change profile just click change profile and upload an image then submit.

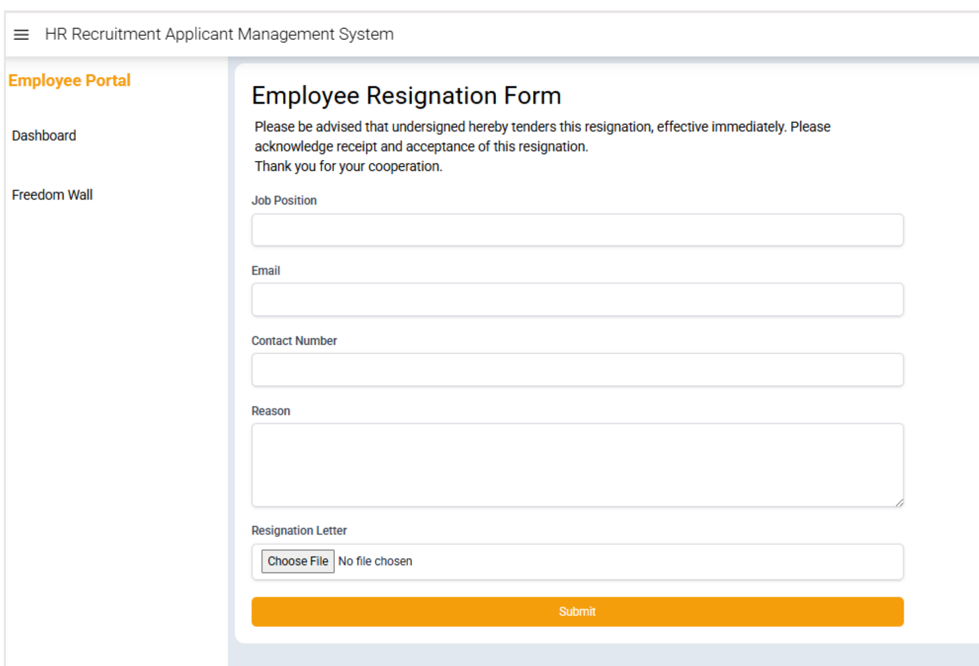


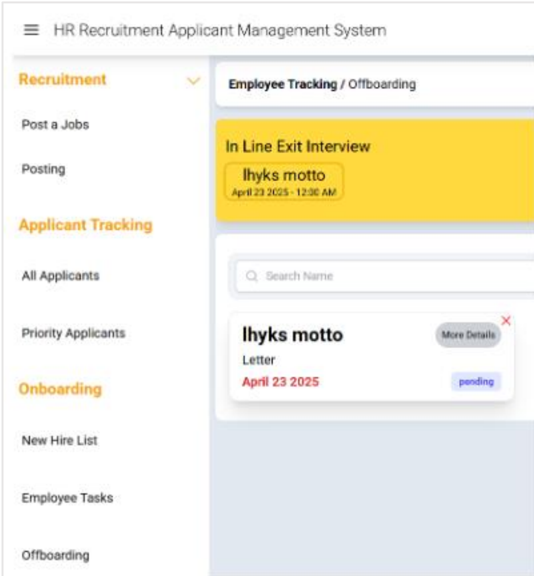
RESIGNATION/OFFBOARDING (for Employee):

If the employee wants to request a Resignation, they must navigate on View Profile then see on right side, click the Resignation form for it to show up



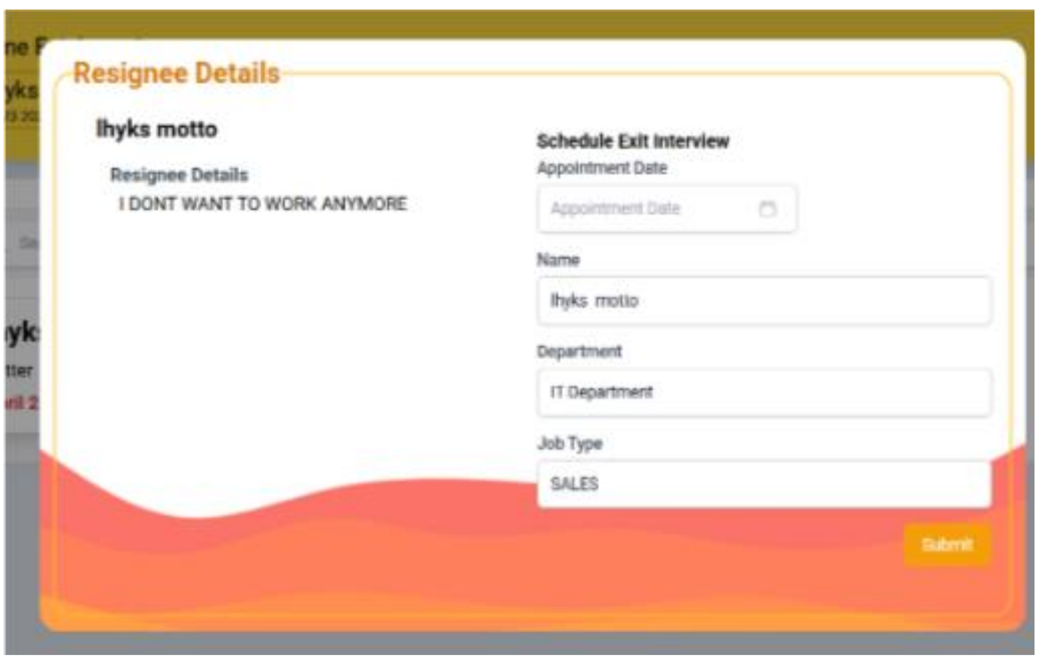
The employee can now fill out the resignation form, including their reason for resigning, and upload their resignation letter.



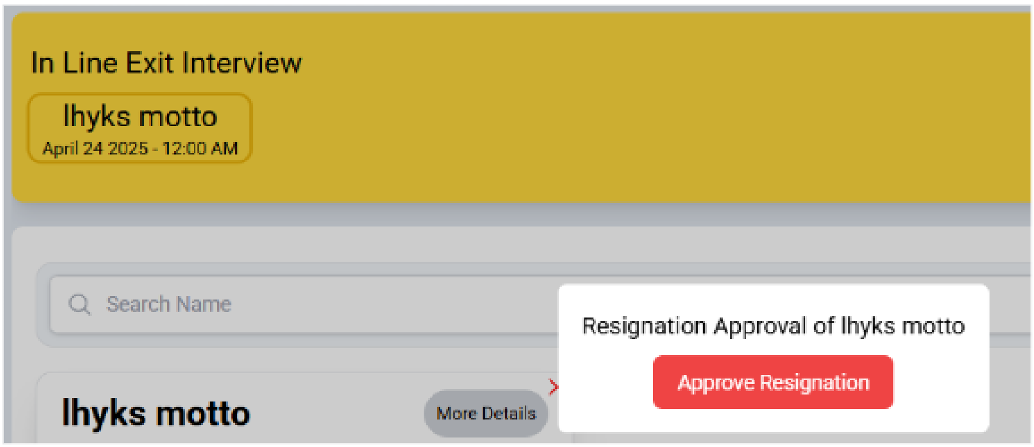


After submission the request is listed on HR offboarding management. The HR can now view the resignation details, and arrange a schedule for the exit interview of the offboarding employee.

If the HR wants to Schedule an Exit interview, simply fill-out the form to Schedule and the schedule is Highlighted and Queued the email of interview is sent to Resignee Email.

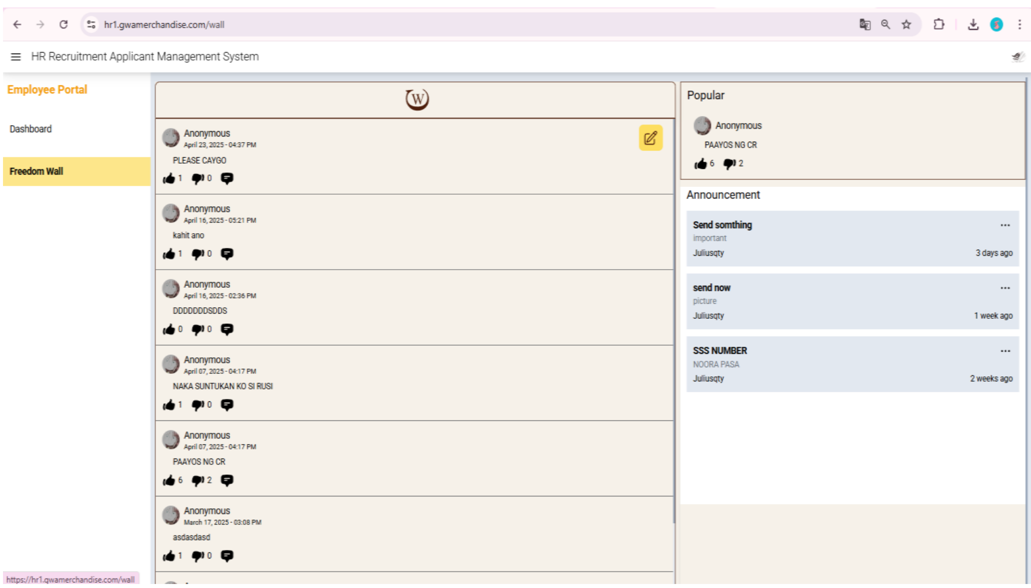


To Approve the Resignation, just click the Name of Employee then cick ”Approve Resignation” the notification was sent to Admin for Account termination of the Resigned Employee.



CONCERN WALL FOR EMPLOYEES (ANONYMOUS FREEDOM WALL) :

In Employee portal to post a Concern or Feedback, the user navigates on “**Freedom Wall**”. Click the Box Pen in Upper Right corner then write their concerns or feedback.



Once the Concern is Posted, other Employees will not be able to see who post which post. And everyone can react Like or Dislike and Comment the post similarly to a X platform (formerly Twitter). The most popular or Liked Post is highlighted in upper Right.

To Logout From Employee portal or HR system, in Upper Right Corner they can Click the Profile Picture then click logout.

6.12. Known Issues and Troubleshooting

To ensure a smooth user experience, we've compiled a list of common issues and frequently asked questions (FAQs) along with their solutions. If you encounter any technical difficulties, first verify that your system meets the minimum requirements, and that all software is up to date. Common troubleshooting steps include clearing your cache, restarting the application, and checking your internet connection. If problems persist, refer to the FAQs for detailed guidance on error messages, data synchronization issues, and configuration settings.

1. File Not Uploading:

Ensure your file meets the required format and size limits. Try refreshing the page and uploading again. If the problem continues, check your internet connection or try using a different browser.

2. Slow Loading:

Slow loading can be caused by a weak internet connection or high server traffic. Refresh the page or try accessing the application at a later time. Clearing your browser's cache may also help improve loading speed.

3. Submission Error:

If you receive an error when submitting a form or data, double-check that all required fields are filled in correctly. If the error persists, try submitting again after refreshing the page. If needed, log out and log back in before retrying. For further assistance, please contact our support team.

Full Name	Contact Number	E-mail Address
Jobert H. Camo	09948415967	jobertcamo@gmail.com
Jay F. Cullo	09813907336	jaycullo@gmail.com
Julius Ervin B. Dela Torre	09459788091	delatorrejuliuservin9@gmail.com
Christian Jay A. Noora	09691213125	jaynoora06@gmail.com
Roselyn A. Rosario	09927207693	roselynrosario6@gmail.com

Table 21. Contact Information for IT Support

6.13. Version Control and Source Code Repository

Repository Link: GitHub - CAPS-2 https://github.com/JobertCamo/caps-2?fbclid=IwY2xjawI2CoNleHRuA2FlbQIxMAABHVMROSbtZH0NytGIswqSey9rrpnU2z0AFfnKMI6V7f5349j5aaacxlLqoQ_aem_XEGXFnHciU5-Igx5KCv08A

6.14. DevOps and Continuous Integration/Continuous Deployment (CI/CD)

The Laravel CI/CD Pipeline is implemented for the system to automate building, testing, and deploying a Laravel application using GitHub Actions. It installs dependencies, runs tests with PHPUnit and Pest, and deploys to staging or production based on the branch. Notifications ensure a smooth and efficient development process.

name: Laravel CI/CD Pipeline

```
# Trigger the workflow on push to specific branches or on pull requests
on:
  push:
    branches:
      - main
      - staging
  pull_request:
    branches:
      - main
      - staging

jobs:
  build:
    name: Build
    runs-on: ubuntu-latest

    steps:
      # Checkout the repository
      - name: Checkout Repository
        uses: actions/checkout@v3

      # Set up PHP with required version and extensions
      - name: Set up PHP
        uses: shivammathur/setup-php@v2  # Correct version
        with:
          php-version: '8.2' # Specify your PHP version
          extensions: mbstring, bcmath, xml, ctype, json, tokenizer, curl, gd,
openssl
          coverage: none
```

```yaml
    # Install Composer dependencies
    - name: Install Dependencies
      run: |
        composer install --prefer-dist --no-progress --no-suggest --no-interaction
    # Create or copy .env file
    - name: Create .env File
      run: cp .env.example .env

    # Generate application key
    - name: Generate APP Key
      run: |
        php artisan key:generate

    # Build assets (if applicable, e.g., using Laravel Mix)
    - name: Build Assets
      run: |
        npm install
        npm run build

    # Notify Build Success (Optional: GitHub automatically shows build status)
    # You can integrate with other notification services like Slack or Email here
    - name: Notify Build Success
      if: success()
      run: echo "Build succeeded!"

    # Notify Build Failure
    - name: Notify Build Failure
      if: failure()
      run: echo "Build failed!"

  test:
    name: Run Tests
    runs-on: ubuntu-latest
    needs: build
```

```yaml
    steps:
      # Checkout the repository
      - name: Checkout Repository
        uses: actions/checkout@v3

      # Set up PHP
      - name: Set up PHP
        uses: shivammathur/setup-php@v2  # Correct version
        with:
          php-version: '8.2'
          extensions: mbstring, bcmath, xml, ctype, json, tokenizer, curl, gd,
openssl
          coverage: none

      # Install Composer dependencies
      - name: Install Dependencies
        run: |
          composer install --prefer-dist --no-progress --no-suggest --no-
interaction

      - name: Set environment variables
        run: |
          echo "APP_KEY=base64:YOUR_GENERATED_KEY_HERE" >>
$GITHUB_ENV  # Replace with your generated APP_KEY
        # Run Tests
      - name: Run Tests
        run: |
          php artisan migrate --env=testing --force
          ./vendor/bin/pest

      # Run PHPUnit tests
      - name: Run Tests
        run: |
          php artisan migrate --env=testing --force
          ./vendor/bin/phpunit --coverage-text
```

```yaml
      # Notify Test Success
      - name: Notify Test Success
        if: success()
        run: echo "Tests passed!"

      # Notify Test Failure
      - name: Notify Test Failure
        if: failure()
        run: echo "Tests failed!"

  deploy_staging:
    name: Deploy to Staging
    runs-on: ubuntu-latest
    needs: test
    if: github.ref == 'refs/heads/staging'

    steps:
      # Checkout the repository
      - name: Checkout Repository
        uses: actions/checkout@v3

      # Deploy to Staging Server
      - name: Deploy to Staging
        env:
          SSH_PRIVATE_KEY: ${{ secrets.STAGING_SSH_KEY }}
          SERVER_USER: ${{ secrets.STAGING_SERVER_USER }}
          SERVER_IP: ${{ secrets.STAGING_SERVER_IP }}
          DEPLOY_PATH: /var/www/staging
        run: |
          mkdir -p ~/.ssh
          echo "$SSH_PRIVATE_KEY" > ~/.ssh/id_rsa
          chmod 600 ~/.ssh/id_rsa
          ssh-keyscan $SERVER_IP >> ~/.ssh/known_hosts
          rsync -avz --exclude='.git*' --exclude='storage/logs/*' ./
$SERVER_USER@$SERVER_IP:$DEPLOY_PATH
```

```
          ssh $SERVER_USER@$SERVER_IP "cd $DEPLOY_PATH &&
composer install --no-dev --optimize-autoloader && php artisan migrate --
force && php artisan config:cache && php artisan route:cache"

      # Notify Staging Deployment
      - name: Notify Staging Deployment
        if: success()
        run: echo "Deployed to Staging successfully!"

  deploy_production:
    name: Deploy to Production
    runs-on: ubuntu-latest
    needs: test
    if: github.ref == 'refs/heads/main'

    steps:
      # Checkout the repository
      - name: Checkout Repository
        uses: actions/checkout@v3

      # Deploy to Production Server
      - name: Deploy to Production
        env:
          SSH_PRIVATE_KEY: ${{ secrets.PRODUCTION_SSH_KEY }}
          SERVER_USER: ${{ secrets.PRODUCTION_SERVER_USER }}
          SERVER_IP: ${{ secrets.PRODUCTION_SERVER_IP }}
          DEPLOY_PATH: /var/www/production
        run: |
          mkdir -p ~/.ssh
          echo "$SSH_PRIVATE_KEY" > ~/.ssh/id_rsa
          chmod 600 ~/.ssh/id_rsa
          ssh-keyscan $SERVER_IP >> ~/.ssh/known_hosts
          rsync -avz --exclude='.git*' --exclude='storage/logs/*' ./
$SERVER_USER@$SERVER_IP:$DEPLOY_PATH
          ssh $SERVER_USER@$SERVER_IP "cd $DEPLOY_PATH &&
composer install --no-dev --optimize-autoloader && php artisan migrate --
force && php artisan config:cache && php artisan route:cache"
```

```yaml
# Notify Production Deployment
- name: Notify Production Deployment
  if: success()
  run: echo "Deployed to Production successfully!"
```

6.15. Licensing and Open-Source Libraries

This system project used Gemini API “free tier” to implement AI Resume Evaluation. This “free tier” API had 600 free trial analyzation and generating a response to show the evaluated resume result. This Gemini API is free of charge.

6.16. Performance Metrics and Monitoring

Metrics collected and monitored

This graph displays the average system response times across various user actions, with all scenarios performing between 1 to 2 seconds.

Login and API Request (Applicant) recorded the fastest response times at approximately 1 second, indicating highly efficient server handling.

Dashboard Loading, Updating Task Status, and Submitting Resignation Letter showed response times around 2 seconds, reflecting smooth system navigation even during more complex operations.

Job Application Submission and Document Upload were slightly faster, maintaining a consistent speed close to 1 second.

Overall, the system demonstrates excellent performance, with all user interactions achieving quick and stable response times, well within industry best practices.

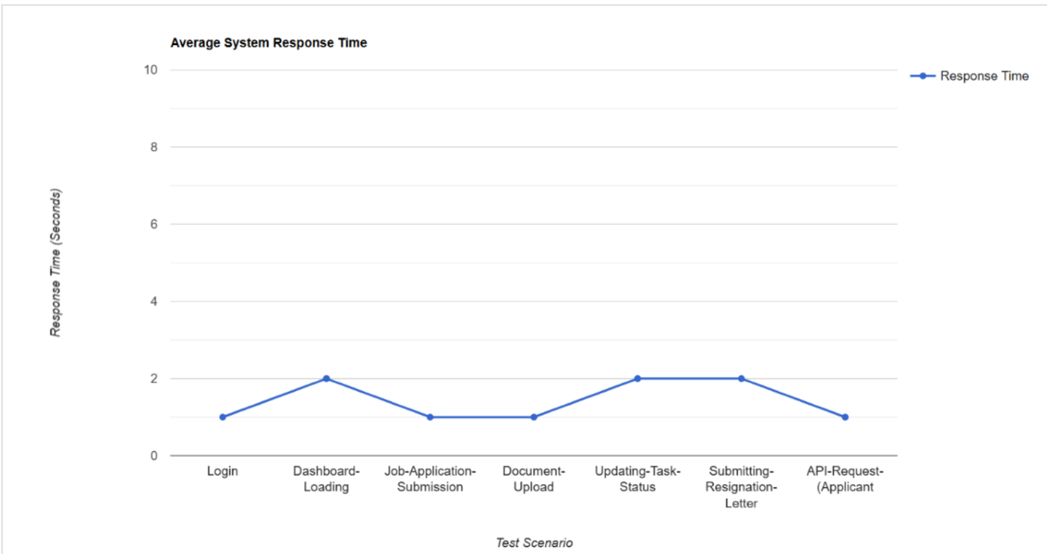


Figure 24. Average System Response Time

Tools and dashboards used for performance analysis

To monitor and analyze system performance during testing and deployment, the following tools and dashboards were utilized:

1. **Laravel Telescope**: Used for real-time monitoring of application requests, response times, database queries, errors, and queue jobs. It provided detailed insights into backend performance and system health.
2. **Server Management Dashboard (Indevfinite Hosting Panel)**: Monitored server resource usage including CPU load, memory consumption, and disk utilization to ensure hosting stability and performance.

3. **Custom Admin Activity Logs**: Recorded user actions such as login attempts, form submissions, document uploads, and task completions, helping the development team track usage patterns and spot potential bottlenecks.

4. **Browser Developer Tools (Google Chrome DevTools)**: Used during frontend testing to measure loading times, inspect network request speeds, and optimize frontend asset performance.

Actions taken based on performance data

Based on the results gathered from the monitoring tools, several actions were implemented to improve system performance:

Database Query Optimization: Indexed frequently accessed tables and refactored inefficient SQL queries, resulting in faster data retrieval for user dashboards and application submissions.

2. **Frontend Asset Optimization**: Minified JavaScript and CSS files, reduced image sizes, and implemented lazy loading techniques to decrease page load times.

3. **Server Configuration Improvements**: Adjusted server cache settings and PHP processing limits to handle a higher number of concurrent requests more efficiently.

4. **Queue Management for Email Notifications**: Implemented background job queues for email dispatching, preventing delays in user interactions while still ensuring timely email delivery.

5. **Session and Cache Management Enhancements**: Improved session storage and implemented server-side caching strategies to maintain consistent system speed during peak usage.

As a result of these actions, system response times improved significantly, maintaining consistent speeds between **1 to 2 seconds** across all major user interactions.

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CERTIFICATE OF SIMILARITY

This certifies that the research titled, **MERCHANDISING MANAGEMENT SYSTEM (HUMAN RESOURCE 1): ONBOARDING AND OFFBOARDING, TASK MANAGEMENT, AND ONLINE JOB POSTING WITH AI RESUME EVALUATION** has been checked and passed with a Turnitin similarity index of **2%**. Given this **27th of May, 2025.**

JOSHUA LOVIC D. ABAYA, RL
Librarian

JOHN PROS B. VALENCIA, RL, MLIS
Library Director

TC2420814

DECLARATION

The researchers of this group certify that this project study does not incorporate, without acknowledgement, any material previously submitted for a degree of diploma in any university and to the best of our knowledge and belief. It does not contain any material previously published or written by another person or ourselves except where due reference is made in the text. Furthermore, also hereby give consent for our Project Study. If accepted, to be made available for photocopying and for interlibrary and for the title and summary to be available to outside organizations.

Signature of Group/ Individual: Date: May 2025

Jobert H. Camo

Jay F. Cullo

Julius Ervin B. Dela Torre

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Roselyn A. Rosario

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Project Adviser

JOBERT CAMO

INFORMATION TECHNOLOGY

CONTACT

0994 841 5967

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Brgy. Pasong Putik Maligaya
Novaliches Quezon City

TECHNICAL SKILLS

- **Web Development**: Laravel, Livewire, Alpine.js, Tailwind CSS
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Database Management**: MySQL

OBJECTIVES

I'm an IT student with experience in backend development and some knowledge of frontend. I'm looking for a role where I can improve my skills, work on real projects, and learn more about building websites and systems.

EDUCATION

BESTLINK COLLEGE OF THE PHILIPPINES

Bachelor of Science in Information Technology, Major in Network Administration

2021-2025

ACCESS COMPUTER COLLEGE

TVL - Home Economics

2020 - 2021

CERTIFICATION

- Truline Certificate
  Completed training on emerging technologies including basics of Artificial Intelligence (AI), Java programming, and Python scripting, awarded by Truline Technology Center.

JAY CULLO
INFORMATION TECHNOLOGY

CONTACT

09813907336

jaycullo08@gmail.com

Canumay West, Valenzuela City

SKILLS

- MS Word
- Photo/Video Editing

OBJECTIVES

Demonstrated adaptability in a rapidly changing work environment, adjusting priorities and workflows to meet evolving business needs.

EDUCATION

BESTLINK COLLEGE OF THE PHILIPPINES
Bachelor of Science in Information Technology, Major in Network Administration

2021-2025

BESTLINK COLLEGE OF THE PHILIPPINES
ICT - Information and Communication Technology
2019 - 2020

CERTIFICATION

JVD EVENT AND TRAVEL MANAGEMENT COMPANY SESSION 2:
For his/her active participation during the conduct of a three days live in seminars with the theme “Techscape: Discover Innovation in the Cool Climes of BSIT”
November 27, 2023

TRUELINE HOLDINGS AND DEVELOPMENT PTE. LTD.
For participating 8 modules learning with a theme of " Information Management in the Digital Age”
November 29, 2024

BESTLINK COLLEGE OF THE PHILIPPINES (RESEARCH & DEV) For participating in the research forum 2024 with the theme “Cultivating a multidisciplinary Research Culture with Artificial Intelligence: Integration, Innovation, and Impacts”
September 8, 2024

JULIUS ERVIN B.
DELA TORRE
INFORMATION TECHNOLOGY

CONTACT

+63 9459788091

delatorrejuliuservin9@gmail.com

3 Cassanova Drive Barangay Culiat Q.C.

TECHNICAL SKILLS

Frontend Development:

- Languages: HTML, CSS, JavaScript
- Frameworks & Libraries: Tailwind CSS
- Basic Graphic Designing & Editing

Version Control:

- Git / Github

PC TROUBLESHOOTING & HARDWARE:

- PC Reset & Software Troubleshooting
- Laptop Disassembly & Assembly
- Basic Hardware Diagnostics

OTHERS

- Microsoft Office (MS Word, Excel & PowerPoint)
- Encoder
- Multi Tasking
- File Management

OBJECTIVES

Hi! I'm Julius, i am willing to do everything I can to achieve my dreams, enjoying and dedicating all my effort and time to surpass life's challenges. Now, as an applicant to your company, I promise to sincerely perform any job assigned to me and do my best to contribute to your company.

EDUCATION

BESTLINK COLLEGE OF THE PHILIPPINES

Bachelor of Science in Information Technology, Major in Network Administration

2021-2025

SAN FRANCISCO HIGH SCHOOL

Information and Communication Technologies

2020 - 2021

CERTIFICATION

Certificate of Participation

April 20, 2023
“STAY & LEARN” BITZ 2023 Session 2: Understanding the Innovators’ Role in Digital Transformation

TRULINE HOLDINGS AND DEVELOPMENT PTE LTD
October 05, 2024
Information Management in the Digital Age

“I hereby certify that the above information is true and correct to the best of my knowledge and belief.”

CHRISTIAN JAY NOORA

INFORMATION TECHNOLOGY

CONTACT

09691213125

jaynoora06@gmail.com

blk 13 Lot 08 whispering palm subd. Llano cal. city

SKILLS

- Hardware & Software Maintenance
- Installation & Maintenance of Operating Systems (Windows)
- Basic Network Troubleshooting
- Computer Hardware and Software Troubleshooting
- Problem-Solving & Analytical Thinking
- Effective Communication & Team Collaboration

OBJECTIVES

I'm a passionate and driven 4th-year IT student at Bestlink College of the Philippines (BCP), specializing in Network Administration. I'm eager to gain hands-on experience through an On-the-Job Training (OJT) placement, where I can apply and refine my skills in IT support, help desk assistance, hardware & software troubleshooting, and network administration. I look forward to contributing to a dynamic IT team while learning from real-world challenges.

EDUCATION

BESTLINK COLLEGE OF THE PHILIPPINES

Bachelor of Science in Information Technology, Major in Network Administration

2021-2025

ACCESS COMPUTER COLLEGE

ICT - Information and Communication Technology

2020 - 2021

CERTIFICATION

- Truline Certificate
  Completed training on emerging technologies including basics of Artificial Intelligence (AI), Java programming, and Python scripting, awarded by Truline Technology Center.

ROSELYN ROSARIO

INFORMATION TECHNOLOGY

CONTACT

09927207693

roselynrosario6@gmail.com

Phase 4C Package 6 Bagong Silang, Caloocan City

SKILLS

- Graphic Design
- Layout Design
- MS Word, Excel
- Photo Editing

OBJECTIVES

To be able to adapt to challenges and acquire new skills to which I can contribute my knowledge in this field, and my dedication for improvement.

EDUCATION

BESTLINK COLLEGE OF THE PHILIPPINES

Bachelor of Science in Information Technology, Major in Network Administration

2021-2025

ACCESS COMPUTER COLLEGE

ICT - Information and Communication Technology

2020 - 2021

CERTIFICATION

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  Completed training on emerging technologies including basics of Artificial Intelligence (AI), Java programming, and Python scripting, awarded by Truline Technology Center.